质谱仪的组成部分

进样系统 → 离子化方式 → 质量分析器 → 检测器 → 数据处理系统

电喷雾离子化
ESI
大气压化学离子化APCI
大气压光离子化 *APPI*
基质辅助激光解析电离
(MALDI)

Quadropole
Ion Trap
TOF
Magnetic
FT-ICR
Obitrap

电子倍增器

二、离子化的方法

电子轰击电离 Electron Impact Ionization，EI

化学离子化 Chemical Ionization，CI

场电离，场解吸 Field Ionization FD，Field Desorption FD

快原子轰击 Fast Atom Bombardment，FAB

基质辅助激光解析电离 Matrix-Assisted Laser Desorption Ionization，MALDI

电喷雾电离 Electrospray Ionization，ESI

大气压化学电离 Atmospheric Pressure Chemical Ionization，APCI

电喷雾电离（ESI）和

大气压化学电离（APCI）

电喷雾电离（ESI）第一步：离子的产生

待测物质在溶液中的状态（溶液化学）

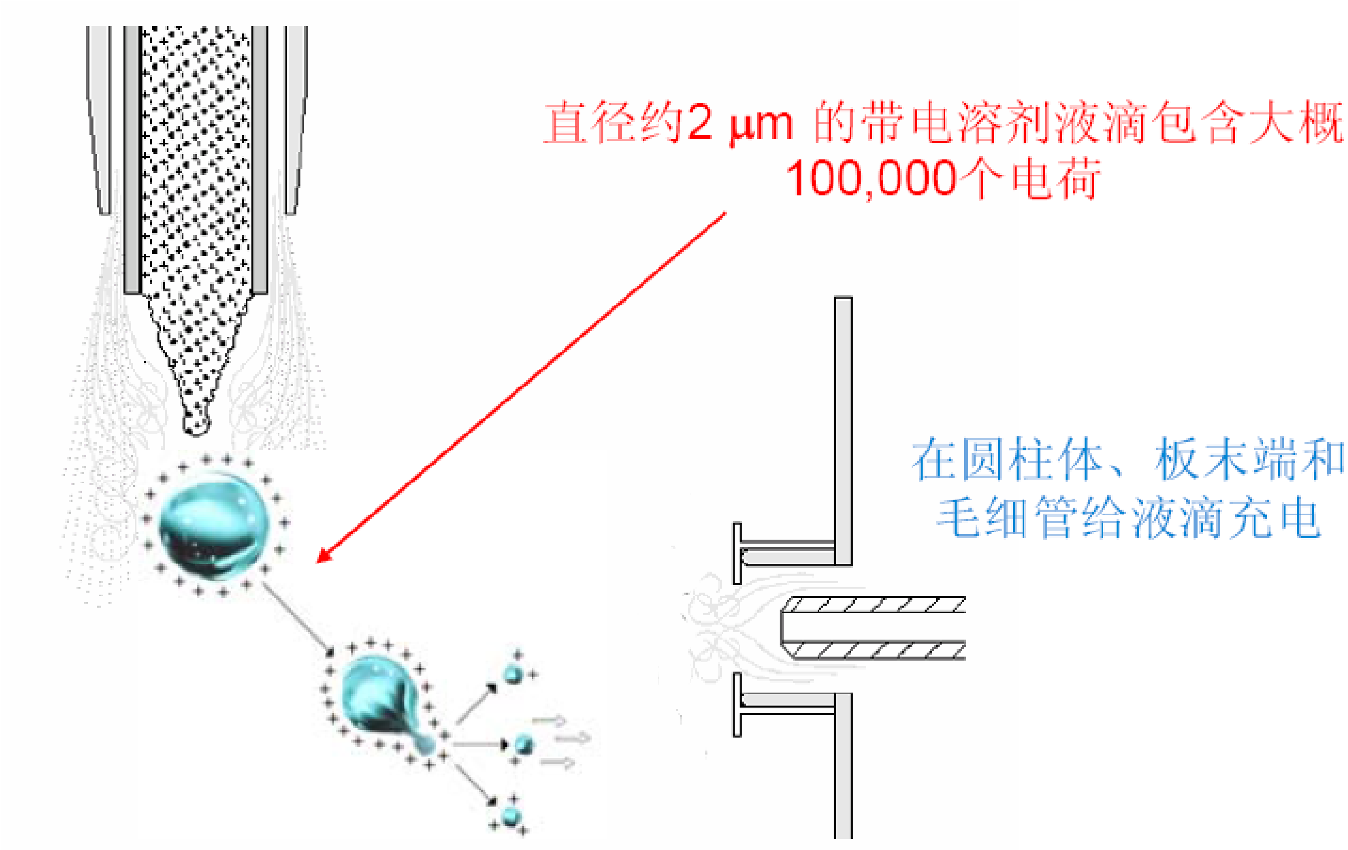
$B + RCOOH \longrightarrow BH^{+} + RCOO^{-}$

有机碱

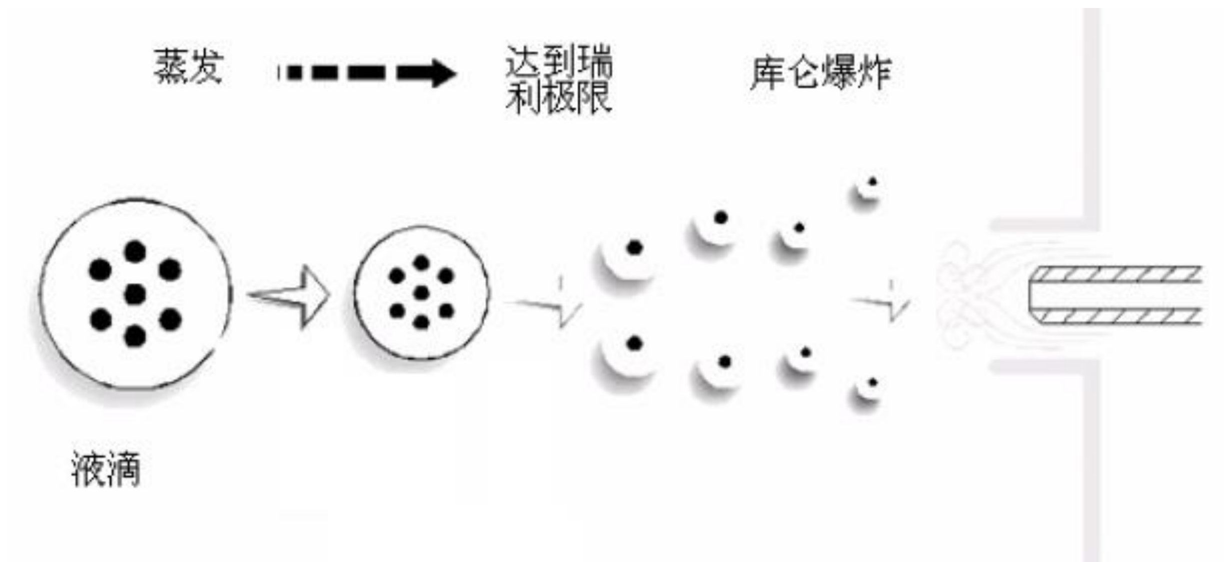
$HA + NH3 \longrightarrow A^{-} + NH_{4}$

有机酸

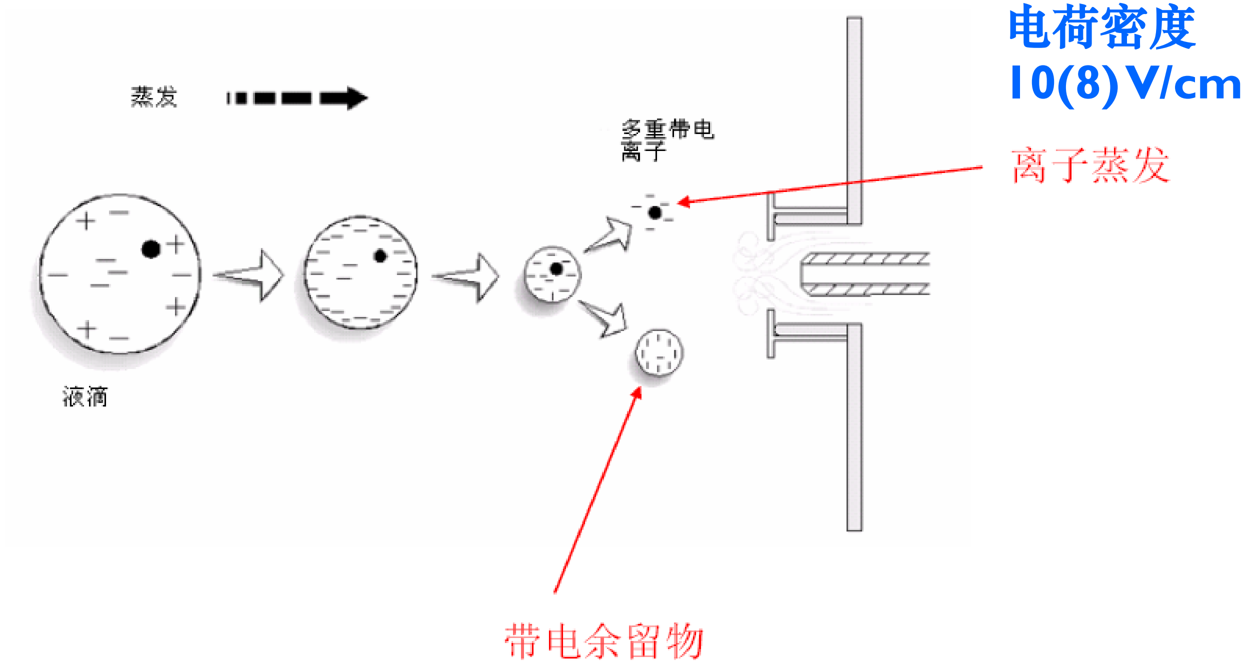
第二步：雾化



第三步：去溶剂化

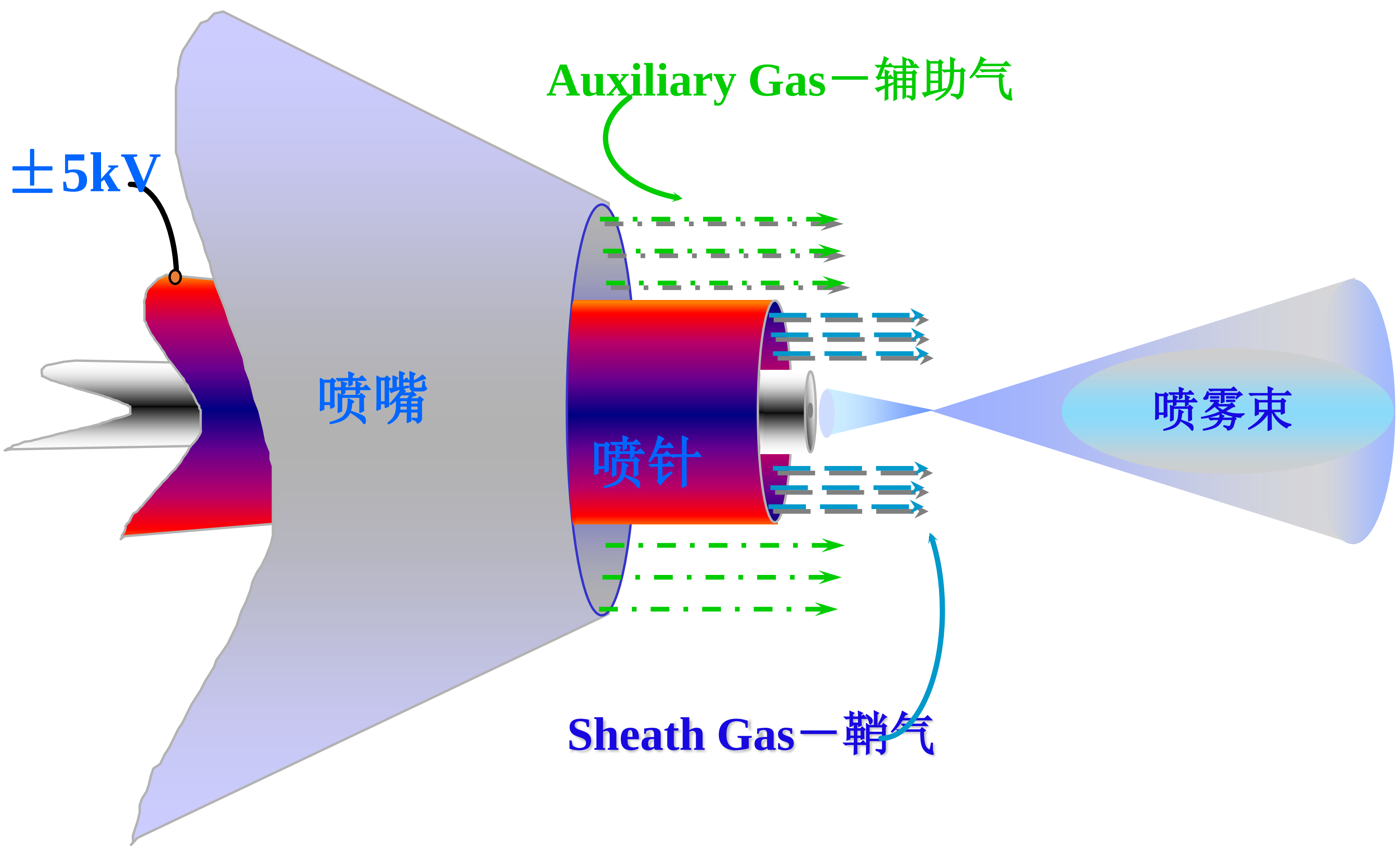


第四步: 离子从溶剂的解吸附（气相离子的产生）

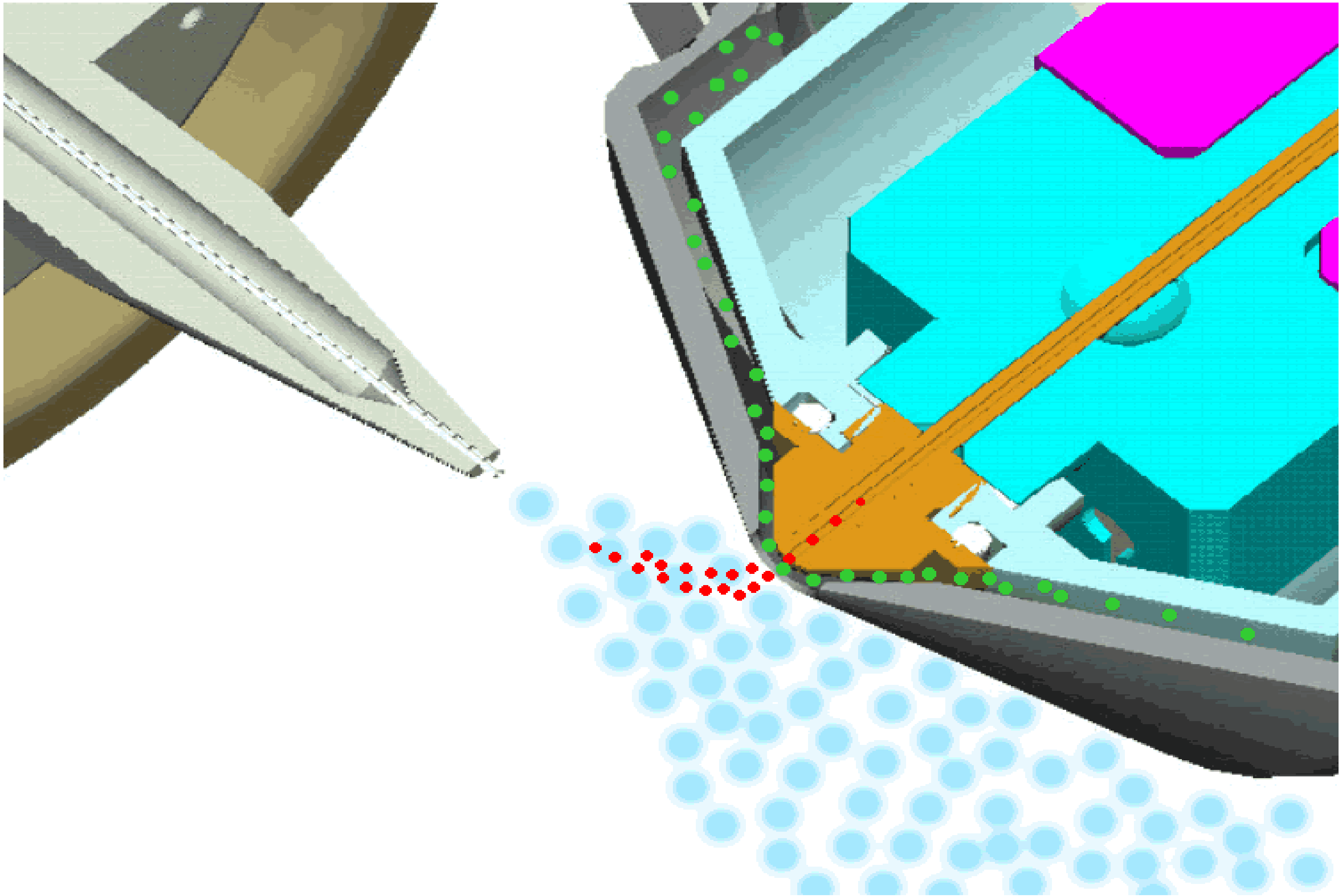


~为了把 ESI 灵敏度最大化，使用挥发性的缓冲液，不形成离子对。

ESI 喷嘴横截面



ESI



大气压化学电离 (APCI)

- 通过电晕放电离子化
- 大气压化学电离有三个过程

1. 在放电电极电晕放电作用下，氮气载气和气化的液相色谱溶剂发生反应，产生**预反应离子**。

$$O_2 + e^- \rightarrow O_2^{+\cdot} + 2e^-$$
$$N_2 + e^- \rightarrow N_2^{+\cdot} + 2e^-$$

2. 通过一系列复杂反应，预反应离子与溶剂分子反应形成**溶剂离子** H_3O^+和CH_3OH^+

3. 溶剂离子与被分析物分子反应，在正电离模式下，形成正的分子离子；在负的电离模式下，形成负的分子离子

$$H_3O^+ + \text{Analyte} \rightarrow (\text{Analyte} + H)^+ + H_2O$$
$$OH^- + \text{Analyte} \rightarrow (\text{Analyte} - H)^- + H_2O$$

大气压光电离 (APPI)

- 通过APCI探头去溶剂化、气化
- 借助紫外光灯源离子化
- 电离过程分两步：

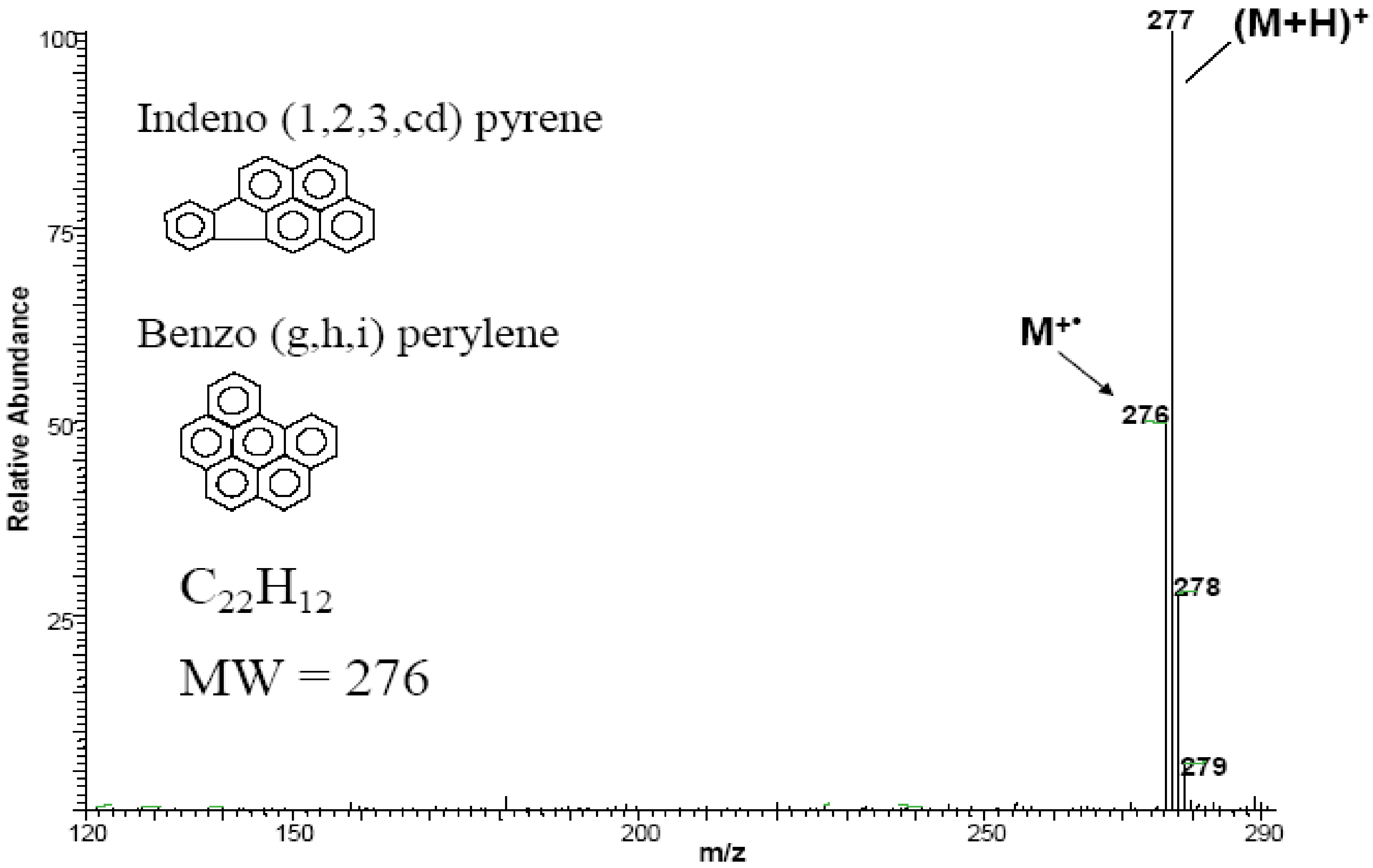
1. 被分析物质与紫外光源发生相互作用（氪灯发射10.0到10.6电子伏特的光子）。如果被分析物质的电离势能（IP）小于光子能量，那么被分析物质被电离成分子离子

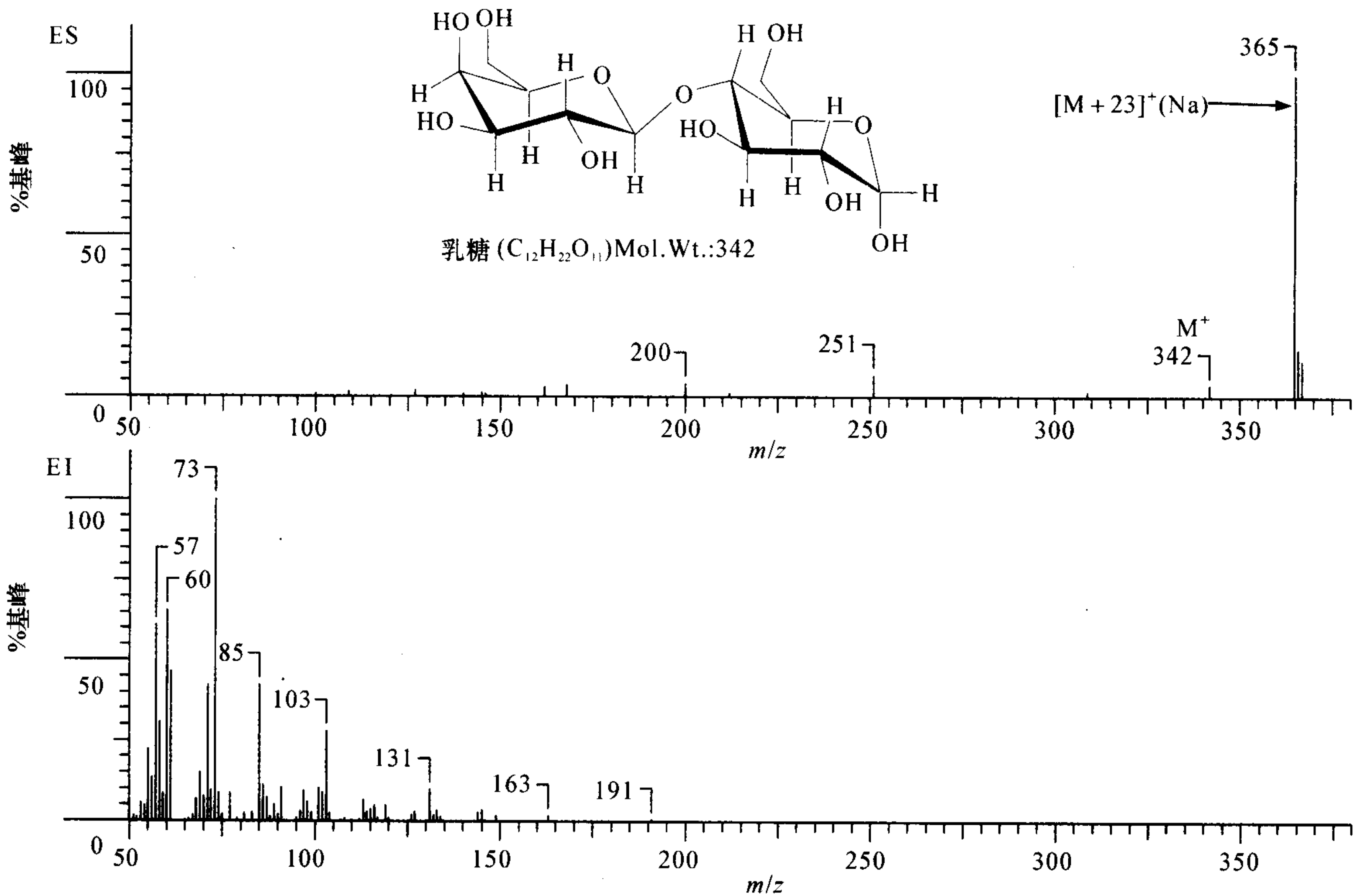
$$M + h\nu \rightarrow M^{+} + e^{-}$$

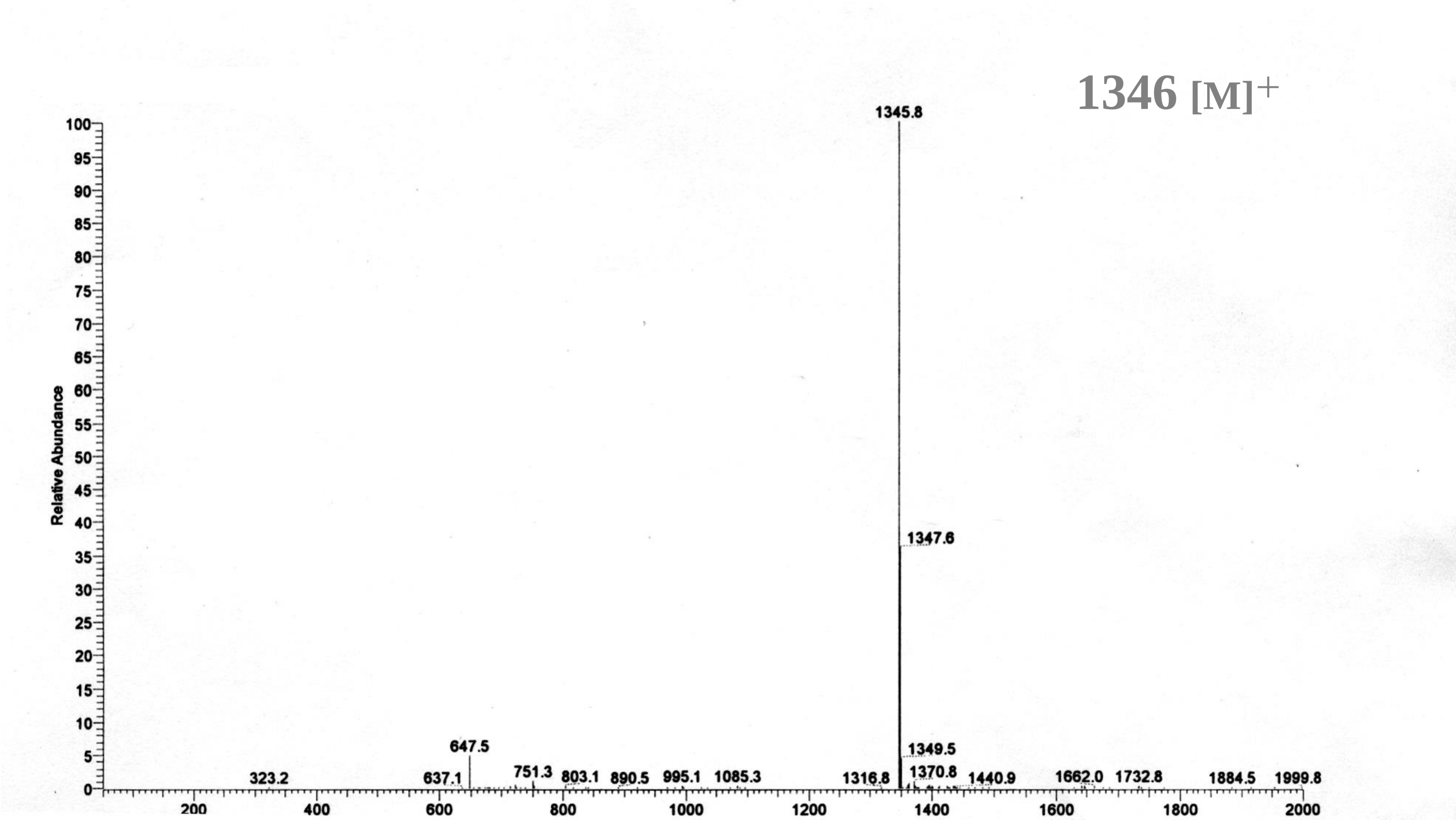
2. 如果在质子溶剂介质中，分子离子会接受质子而形成[M+H]+离子

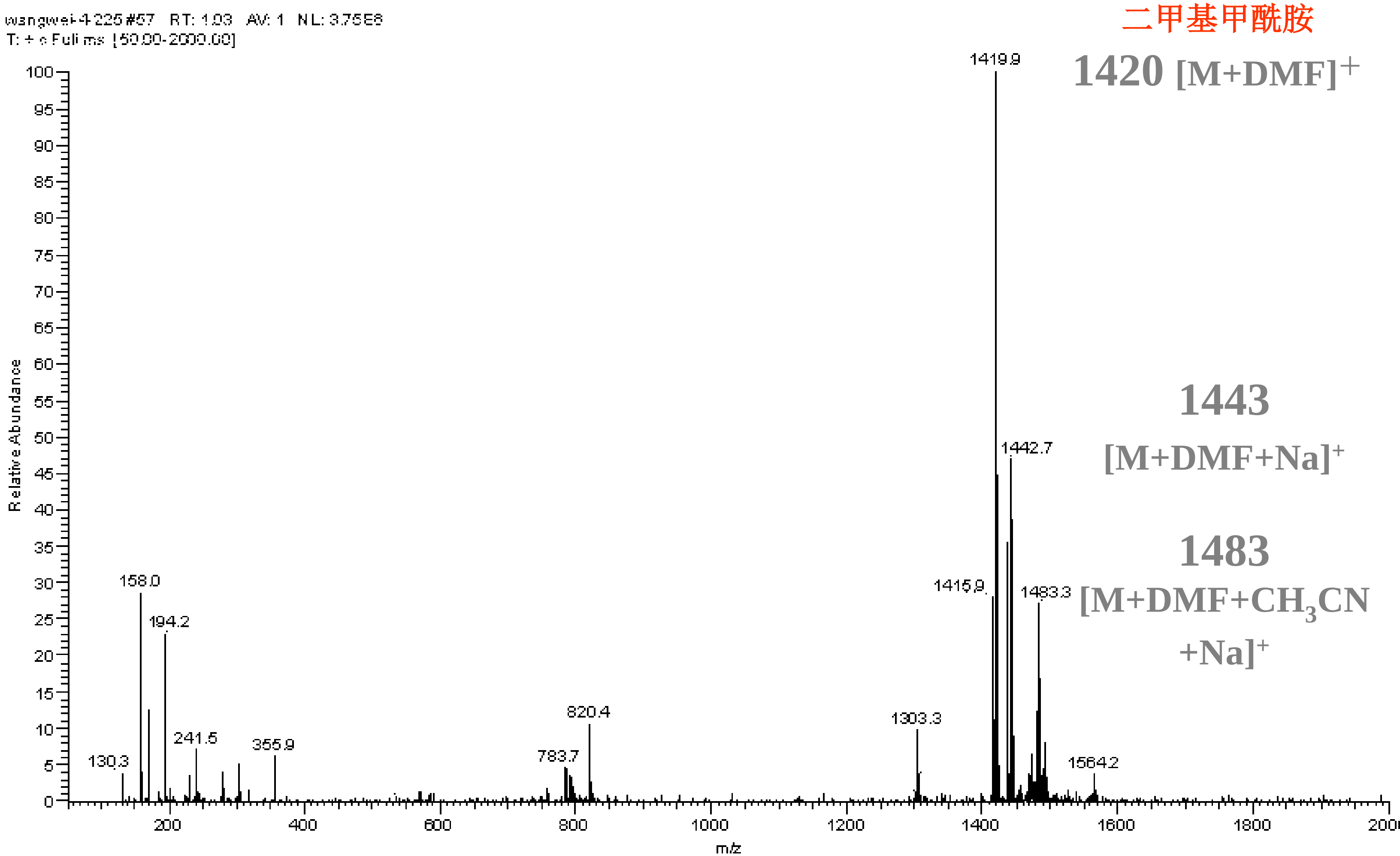
$$M^{+} + S \rightarrow [M+H]^{+} + [S\text{-}H]$$

PAHs APPI 质谱图




ES
EI
%基峰
乳糖 ($C_{12}H_{22}O_{11}$)Mol.Wt.:342
365
[M+23]⁺(Na)
M⁺
342
200
251
73
57
60
85
103
131
163
191
m/z
0
50
100
150
200
250
300
350


1346 [M]+
1345.8
1347.6
1349.5
1370.8
1440.9
1662.0
1732.8
1884.5
1999.8
1316.8
1085.3
995.1
890.5
803.1
751.3
647.5
637.1
323.2
Relative Abundance



二甲基甲酰胺
1420 [M+DMF]$^{+}$
1443
[M+DMF+Na]$^{+}$
1483
[M+DMF+CH_3CN
+Na]$^{+}$
wangwei-4-225 #57 RT: 1.03 AV: 1 NL: 3.75E6
T: + c Full ms [50.00-2000.00]
Relative Abundance
m/z
1419.9
1442.7
1415.9
1483.3
158.0
194.2
130.3
241.5
355.9
783.7
820.4
1303.3
1564.2

离子源选择 (需要考虑的化学因素)

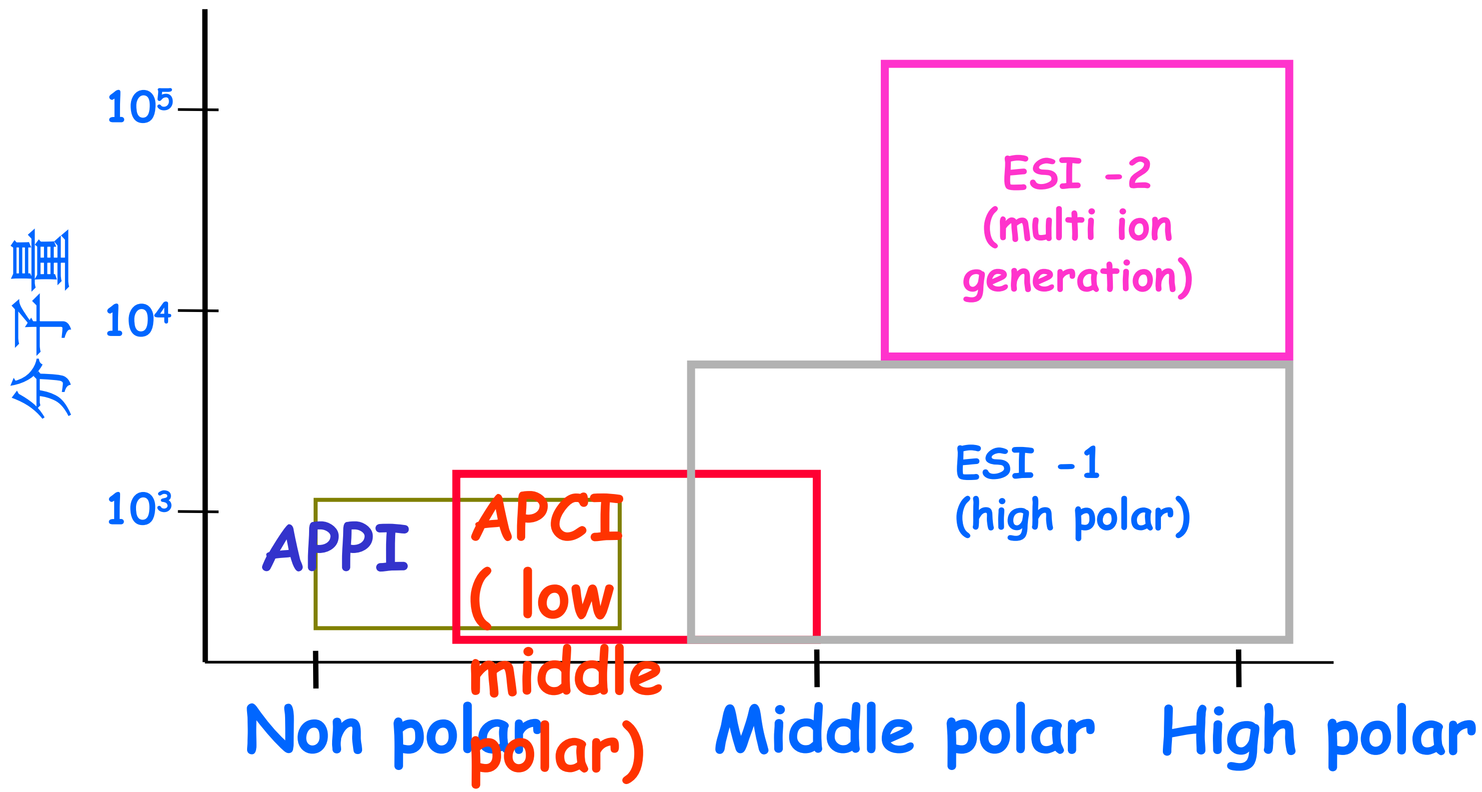
ESI (H-ESI):

- ✓ 离子在液态中形成 ***(Ions formed by solution chemistry?)***
- ✓ 对热不稳定化合物首选 ***(Good for thermally labile analytes)***
- ✓ 对极性或半极性化合物首选 ***(Good for polar / semi-polar analytes)***
- ✓ 可形成多电荷离子分析蛋白质、多肽等大分子物质 ***（Good for large molecules (proteins / peptides)***

APCI/APPI:

- ✓ 离子从气态中形成***（Ions formed by gas phase chemistry）***
- ✓ 对易挥发、热稳定的化合物首选***（Good for volatile / thermally stable）***
- ✓ 对弱极性或半极性化合物首选 ***（Good for non-polar / semi-polar）***
- ✓ 只可形成单电荷离子，可分析小分子化合物 ***（Good for small molecules (steroids)***
- ✓ **APPI**对化学结构中有共轭双键的化合物分析有利 ***（Good for ions containing a chromophore (APPI)***

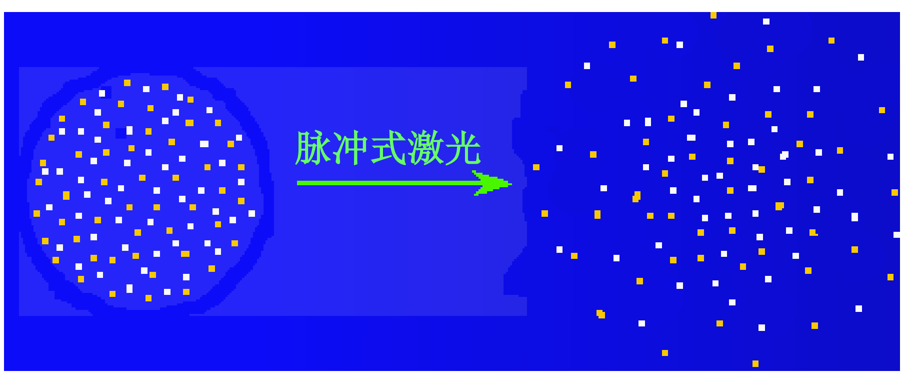
大气压下离子化适用范围



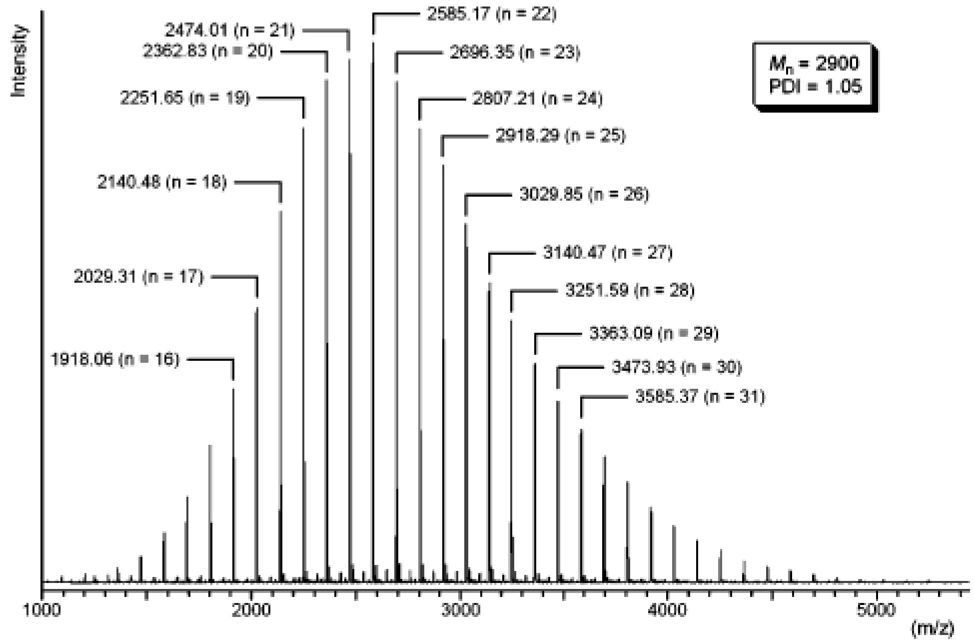
离子源：基质辅助激光解析电离 (MALDI)

MALDI可使热敏感或不挥发的化合物由固相直接得到离子。

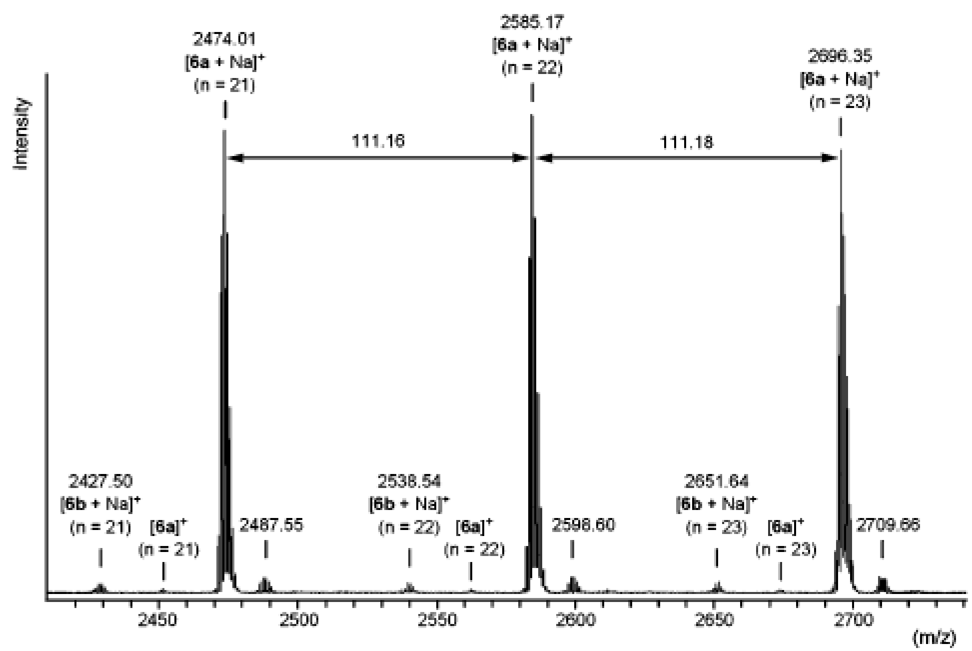
待测物质的溶液与基质的溶液混合后蒸发，使分析物与基质成为晶体或半晶体，用一定波长的脉冲式激光进行照射时，基质分子能有效的吸收激光的能量，使基质分子和样品分子进入气相并得到电离。



MALDI适用于生物大分子，如肽类，核酸类化合物。可得到分子离子峰，无明显碎片峰。此电离方式特别适合于飞行时间质谱计。



Poly(*N*-vinylpyrrolidone):



进样系统 → 离子源 → 质量分析器 → 检测器

进样系统：
1. 气体扩散
2. 直接进样
3. 气相色谱

离子源：
1. 电子轰击
2. 化学电离
3. 场致电离
4. 激光

质量分析器：
1. 单聚焦
2. 双聚焦
3. 飞行时间
4. 四极杆

将引入的样品转化成为碎片离子的装置。根据样品离子化方式和电离源能量高低，通常可将电离源分为：

气相源：先蒸发再激发，适于沸点低于$500°C$、对热稳定的样品的离子化，包括电子轰击源、化学电离源、场电离源、火花源；

解吸源：固态或液态样品不需要挥发而直接被转化为气相，适用于分子量高达10^5的非挥发性或热不稳定性样品的离子化。包括场解吸源、快原子轰击源、激光解吸源、离子喷雾源和热喷雾离子源等。

硬源：离子化能量高，如EI。伴有化学键的断裂，谱图复杂，可得到分子官能团的信息；

软源：离子化能量低，如场解吸源。产生的碎片少，谱图简单，可得到分子量信息。

应根据分子电离所需能量的不同来选择不同电离源。

a) 电子轰击源*(Electron Impact，EI)*

作用过程：

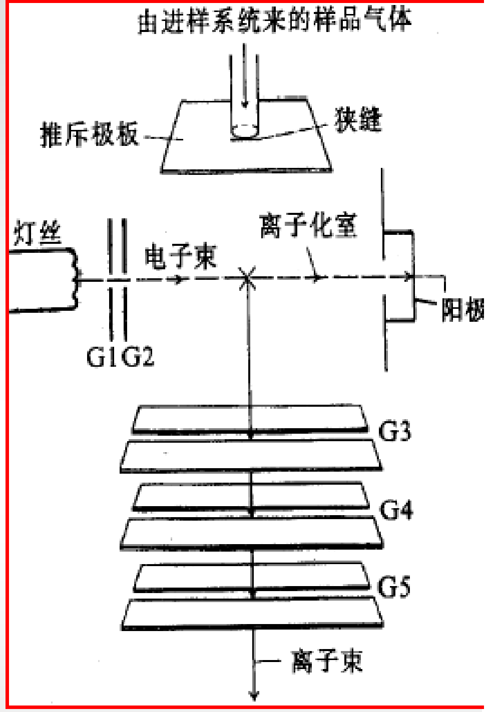
采用高速(高能)电子束冲击样品，从而产生电子和分子离子M^+，M^+继续受到电子轰击而引起化学键的断裂或分子重排，瞬间产生多种离子。

水平方向：灯丝与阳极间(*70V*电压)—高能电子—冲击样品—正离子

垂直方向：*G3-G4*加速电极(低电压)---较小动能---狭缝准直*G4-G5*加速电极(高电压)---较高动能---狭缝进一步准直--离子进入质量分析器。

特　点：

使用最广泛，谱库最完整；电离效率高；结构简单，操作方便；但分子离子峰强度较弱或不出现（因电离能量最高）。



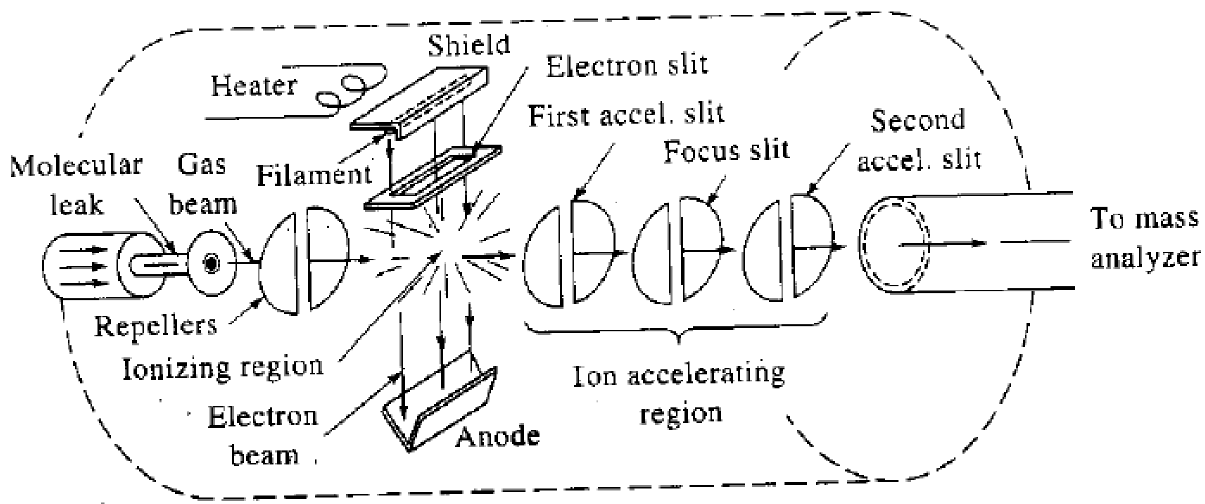


Figure 20-3 An electron-impact ion source. *(From R. M. Silverstein, G. C. Bassler, and T. C. Morrill, Spectrometric Identification of Organic Compounds, 5th ed., p. 4. New York: Wiley, 1991. Reprinted by permission of John Wiley & Sons, Inc.)*

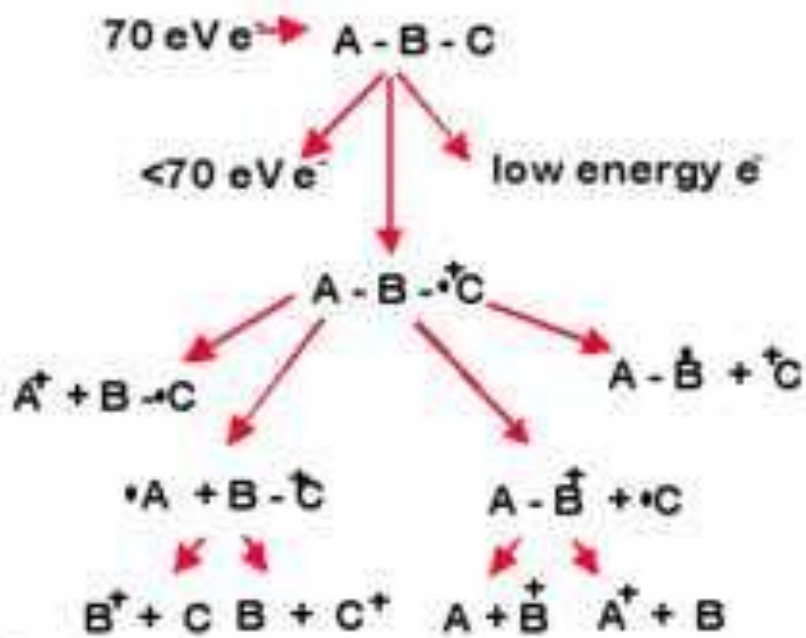
Electron impact process
撞击

An electron (70 eV) strikes a neutral molecule.

This knocks out a second electron, producing a molecular ion - with some additional energy.

This additional energy is internalized via a series of vibrations, rotations and molecular rearrangements

This results in fragmentation of the molecule.



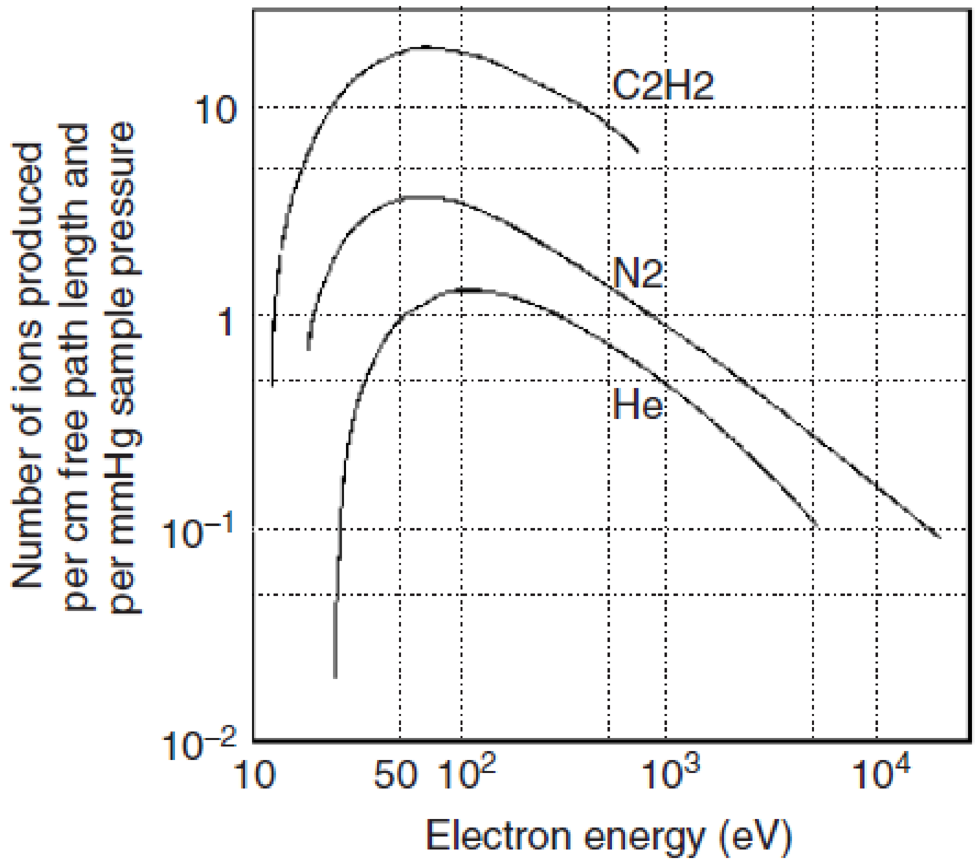


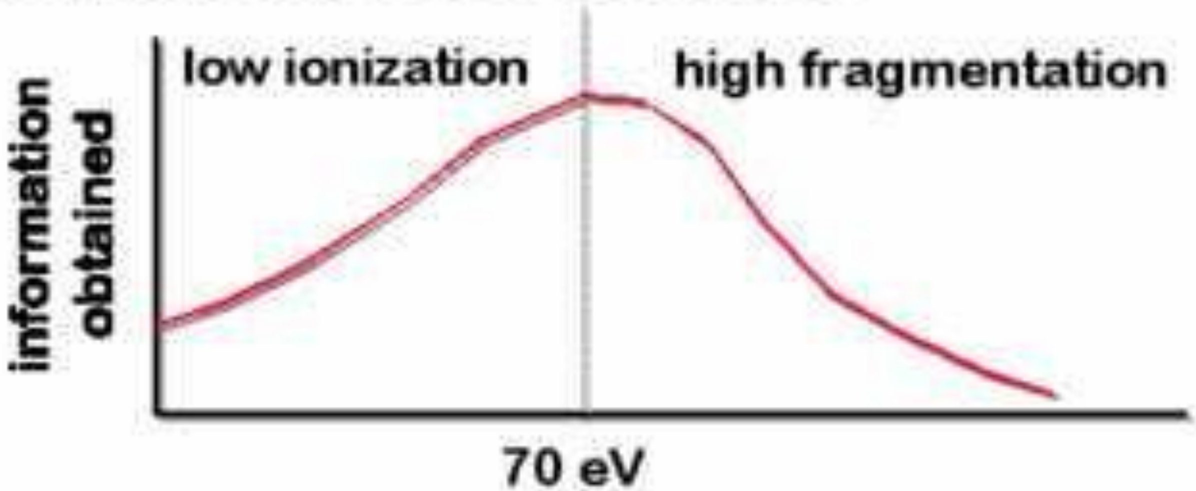
Figure 1.2
Number of ions produced as a function of the electron energy. A wide maximum appears around 70 eV.

Electron impact process

70 eV electrons are typically used.

At lower energies, you only get a small number of ions being produced.

At higher energies the fragmentation is too great and little information is obtained



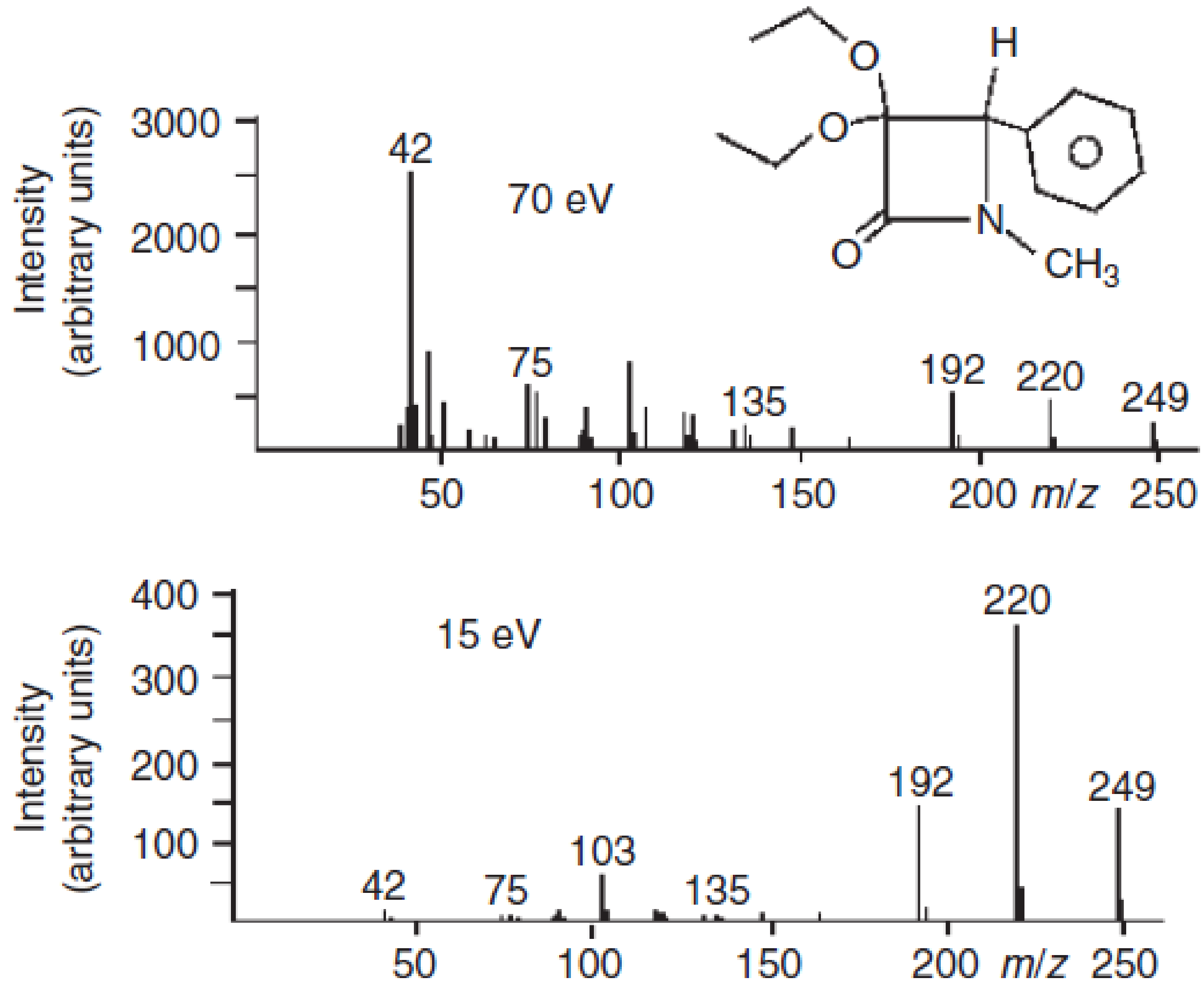
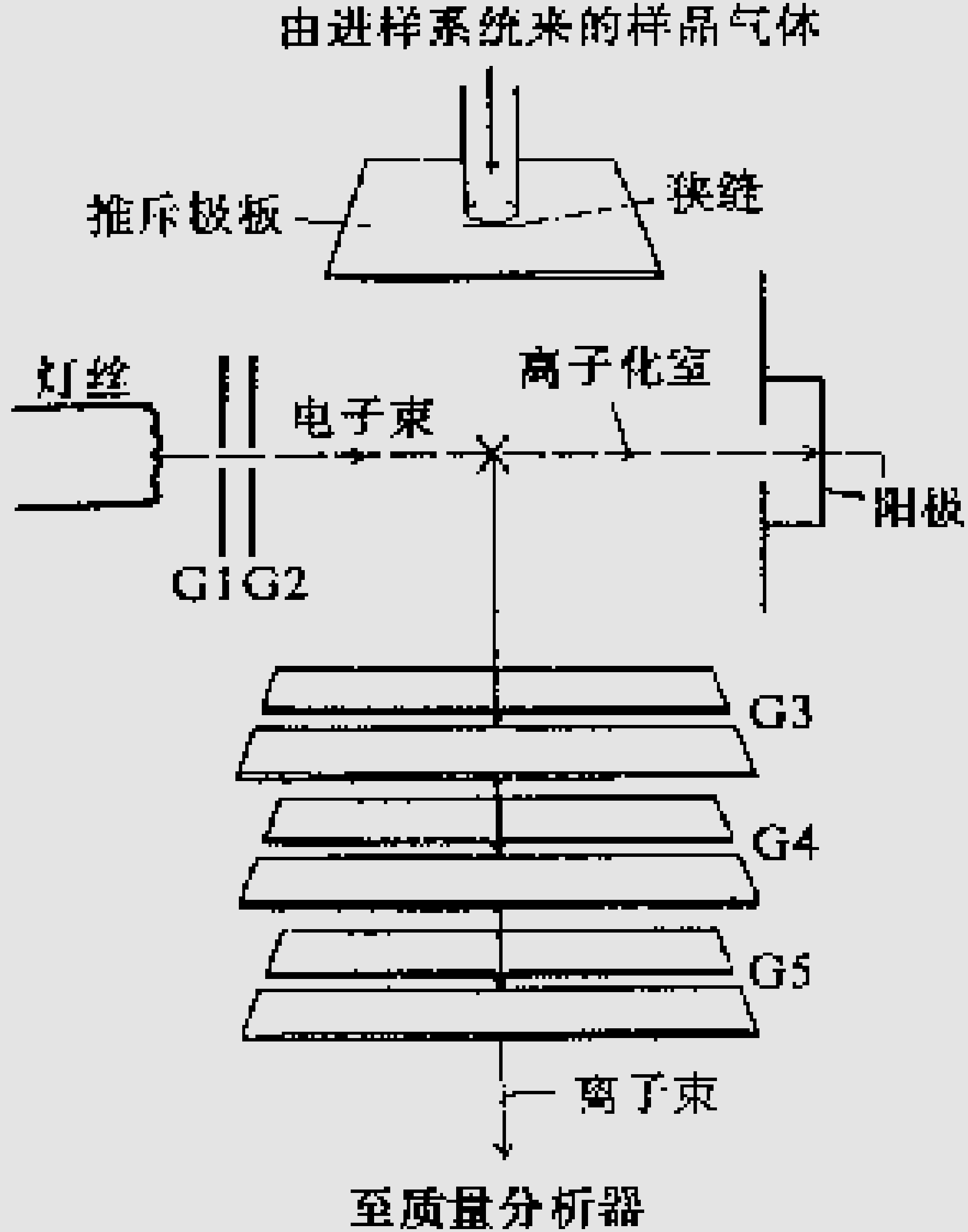
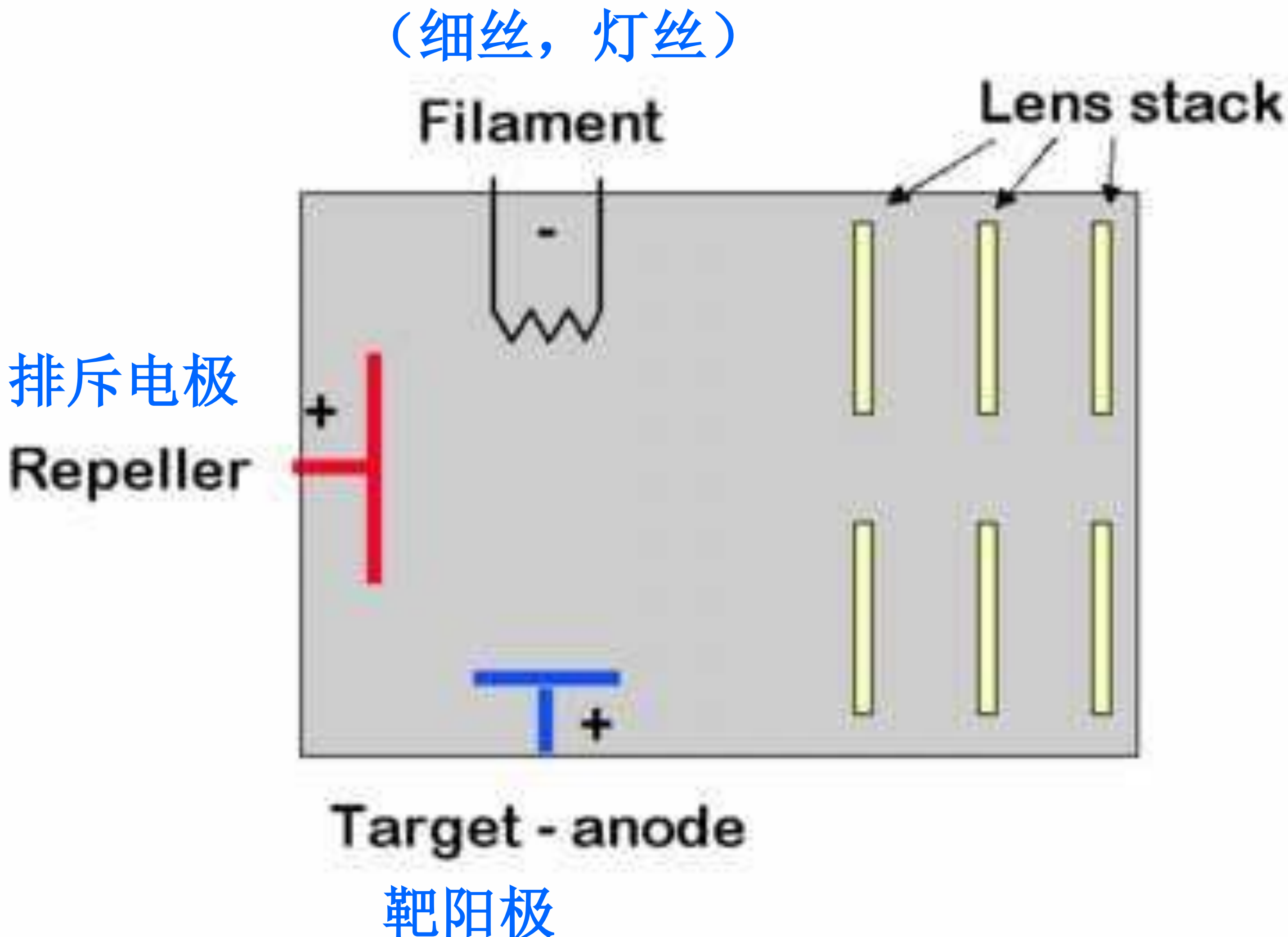


Figure 1.3
Two spectra of β-lactam. While the relative intensity of the molecular ion peak is greater at lower ionization energy, its absolute intensity, as read from the left-hand scale, is actually somewhat reduced.



电子轰击源工作示意图

Electron impact source



Electron impact source

Filament - Typically made of Re. Our source of 70 eV electrons.

Target - anode used in association with the filament to produce electrons.

Repeller - positively charged electrode used to 'push' positive ions out of the ionization source.

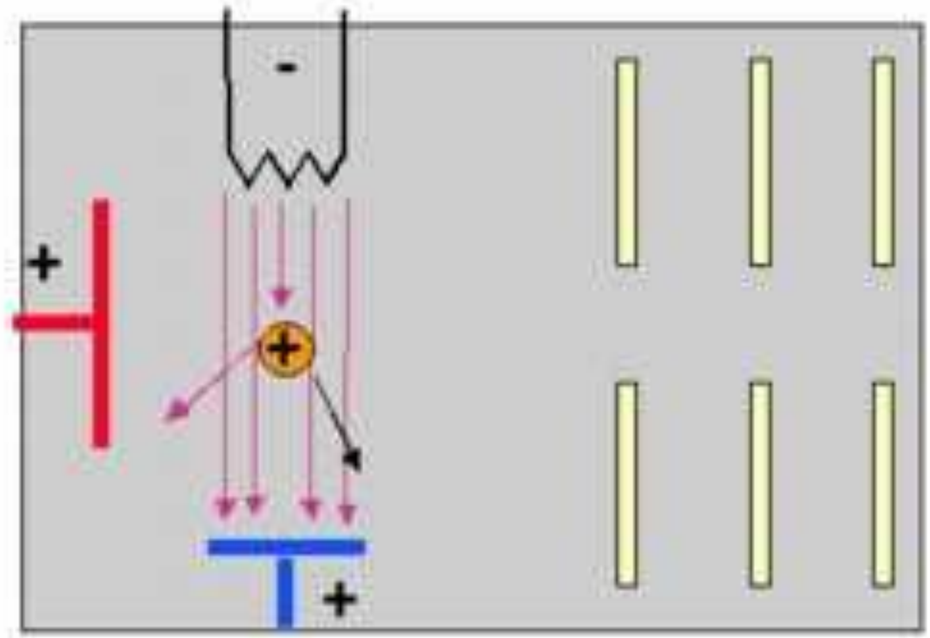
Lens stack - series of increasingly more negative electrodes used to accelerate our ions to constant kinetic energy.

Electron impact source

When a sample molecule enters the source, it passer through the electron beam and is ionized.

The repeller insures that the ion is rapidly pushed out of the source, towards the lens stack.

Any negative ions are pulled into the repeller.

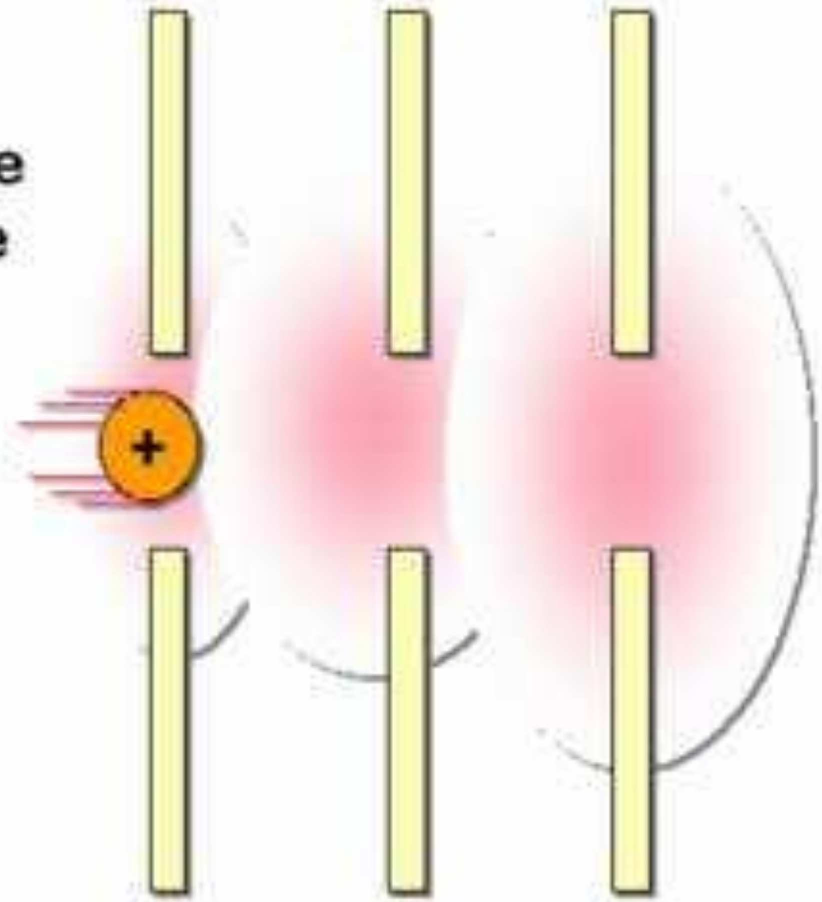


Electron impact source

As an ion accelerates towards the first lens, it comes under the influence of the next, more negative lens. It passes the first lens and accelerates towards the next.

By the final lens, it is traveling so fast that it simply passes directly into the analyzer.

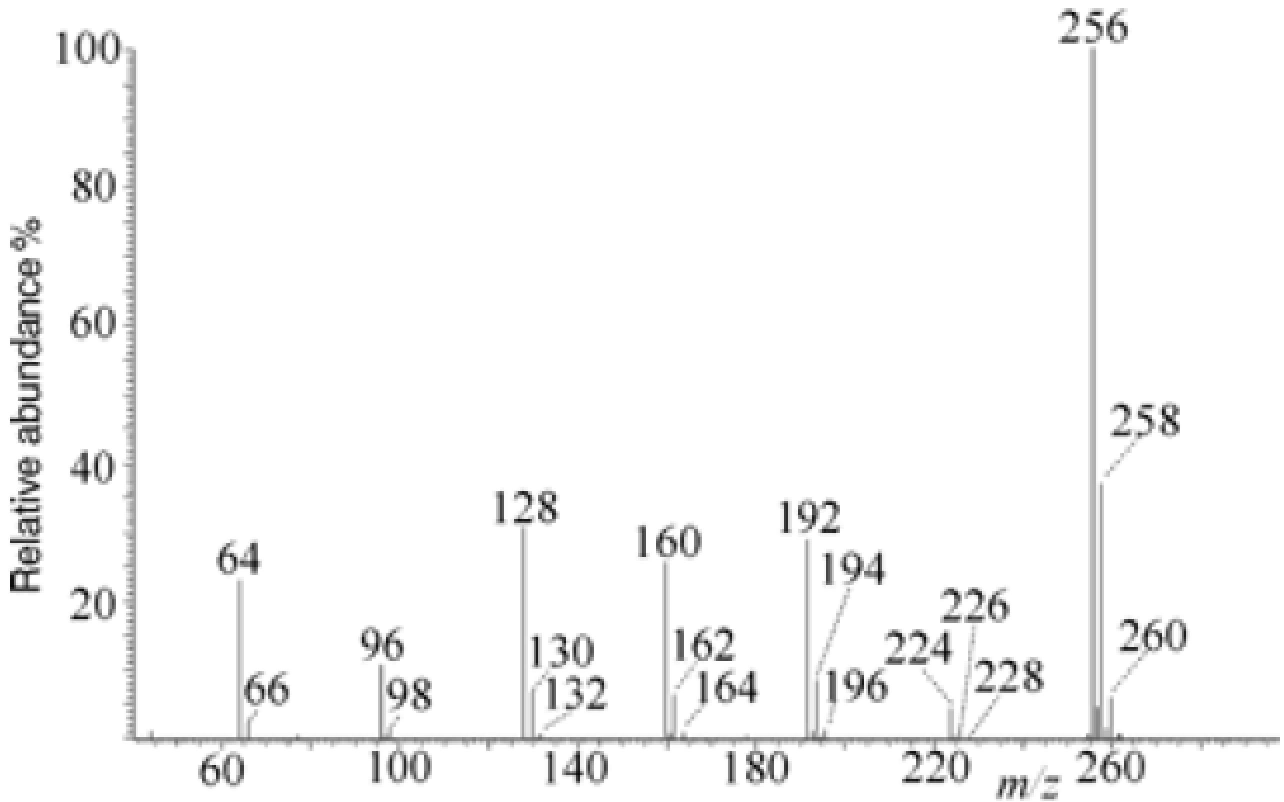
- constant KE



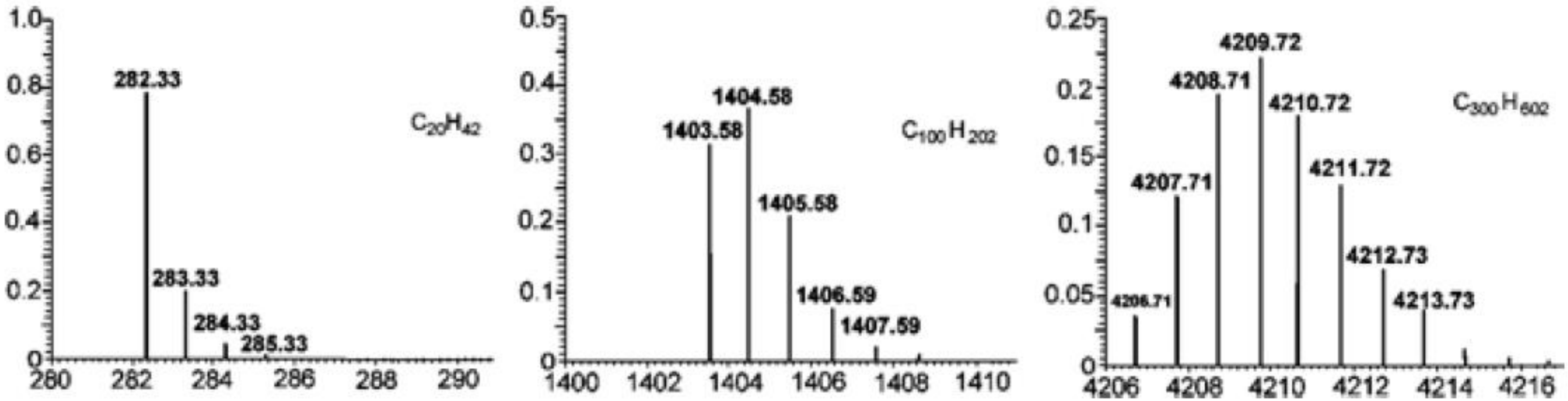
EI的优缺点

- 优点
- 1.级的灵敏度
- 2.有达10万个化合物的数据库可快速检索
- 3.可根据碎片方式鉴定未知物
- 4.从碎片离子判定结构

- 缺点
- 1.质量范围小
- 2.有可能汽化前发生解离
- 3.碎片过多有时看不到分子离子

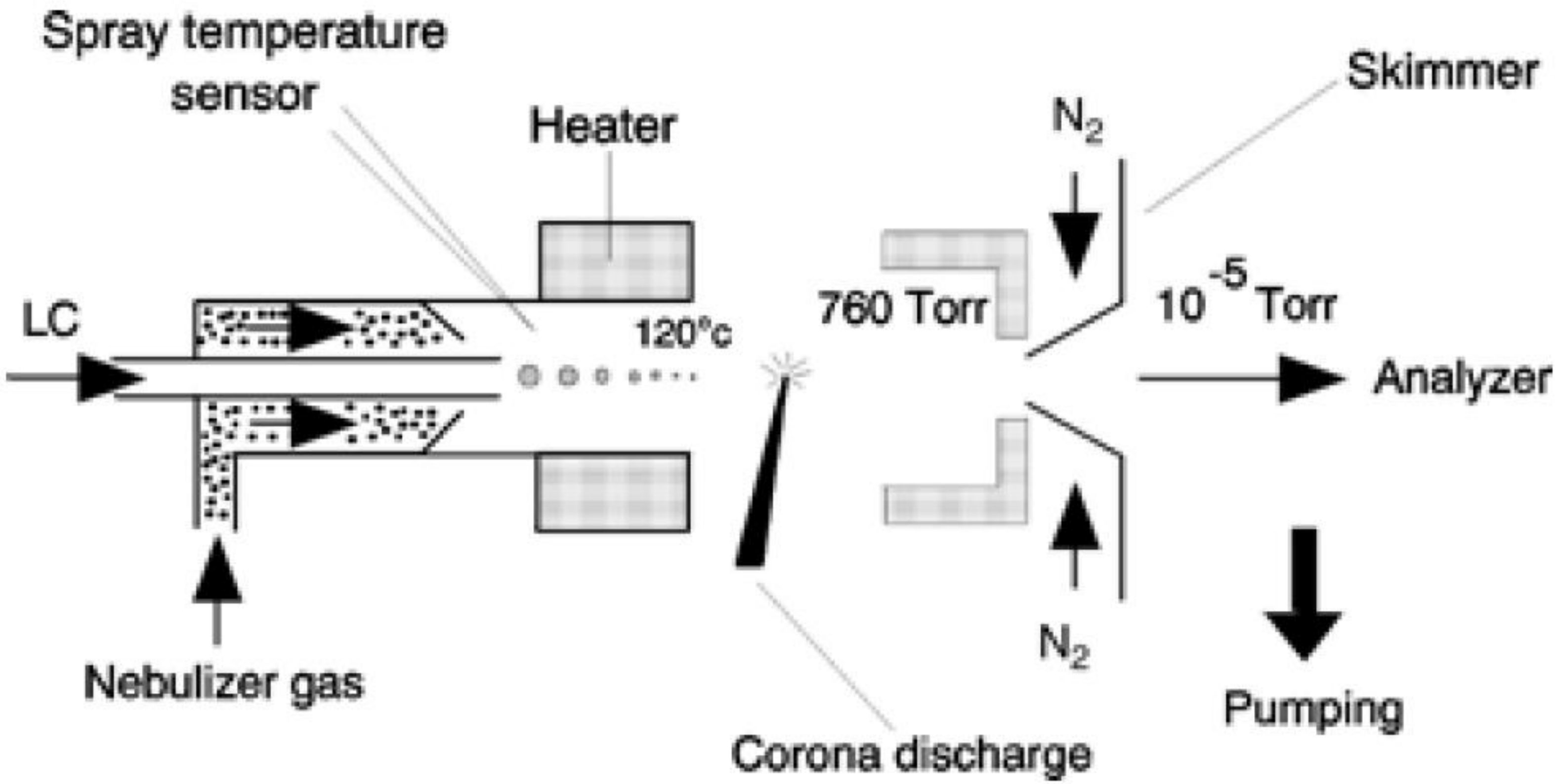


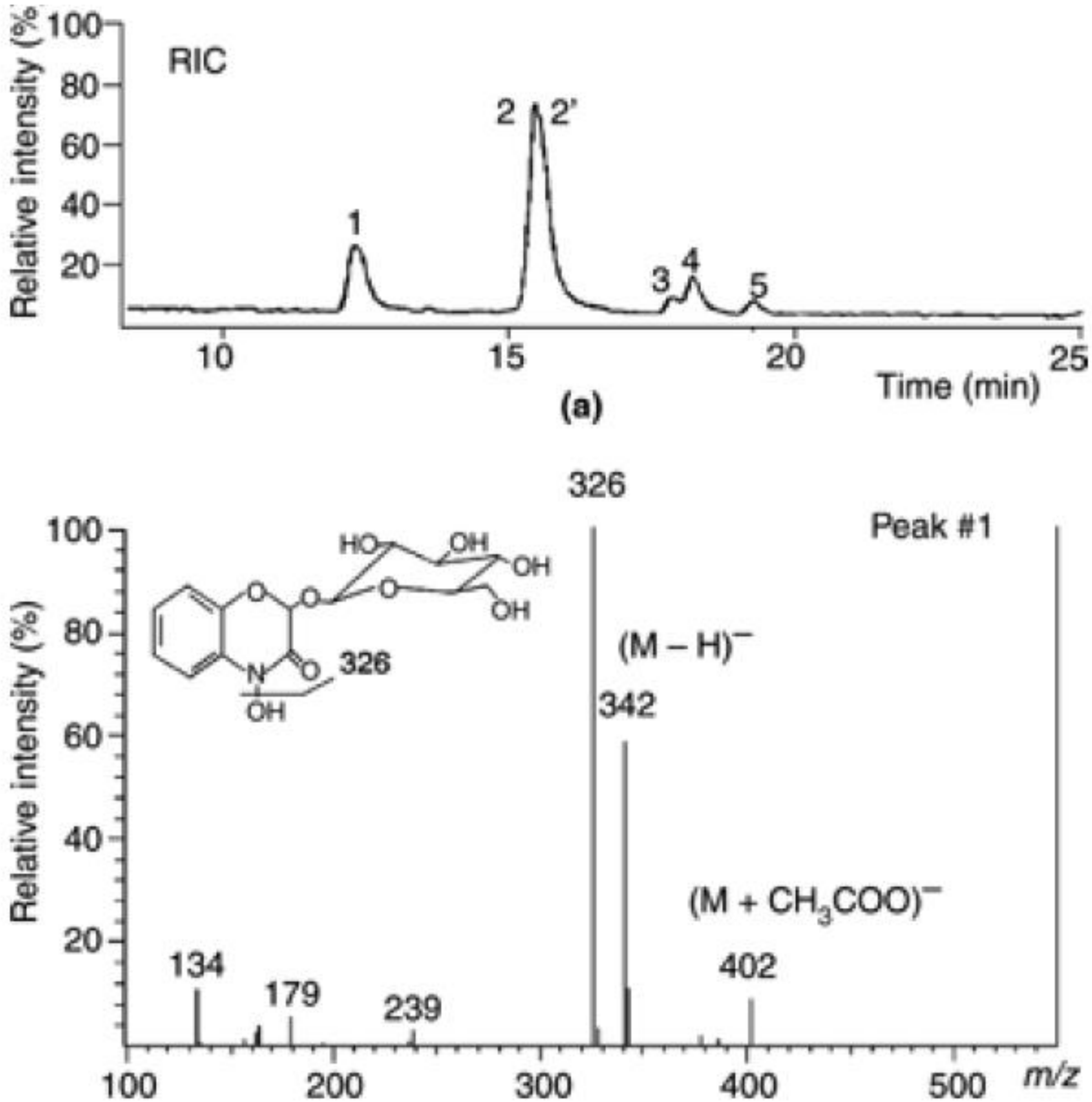
Relative abundance %
100
80
60
40
20
60
100
140
180
220
260
m/z
256
258
260
128
130
132
160
162
164
192
194
196
224
226
228
64
66
96
98



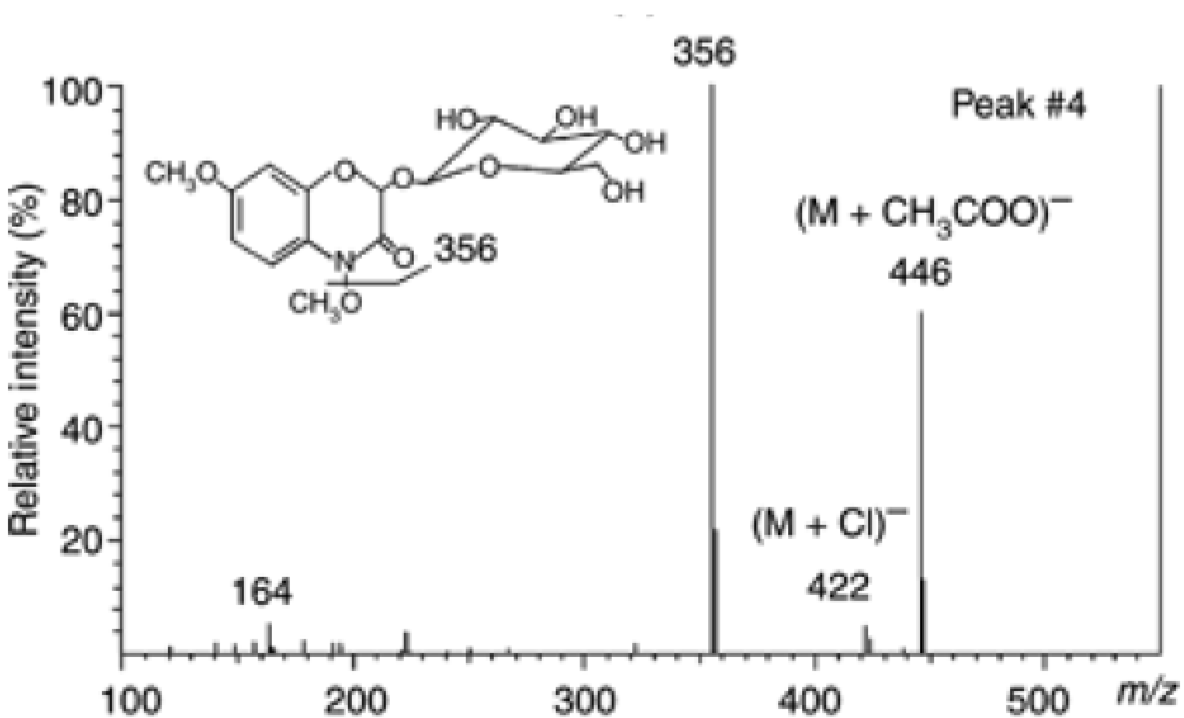
1.0
0.8
0.6
0.4
0.2
0
280
282
284
286
288
290
282.33
283.33
284.33
285.33
C20H42
0.5
0.4
0.3
0.2
0.1
0
1400
1402
1404
1406
1408
1410
1403.58
1404.58
1405.58
1406.59
1407.59
C100H202
0.25
0.2
0.15
0.1
0.05
0
4206
4208
4210
4212
4214
4216
4206.71
4207.71
4208.71
4209.72
4210.72
4211.72
4212.73
4213.73
C300H602

化学电离 Chemical Ionization





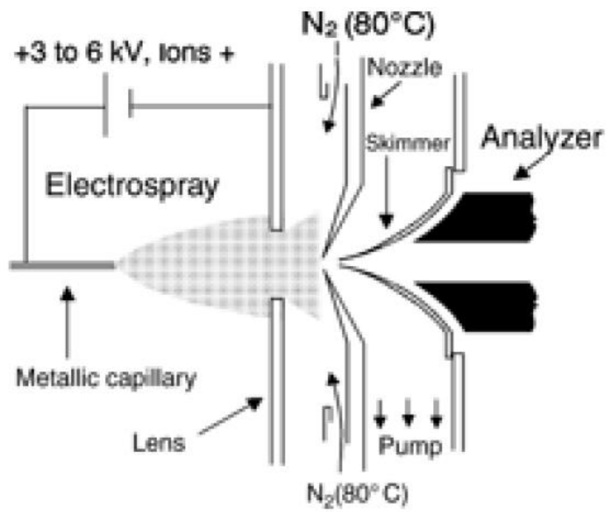
Relative intensity (%)
100
80
60
40
20
RIC
1
2 2'
3
4
5
10
15
20
25
Time (min)
(a)
326
Peak #1
100
80
60
40
20
Relative intensity (%)
HO
OH
OH
O
OH
O
O
O
N
O
OH
326
(M – H)$^-$
342
(M + CH_3COO)$^-$
134
179
239
402
100
200
300
400
500
m/z



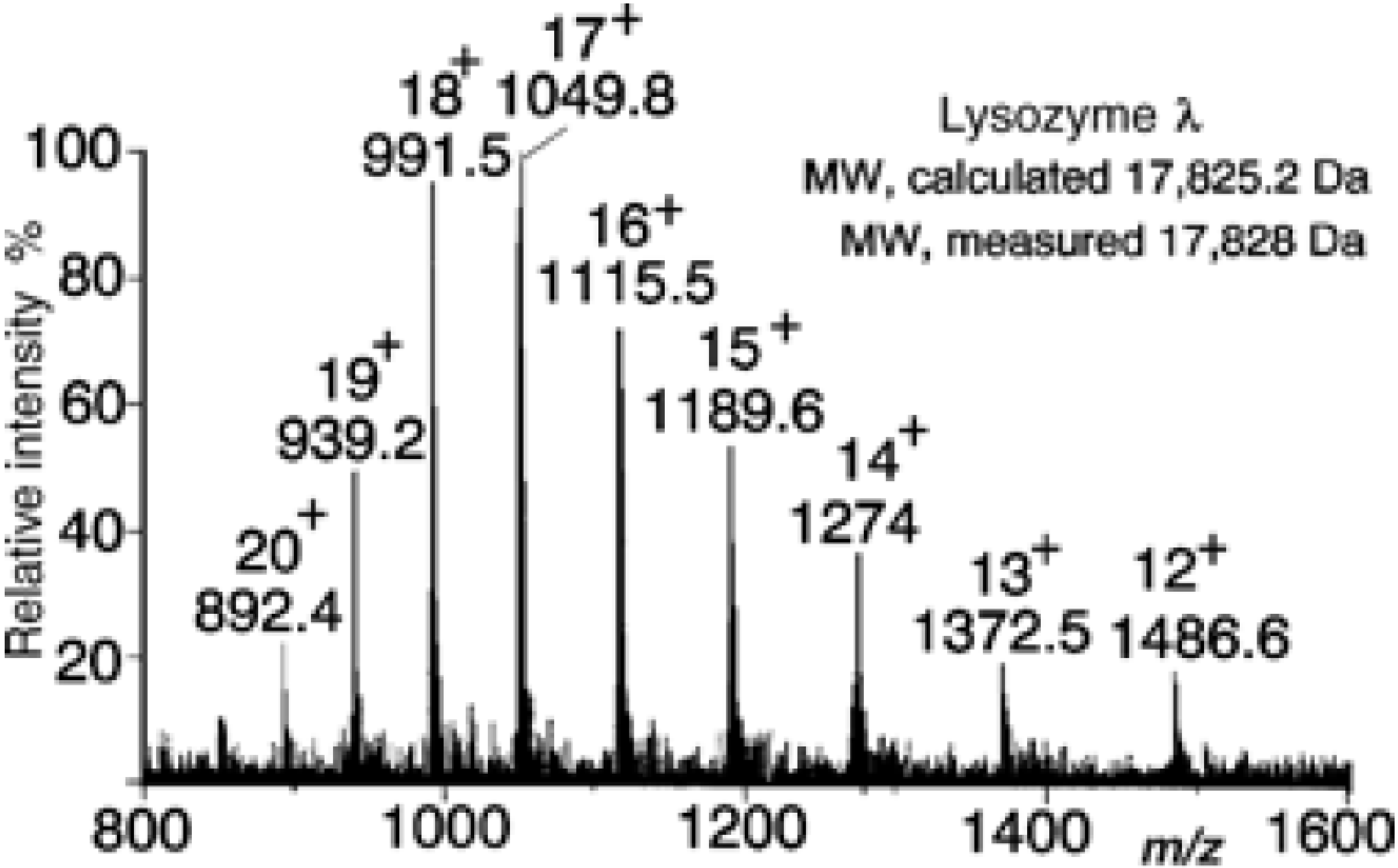
356
Peak #4
HO
OH
OH
OH
CH_3O
O
O
O
356
N
O
CH_3O
$(M + CH_3COO)^-$
446
$(M + Cl)^-$
422
164
Relative intensity (%)
100
80
60
40
20
100
200
300
400
500
m/z

(c)

电喷雾Electrospray Source



(a)



Lysozyme λ
MW, calculated 17,825.2 Da
MW, measured 17,828 Da
Relative intensity %
100
80
60
40
20
800
1000
1200
1400
1600
m/z
20+ 892.4
19+ 939.2
18+ 991.5
17+ 1049.8
16+ 1115.5
15+ 1189.6
14+ 1274
13+ 1372.5
12+ 1486.6

The A+2 elements

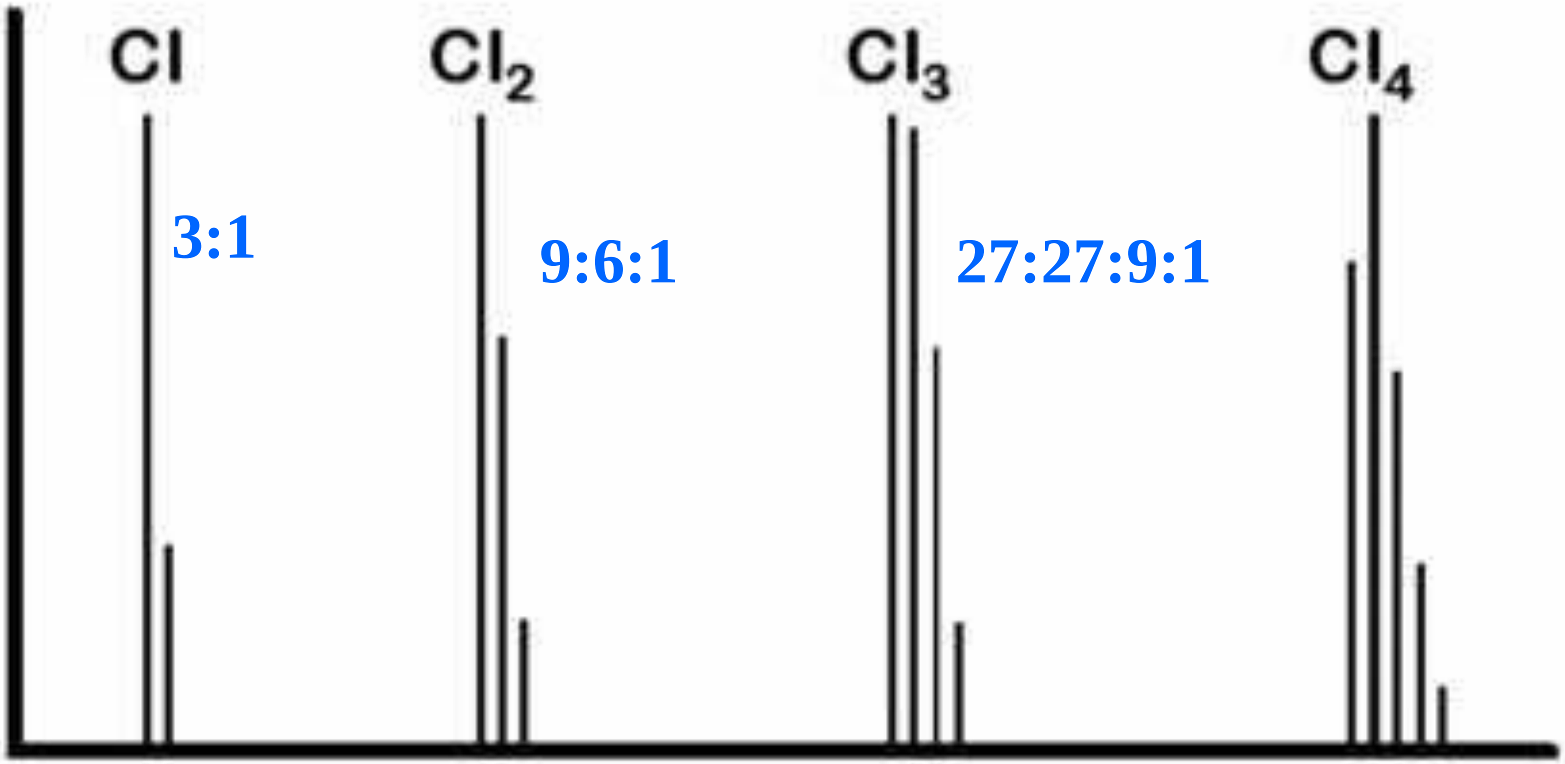
Chlorine and bromine are both common in organic compounds.

They also produce very characteristic patterns due to the relatively high abundance of there isotopes.

Lets look at some of their patterns. You need to be able to identify them if you ever plan of finding the molecular ion of a brominated or chlorinated compound.

Chlorine

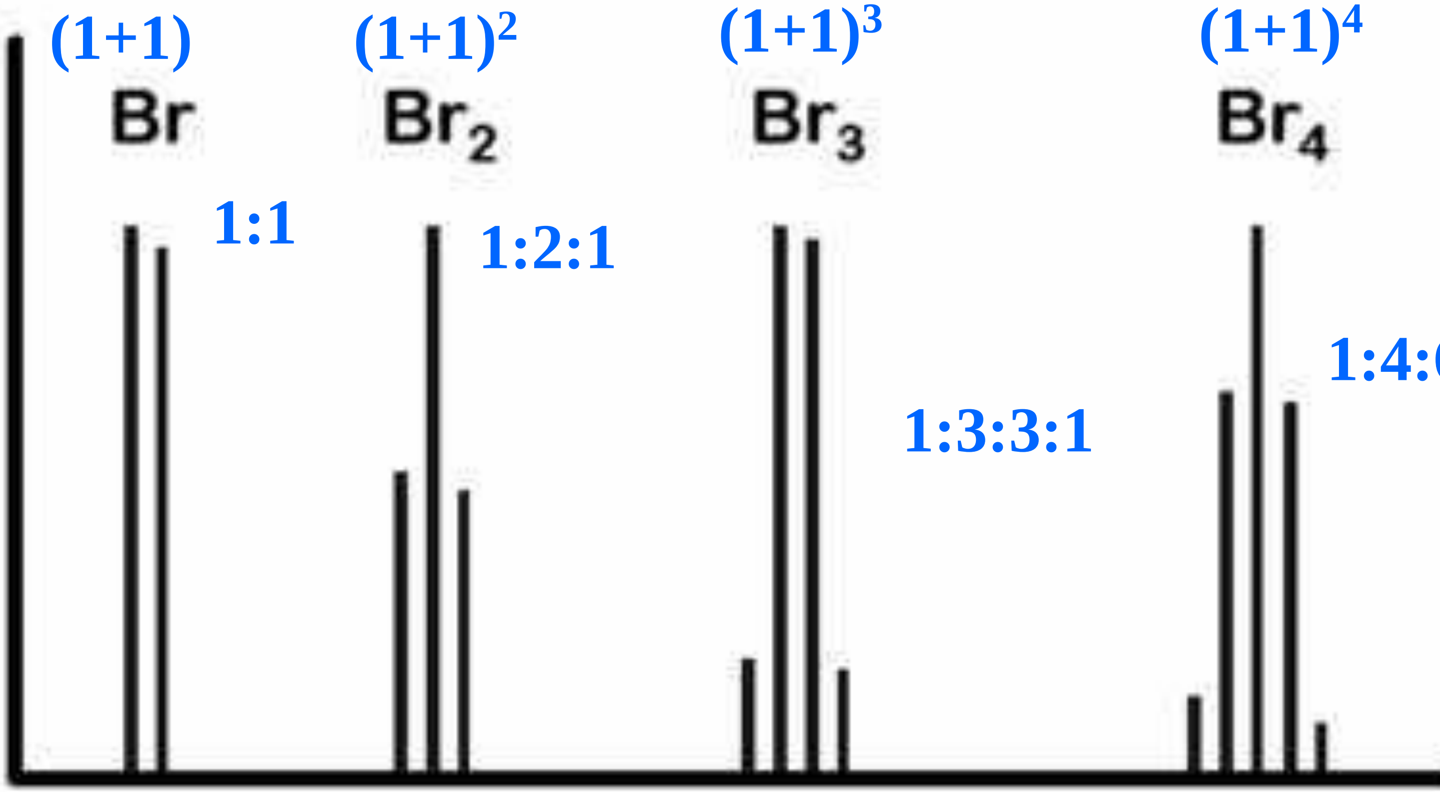
$(a+b)^n$



By the time you have 4 chlorine, the A+2 line is significantly larger than the A lne.

Bromine

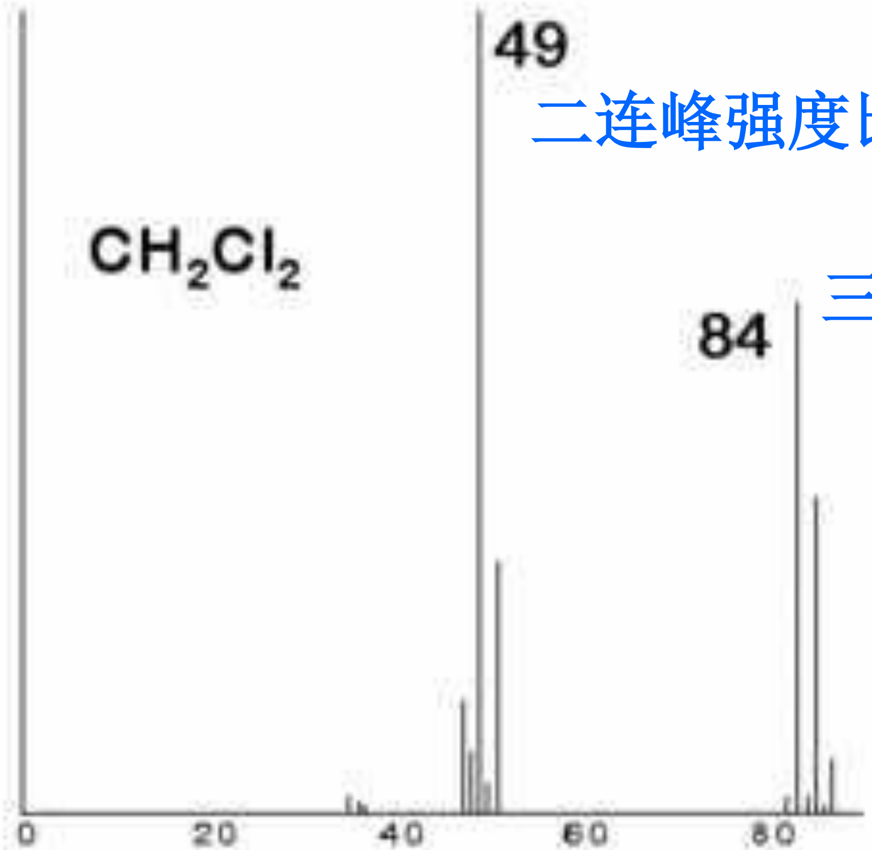
$^{81}Br/^{79}Br = 97.9:100 \backsim 1:1$



Bromine, which is almost a 1:1 mixture of ^{79}Br and ^{81}Br also has a distinctive pattern.

Another example

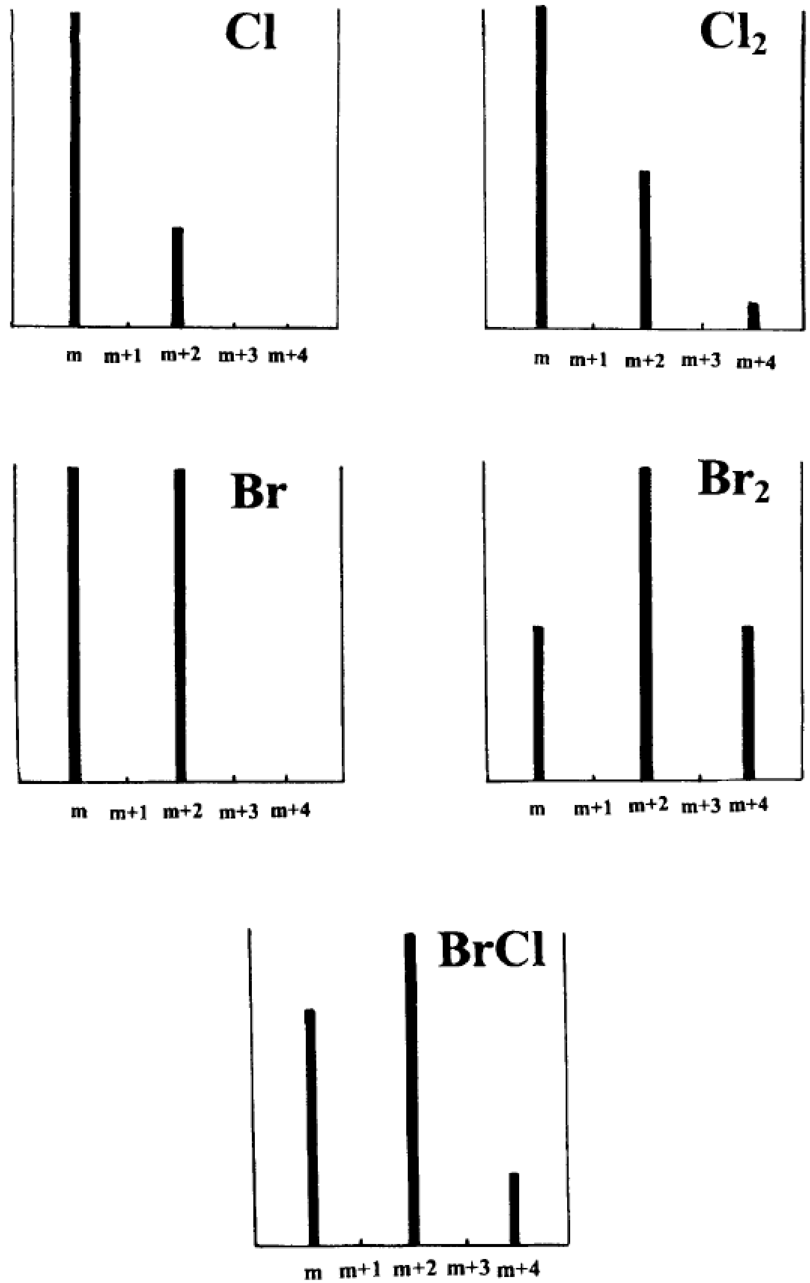
Dichloromethane - a common GC/MS solvent



二连峰强度比为 3:1

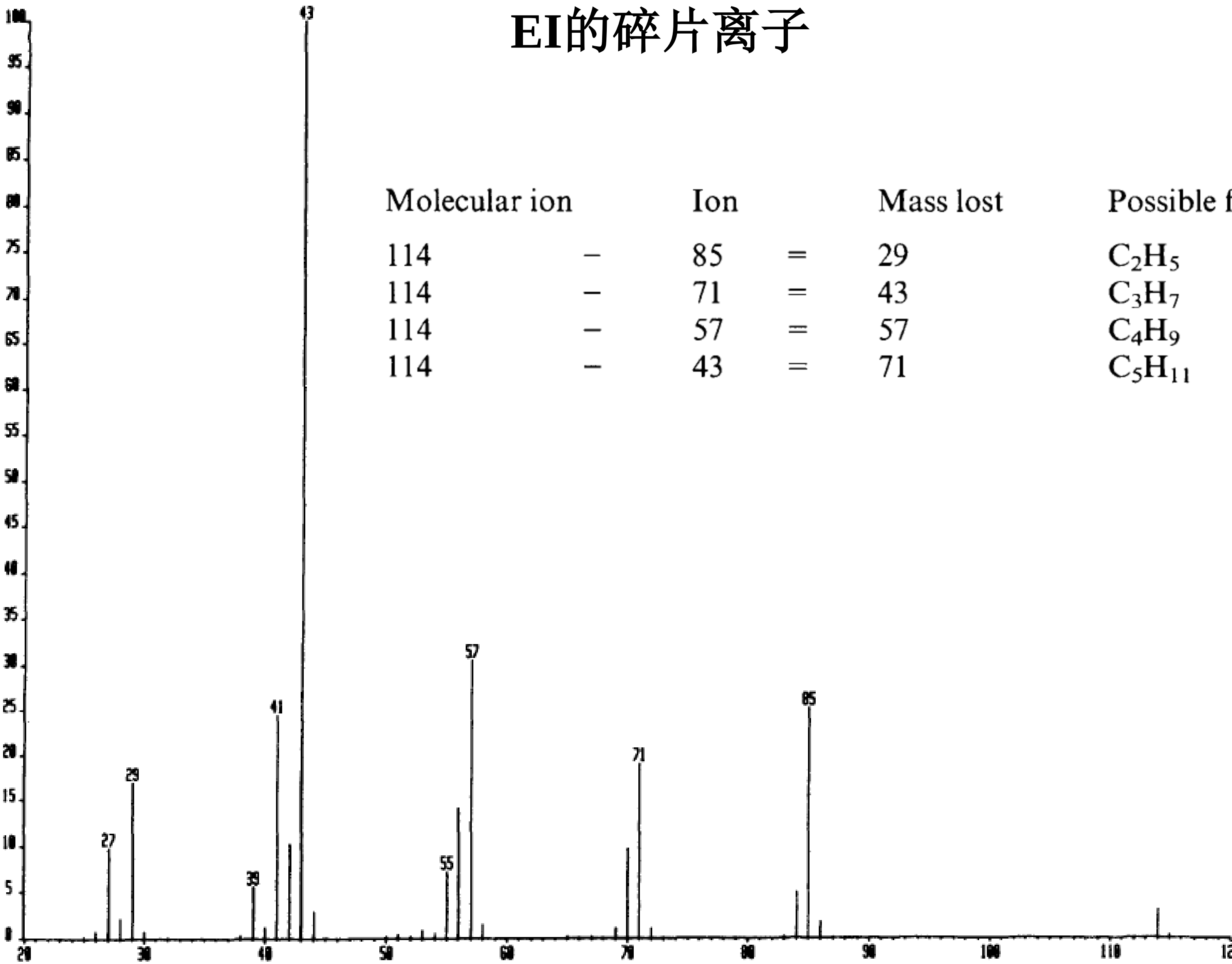
三连峰强度比为 9：6：1

Note the Cl_2 pattern at 84 and the Cl pattern at 49


Cl
m m+1 m+2 m+3 m+4
Cl2
m m+1 m+2 m+3 m+4
Br
m m+1 m+2 m+3 m+4
Br2
m m+1 m+2 m+3 m+4
BrCl
m m+1 m+2 m+3 m+4

EI的碎片离子

Molecular ion		Ion		Mass lost	Possible fragment
114	−	85	=	29	C_2H_5
114	−	71	=	43	C_3H_7
114	−	57	=	57	C_4H_9
114	−	43	=	71	C_5H_{11}



只含有CHO或卤素的分子，主峰为奇数质荷比，分子离子峰为偶数质荷比

In molecules containing only C, H, O and halogens, it is useful to note that the main peaks have odd *m/z ratios, while the molecular* ion has an even *m/z ratio*.

如果分子中含有奇数个氮原子，则相反

***If the molecule contains an odd number of nitrogen* atoms this is reversed: the molecular ion has an odd *m/z ratio, and most of the* peaks have even *m/z ratios*.**

$$CH_3{-}CH_2 \mid CH_2 \mid CH_2 \mid CH_2{-}CH_2{-}CH_2{-}CH_3$$

Knowing that the molecule can split off C_2H_5, C_3H_7 or C_4H_9 and has a molecular weight of 114 provides good evidence for the structure of *n*-octane

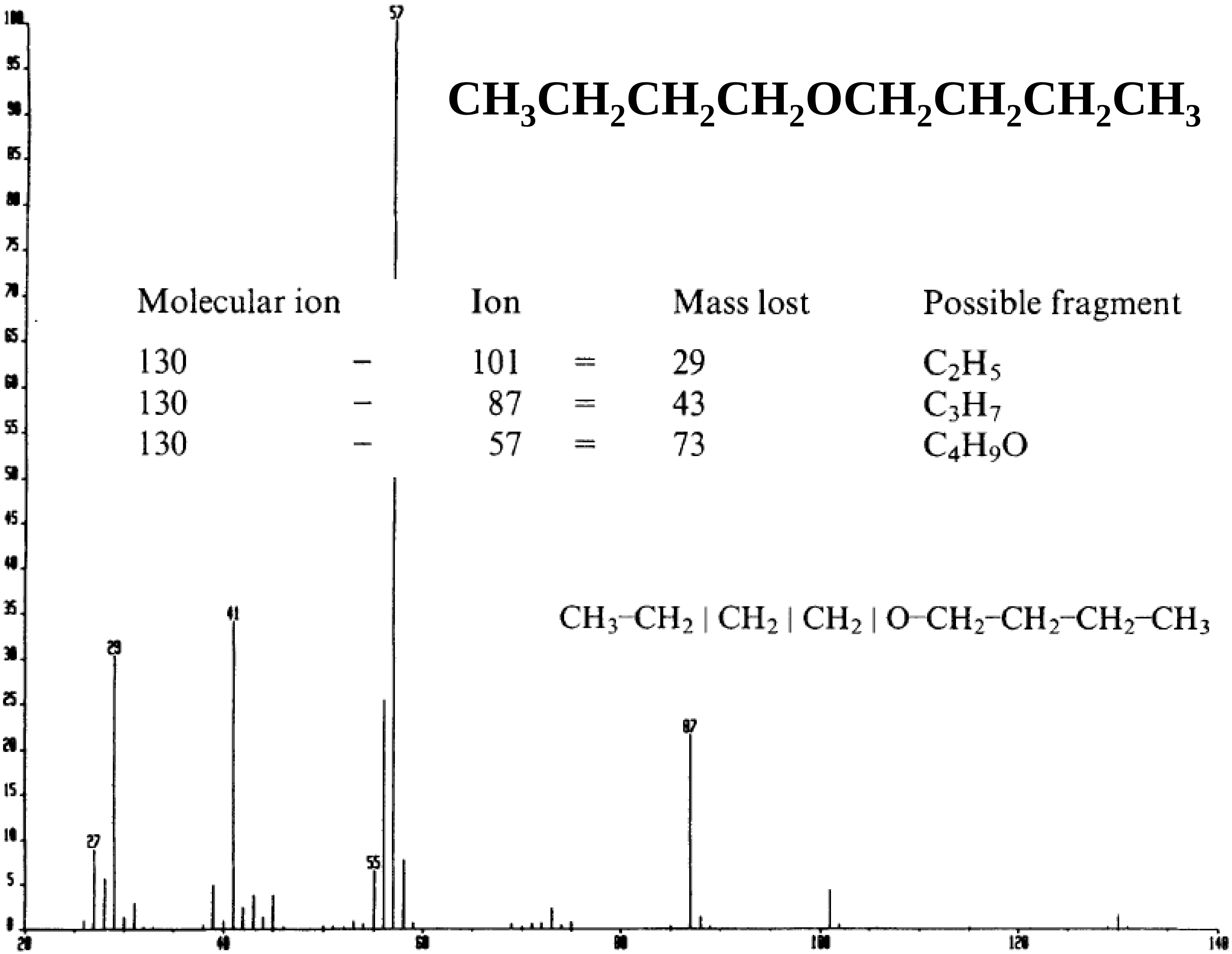
Loss of fragments which increase in mass by 14 *(i.e. CH_2) is characteristic of an* unbranched hydrocarbon chain. Note that octane does not readily lose CH_3.

Possible losses from molecular ions

Mass lost	*Possible group lost*	*Inference*
15	CH_3	
17	OH	Alcohol
18	H_2O	Alcohol
29	C_2H_5 or CHO	–
31	OCH_3	Methyl ester
41	C_3H_5	Propyl ester
43	C_3H_7 or CH_3CO	–
44	CO_2	Anhydride
45	$COOH$	Carboxylic acid
	OC_2H_5	Ethyl ester
55	C_4H_7	Butyl ester
57	C_4H_9 or C_2H_5CO	–
71	C_5H_{11} or C_3H_7CO	–

Mass lost	*Possible group lost*	*Inference*
15	CH_3	
17	OH	Alcohol
18	H_2O	Alcohol
29	C_2H_5 or CHO	–
31	OCH_3	Methyl ester
41	C_3H_5	Propyl ester
43	C_3H_7 or CH_3CO	–
44	CO_2	Anhydride
45	$COOH$	Carboxylic acid
	OC_2H_5	Ethyl ester
55	C_4H_7	Butyl ester
57	C_4H_9 or C_2H_5CO	–
71	C_5H_{11} or C_3H_7CO	–

$CH_3CH_2CH_2CH_2OCH_2CH_2CH_2CH_3$



Molecular ion		Ion		Mass lost	Possible fragment
130	−	101	=	29	C_2H_5
130	−	87	=	43	C_3H_7
130	−	57	=	73	C_4H_9O

$CH_3{-}CH_2 \mid CH_2 \mid CH_2 \mid O{-}CH_2{-}CH_2{-}CH_2{-}CH_3$

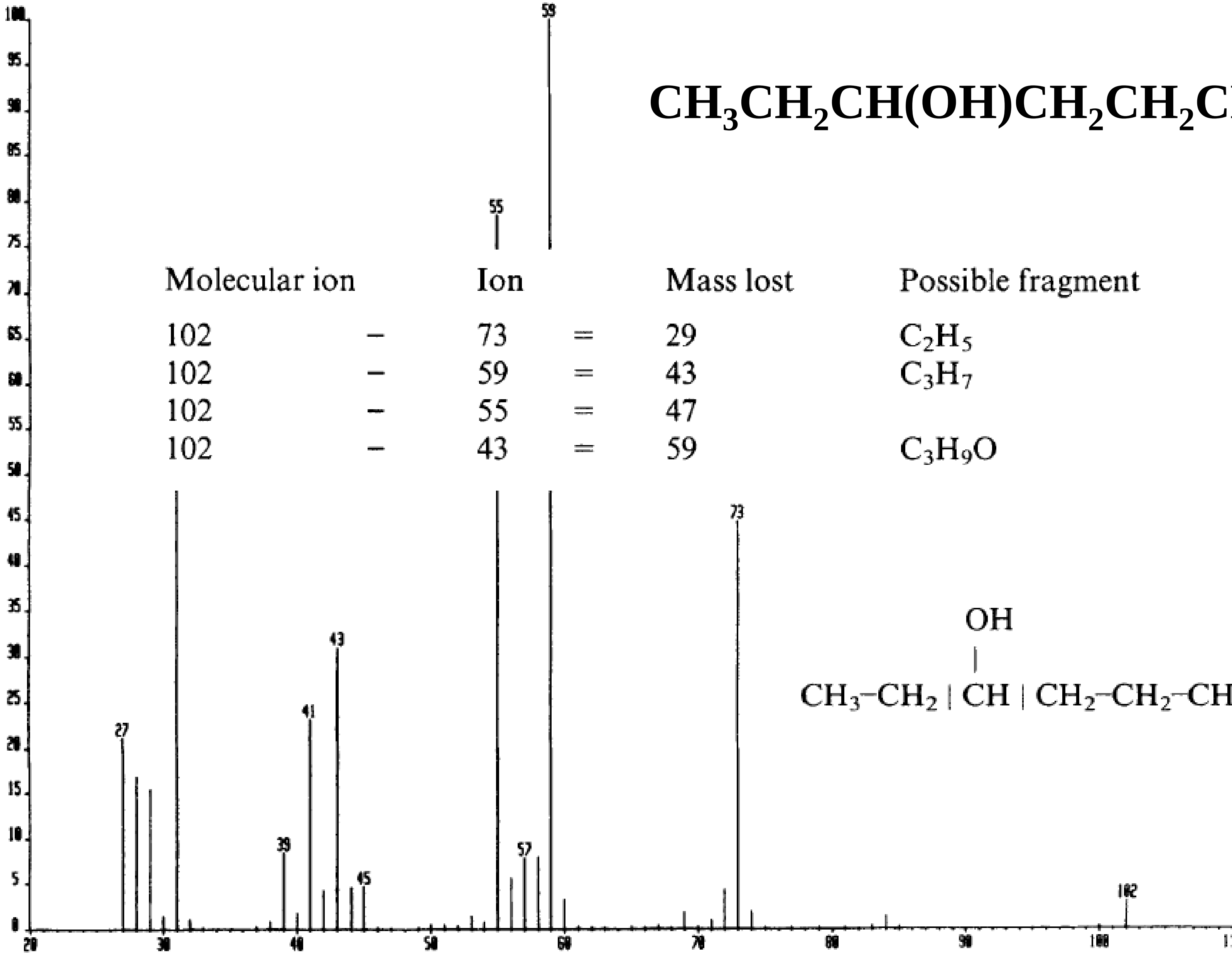
The oxygen atom affects the pattern of splitting of the molecule bystabilising the ion containing oxygen.

The main splitting is that of the C-O bond.

Loss of C_3H_7 also occurs because the ion CH2=O+C_4H_9 is also strongly stabilised by the oxygen.

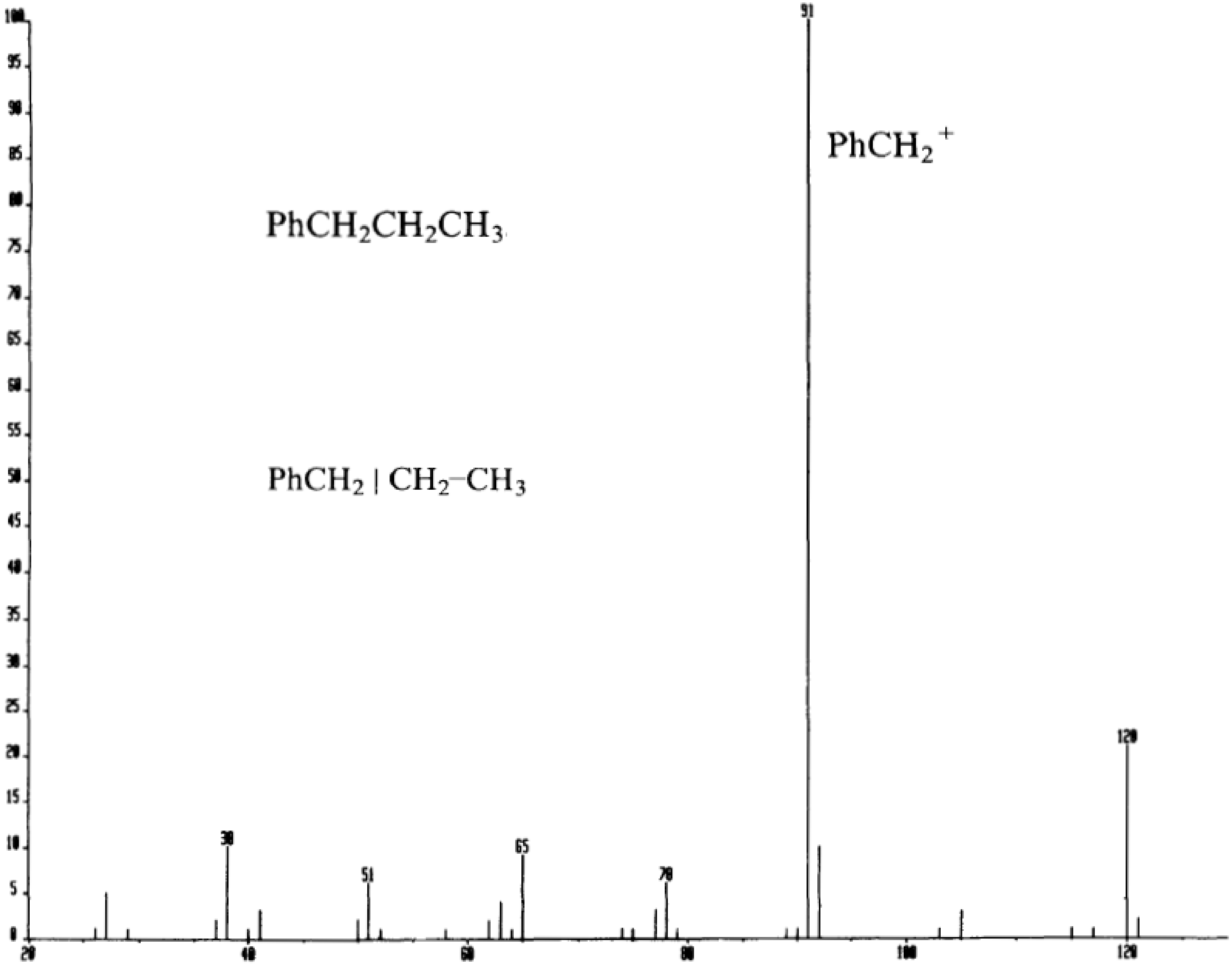
A nitrogen heteroatom gives a similar stabilisation and splitting

$CH_3CH_2CH(OH)CH_2CH_2CH_3$



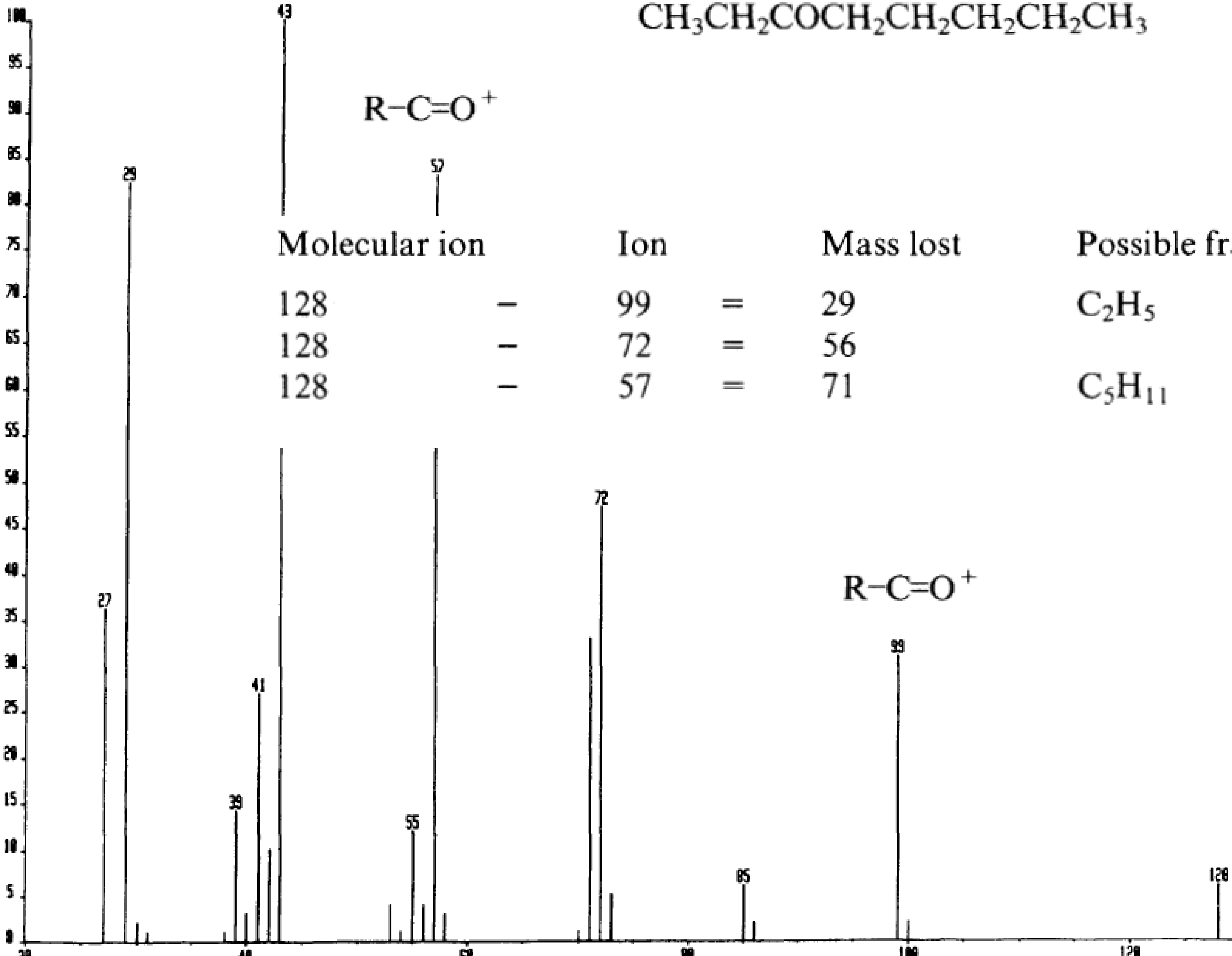
Molecular ion		Ion		Mass lost	Possible fragment
102	−	73	=	29	C_2H_5
102	−	59	=	43	C_3H_7
102	−	55	=	47	
102	−	43	=	59	C_3H_9O

$$\begin{array}{c} \qquad\quad OH \\ \qquad\quad | \\ CH_3{-}CH_2 \mid CH \mid CH_2{-}CH_2{-}CH_3 \end{array}$$



91
PhCH$_2^+$
PhCH$_2$CH$_2$CH$_3$
PhCH$_2$ | CH$_2$–CH$_3$
51
65
120

$CH_3CH_2COCH_2CH_2CH_2CH_2CH_3$



Molecular ion		Ion		Mass lost	Possible fragment
128	−	99	=	29	C_2H_5
128	−	72	=	56	
128	−	57	=	71	C_5H_{11}

It is a general guideline that the mass spectrum of an organic compound containing no element other than carbon, hydrogen, oxygen and the halogens has an even *m/z molecular ion and odd m/z fragment ions*.

A.

The molecule contains an odd number of nitrogen atoms. In this case, the molecular ion will have an odd *m/z ratio, and fragment ions* containing an odd number of nitrogen atoms an even *m/z ratio*.

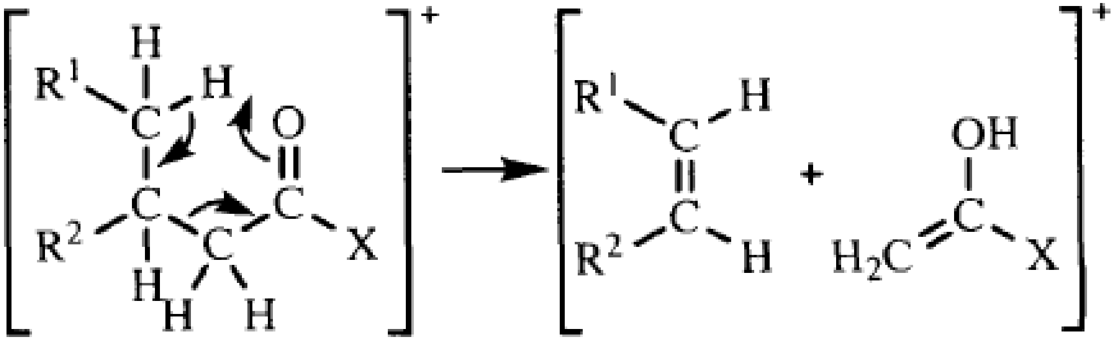
B.

The molecule has undergone an elimination reaction to give a stable product. We have already seen that hexan-3-ol which has a molecular ion at *m/z 102, can lose water to give hexene, which has a molecular ion* at *m/z 84. Carboxylic acids often lose CO2 (mass 44 Daltons) to give an* ion of even *m/z ratio; loss of HC1 (masses 36 and 38 Daltons) and HBr* (masses 80 and 82 Daltons) and loss of ethene (mass 28 Daltons) are also observed.

C.

Such peaks are often small.

McLafferty rearrangement:



The McLaKerty rearrangement occurs whenever we have a carbonyl compound which has a y-hydrogen atom in a position which permits it to migrate to the carbonyl oxygen atom through a cyclic transition state. The reaction can occur with any type of carbonyl group, and the *m/z ratio of the even peak* tells us what other substituents there are on the carbonyl carbon atom .

If we have an amide, so that X = NH2, then the McLafferty rearrangement gives a peak of odd *m/z ratio, 59. If the carbonyl group has a substituent* on the neighbouring carbon atom, then a McLafferty rearrangement gives a different size of even *m/z peak.*

Ion characteristic McLafferty rearrangement

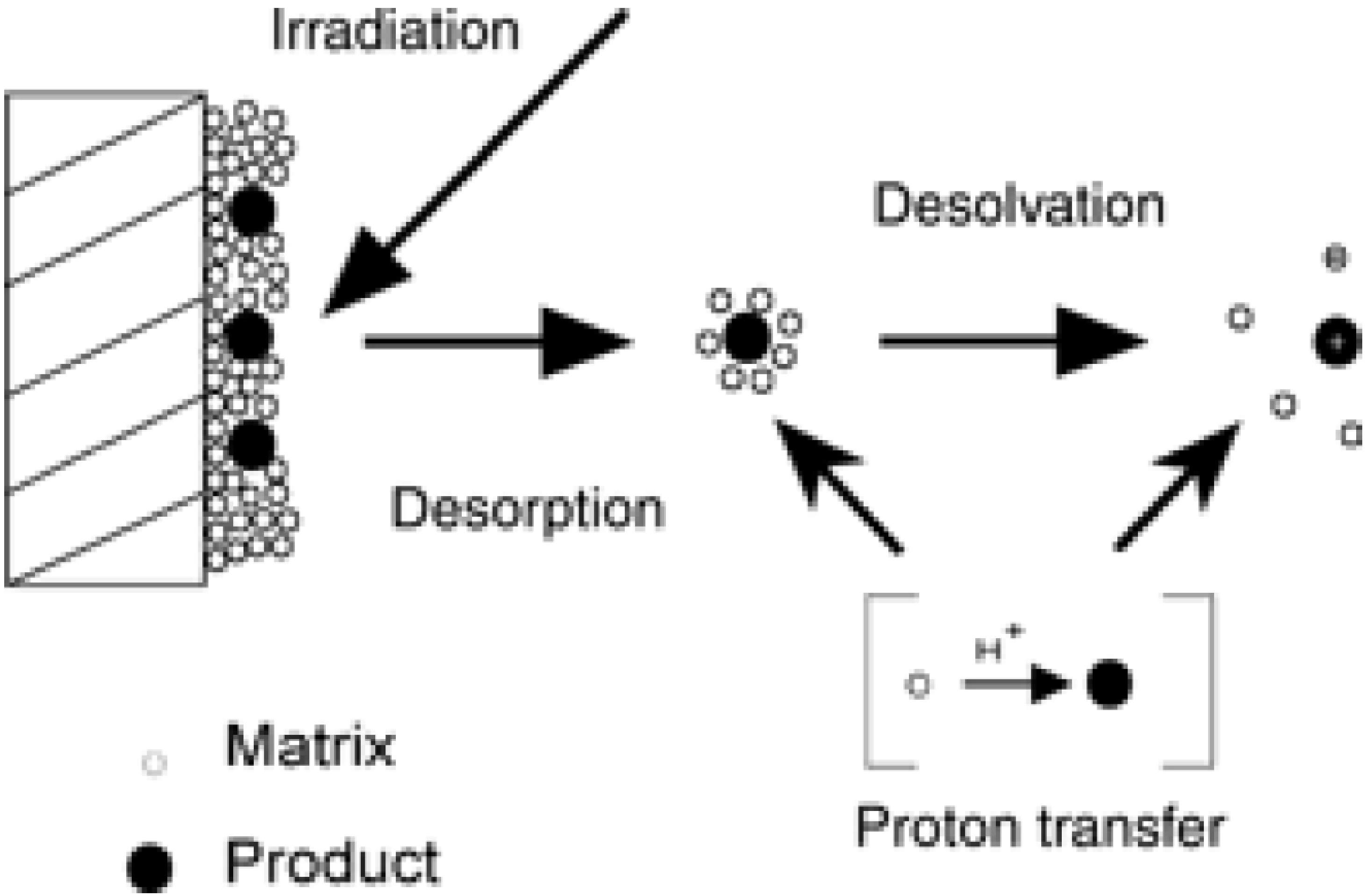
Compound	*X*	*Ion* (m/z)
Aldehyde	H	44
Methyl ketone	CH_3	58
Carboxylic acid	OH	60
Ethyl ketone	CH_3	72
Methyl ester	OCH_3	74
Ethyl ester	OC_2H_5	88

The presence of the halogens. Bromine and chlorine are readily spotted by peak patterns; bromine has two equal peaks two mass units apart, while chlorine has two peaks in the ratio 3 to 1 which are also two mass units apart. Iodine can be spotted from sheer size: it has an atomic weight of 127.

Any major ion of even *m/z ratio. If the molecular ion has an odd m/z* ratio, then the ion contains an odd number of nitrogen atoms; if it has an even *m/z number, then you probably have an elimination reaction or* a McLafferty rearrangement of a carbonyl compound. Look also for ions at *m/z 91 (possibly benzyl) and m/z 77 (possibly phenyl)*.

The main ions in the spectrum, and calculate what size of unit has been split from the molecular ion. From this information you may be able to determine the original structure of the sample.

Matrix Assisted Laser Desorption Source (MALDI)

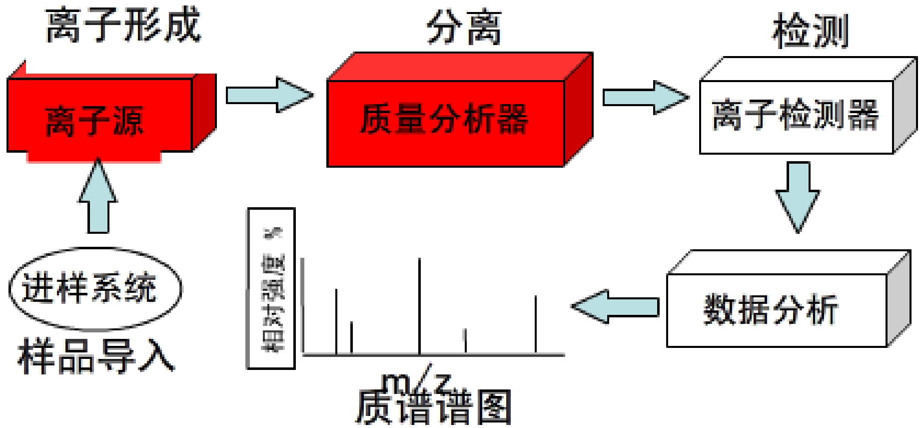


MALDI-TOF ??

基质辅助激光解吸电离飞行时间质谱

- Matrix-assisted laser desorption/ionization time of flight mass spectroscopy
- Developed in 1988 by Professor Franz Hillenkamp and a group of his assistants at the University of Munster at Germany
- They needed an instrument to analyze large biomolecules coupled with low volatility and thermal instability

质谱的基本结构



离子形成

离子源
- EI
- CI（NCI/PCI）
- ESI/ APCI
- FAB
- FD
- FI
- MALDI

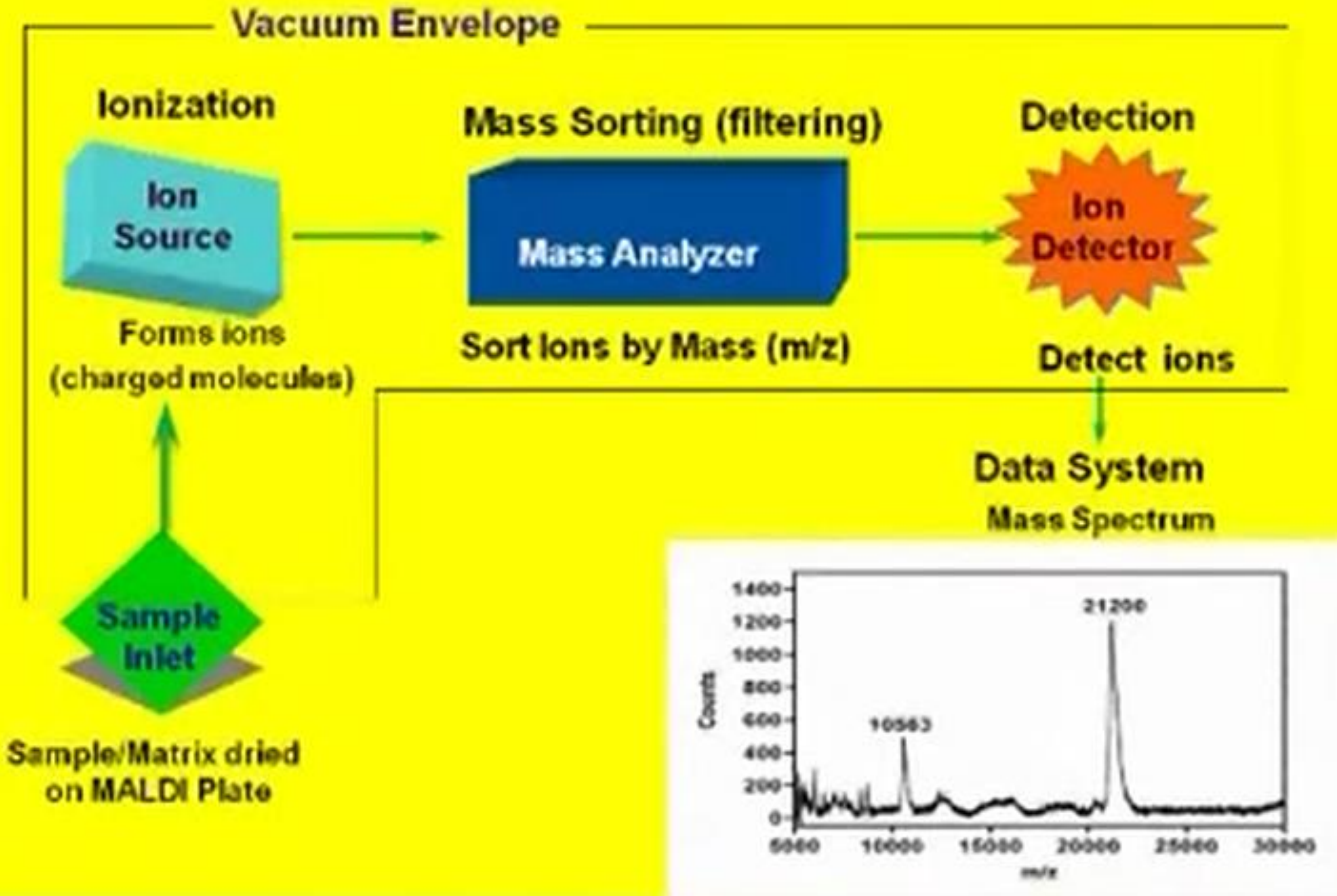
分离器

质量分析器
- Quadrupole
- Ion Trap
- Magnetic Field
- ICR
- TOF

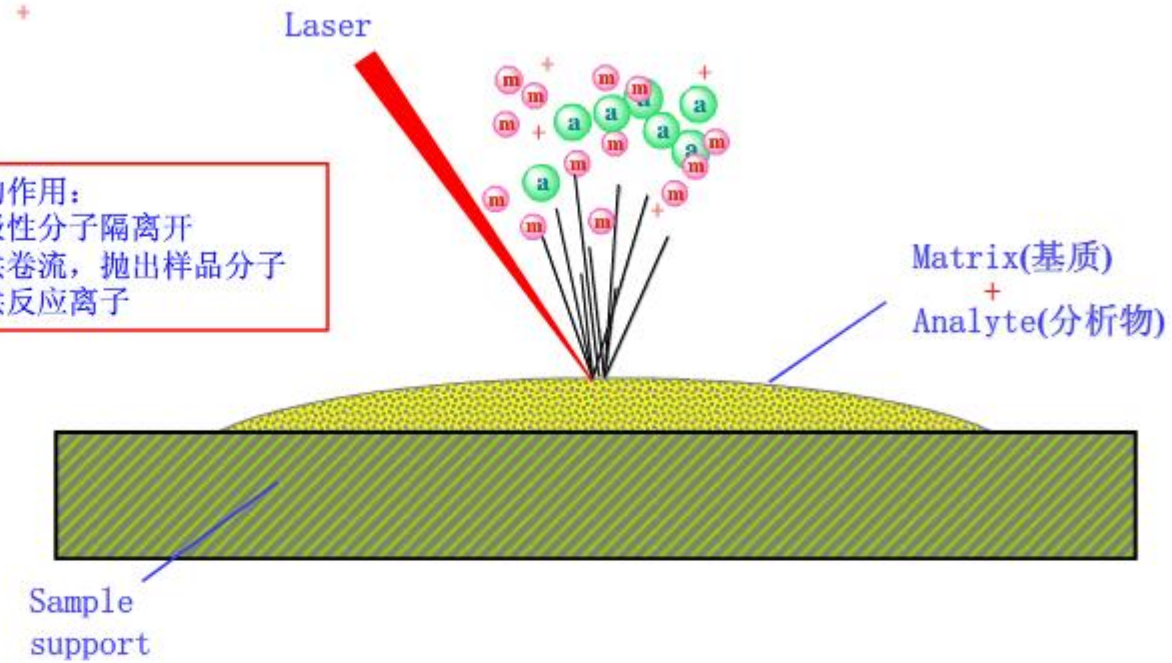
Tandem mass spectrometers (MS/MS):

串联分离器

质量分析器
- Triple Quadrupole (QqQ)
- Quadrupole/time-of-flight (QTOF and QSTAR)
- Tandem time-of-flight (TOF-TOF or TOF2)
- LTQ/Ion trap

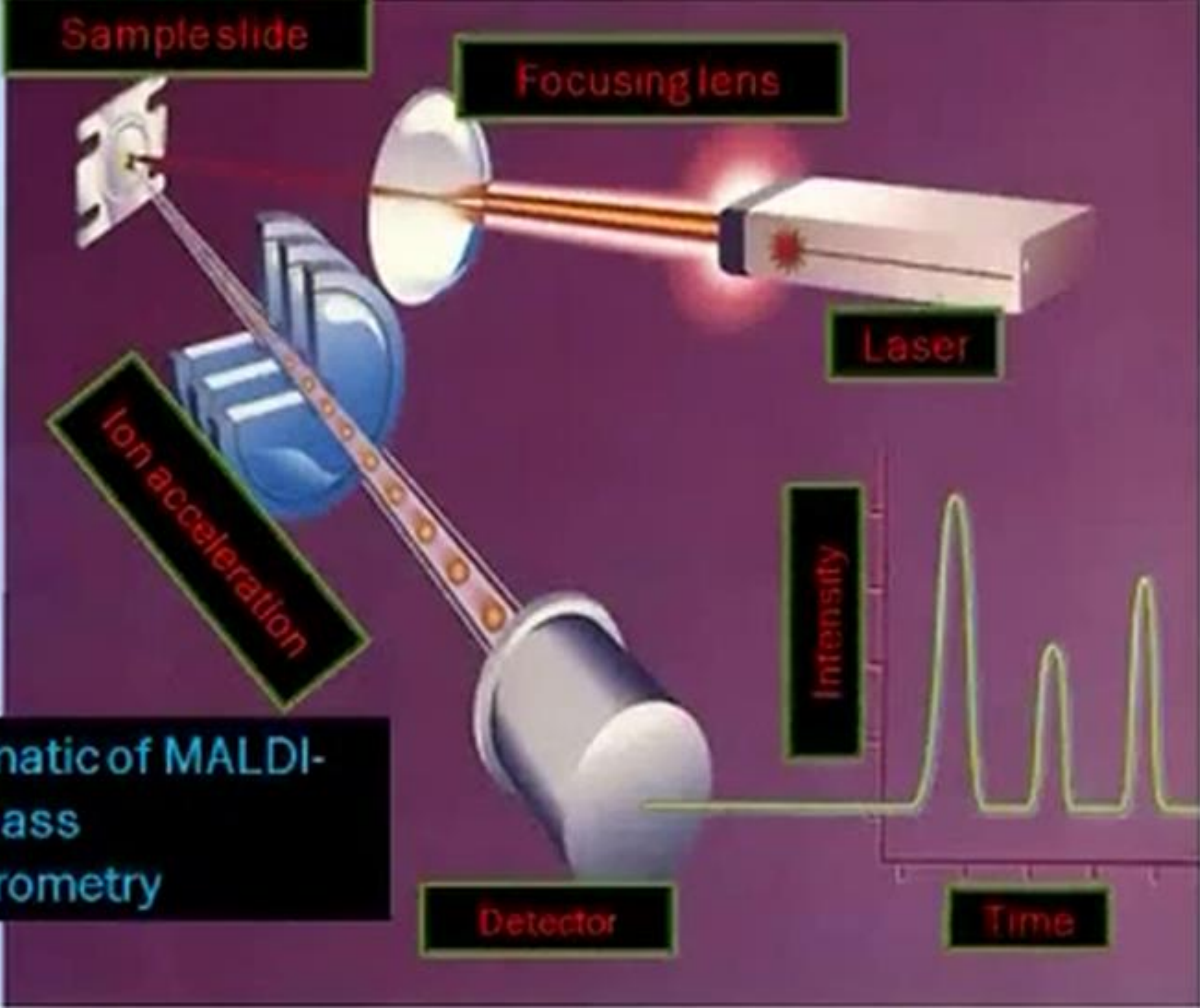

Vacuum Envelope
Ionization
Ion Source
Forms ions (charged molecules)
Mass Sorting (filtering)
Mass Analyzer
Sort Ions by Mass (m/z)
Detection
Ion Detector
Detect ions
Data System
Mass Spectrum
Sample Inlet
Sample/Matrix dried on MALDI Plate
21200
10563
Counts
m/z

MALDI Mechanism

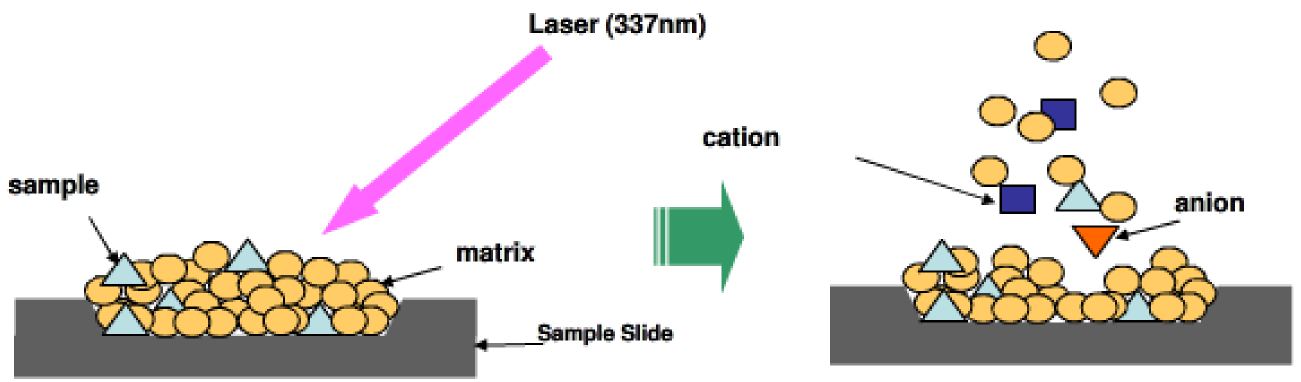


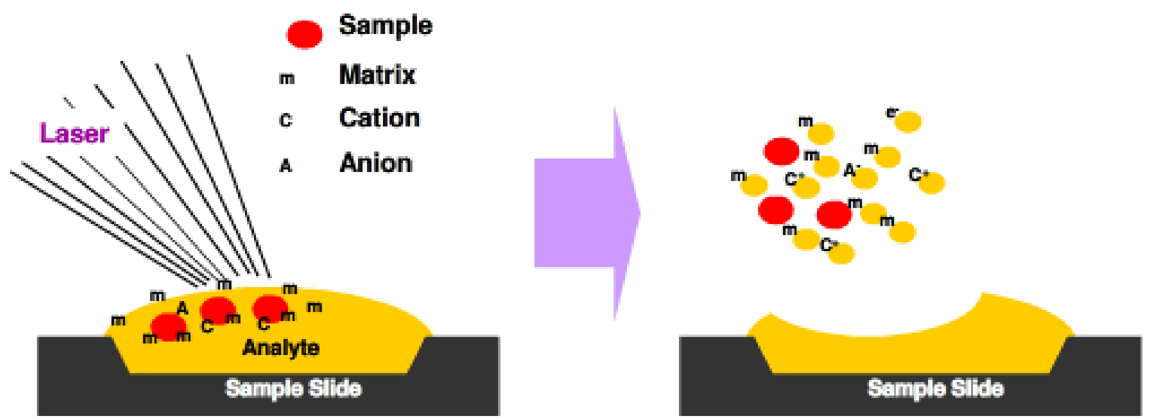
基质的作用：
1. 把极性分子隔离开
2. 提供卷流，抛出样品分子
3. 提供反应离子

诺贝尔奖 — 田中耕一发现使用基质可分析生物大分子 - 蛋白质

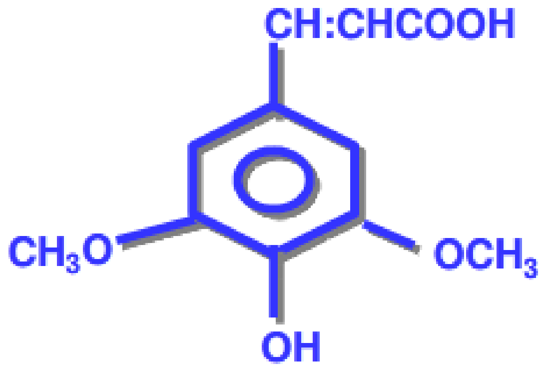


Schematic of MALDI-TOF Mass Spectrometry

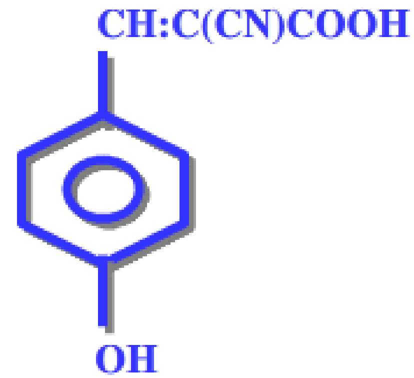

Laser (337nm)
sample
matrix
Sample Slide
cation
anion


Sample
Matrix
Cation
Anion
Laser
Analyte
Sample Slide
Sample Slide

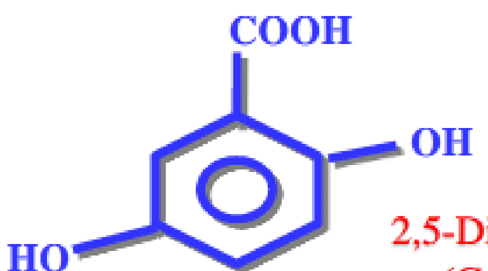
常用基质



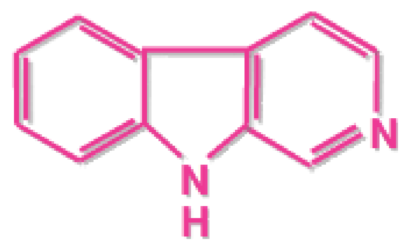
3,5-Dimethoxy-4-hydroxycinnamic acid
(Sinapinic acid, SA)
芥子酸



α−Cyano-4-hydroxycinnamic acid
(CHCA)
肉桂酸

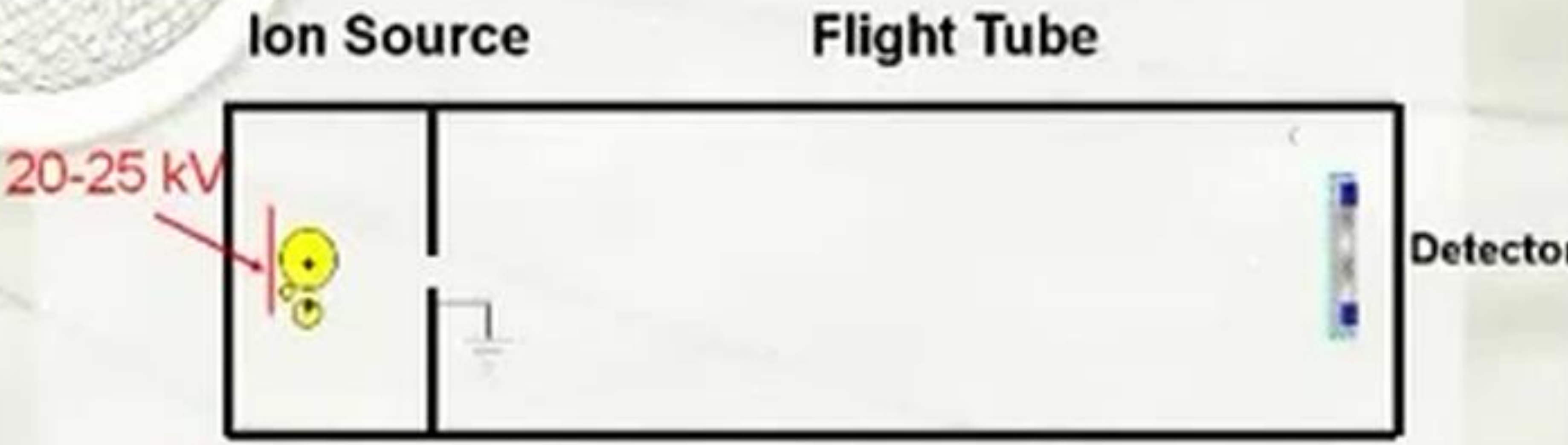


2,5-Dihydroxybenzoic acid
(Gentisic acid, DHB)
2,5-二羟基苯甲酸

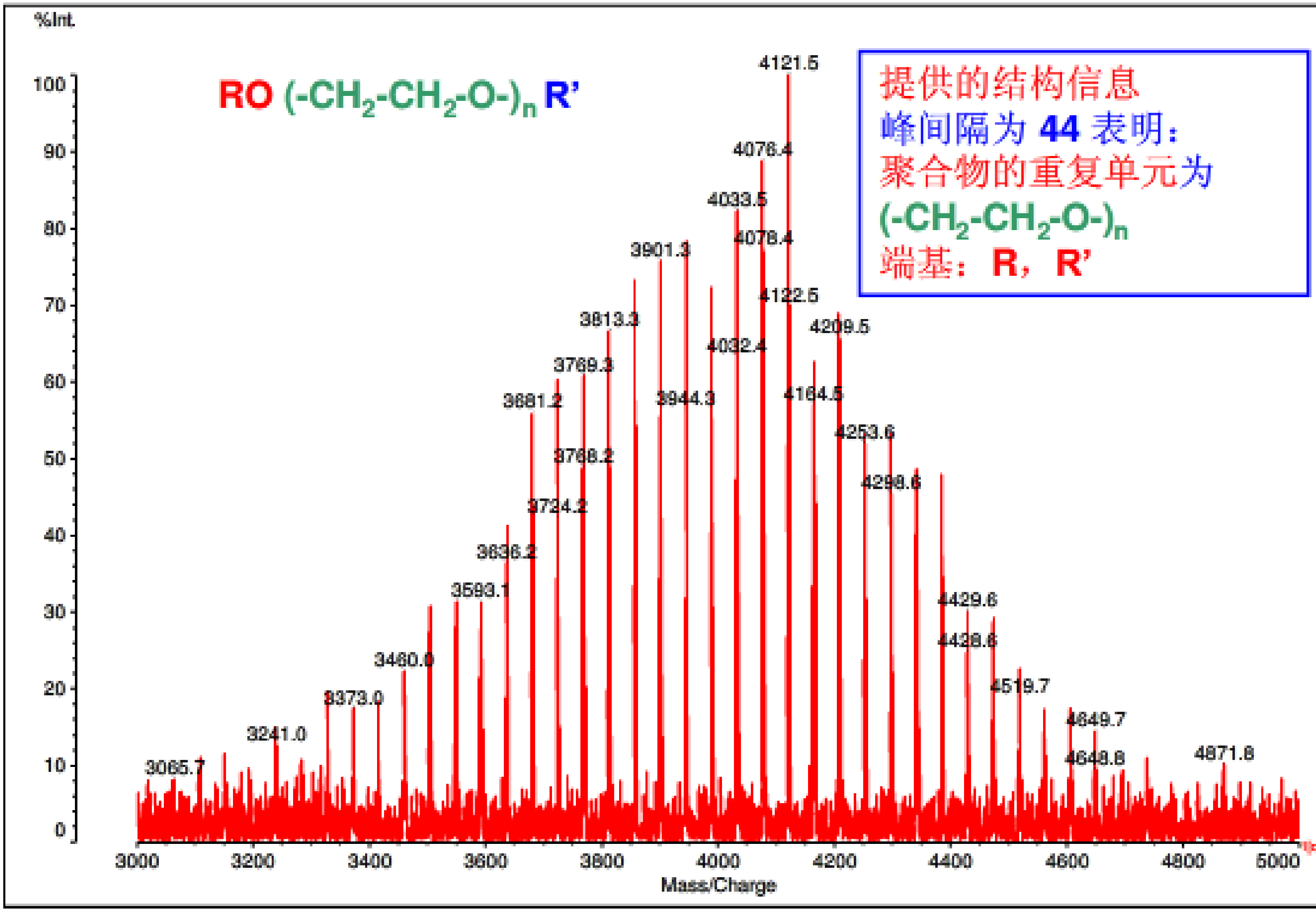


Nor-harmane
2-咔啉
(通用性较好)

Time-of-Flight Mass Analyzer

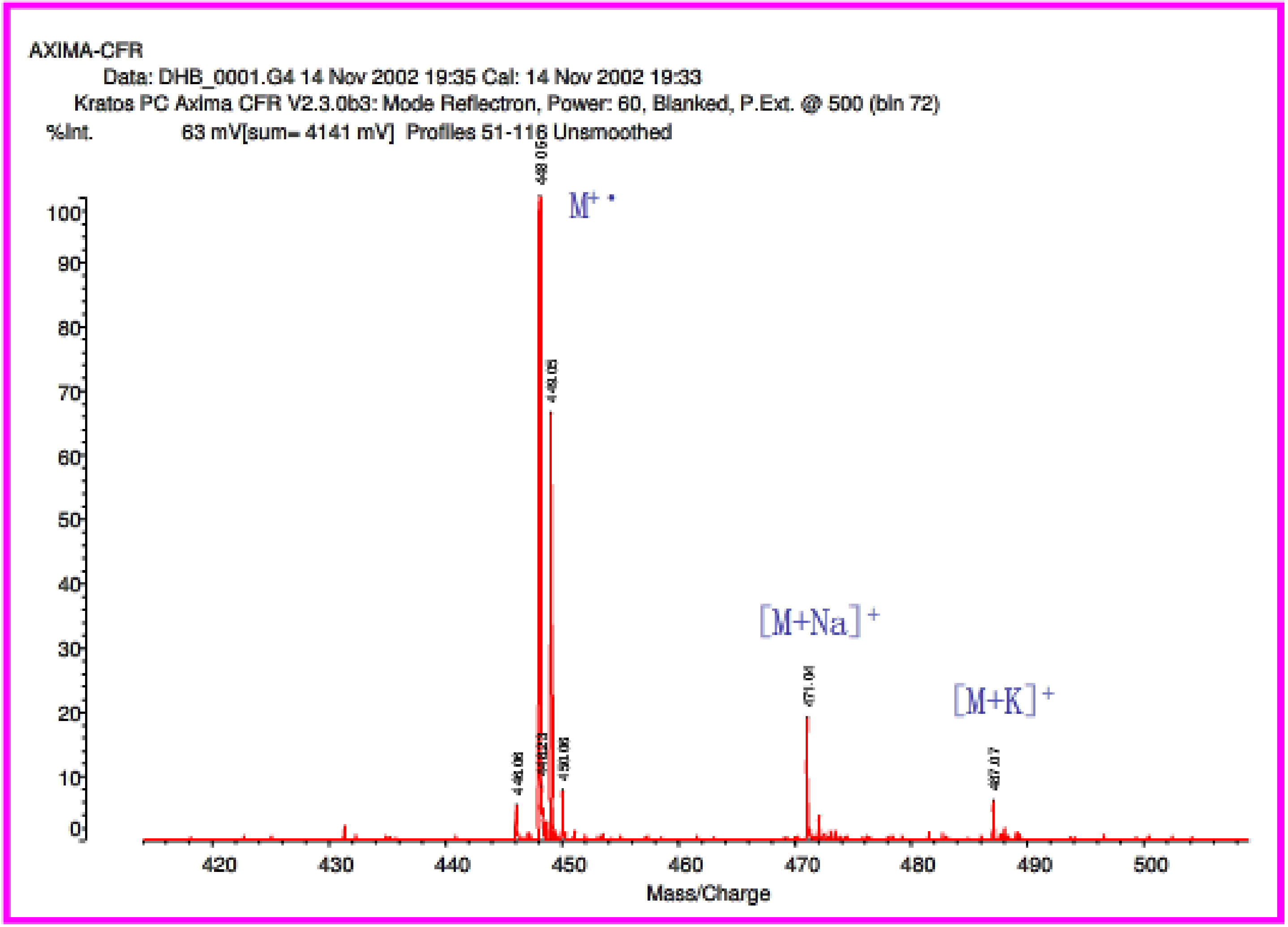


Principle: If ions are accelerated with the same potential at a fixed point and a fixed initial time and are allowed to drift, the ions will separate according to their mass to charge ratios.

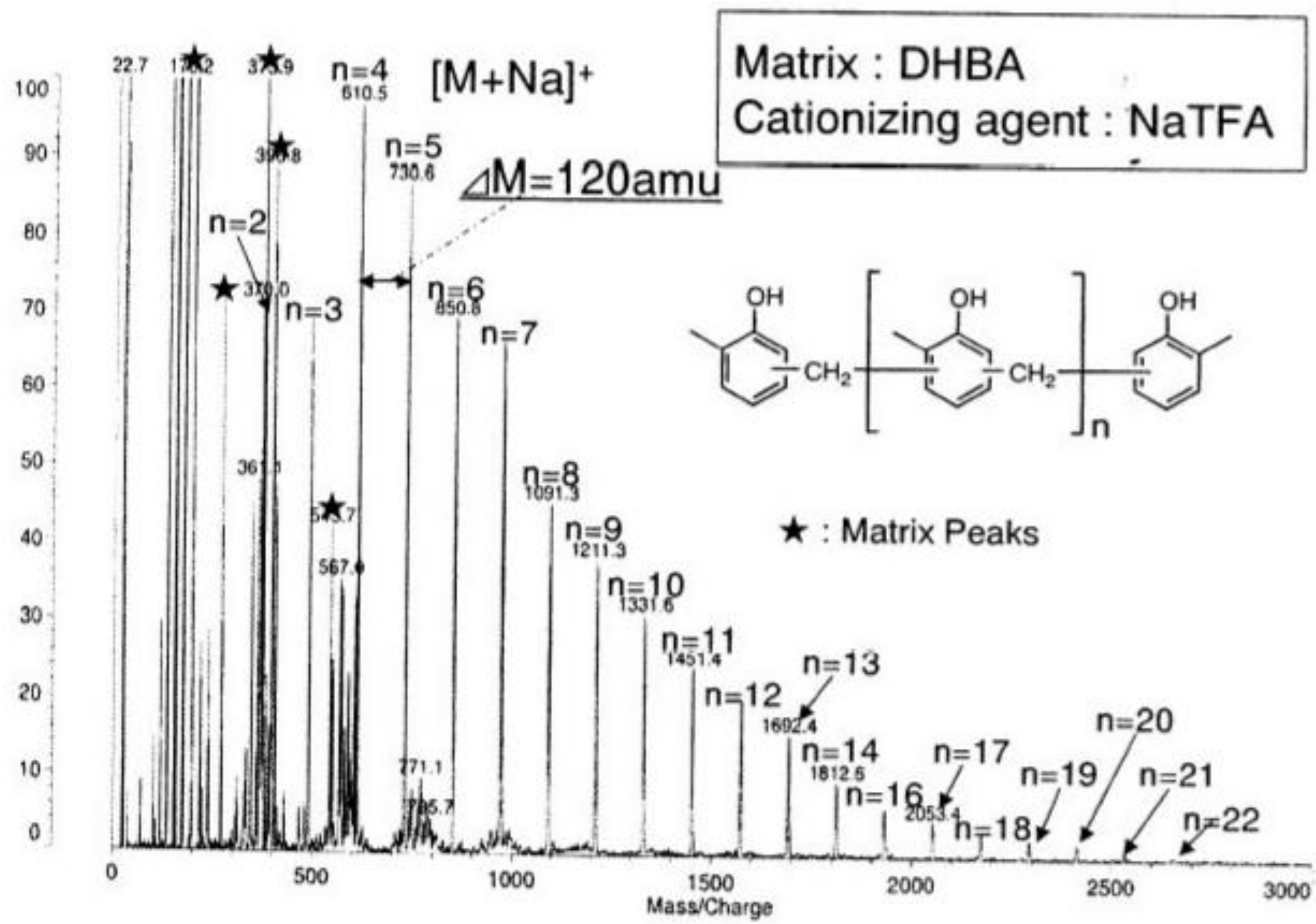


%Int.
RO (-CH2-CH2-O-)n R'
提供的结构信息
峰间隔为 44 表明：
聚合物的重复单元为
(-CH2-CH2-O-)n
端基：R，R'
4121.5
4076.4
4033.5
4078.4
3901.3
4122.5
3813.3
4209.5
4032.4
3769.3
3681.2
3944.3
4164.5
4253.6
3768.2
4298.6
3724.2
3636.2
3593.1
4429.6
4428.6
3460.0
4519.7
3373.0
4649.7
3241.0
4648.8
4871.8
3065.7
Mass/Charge

Small molecules / pharmaceuticals

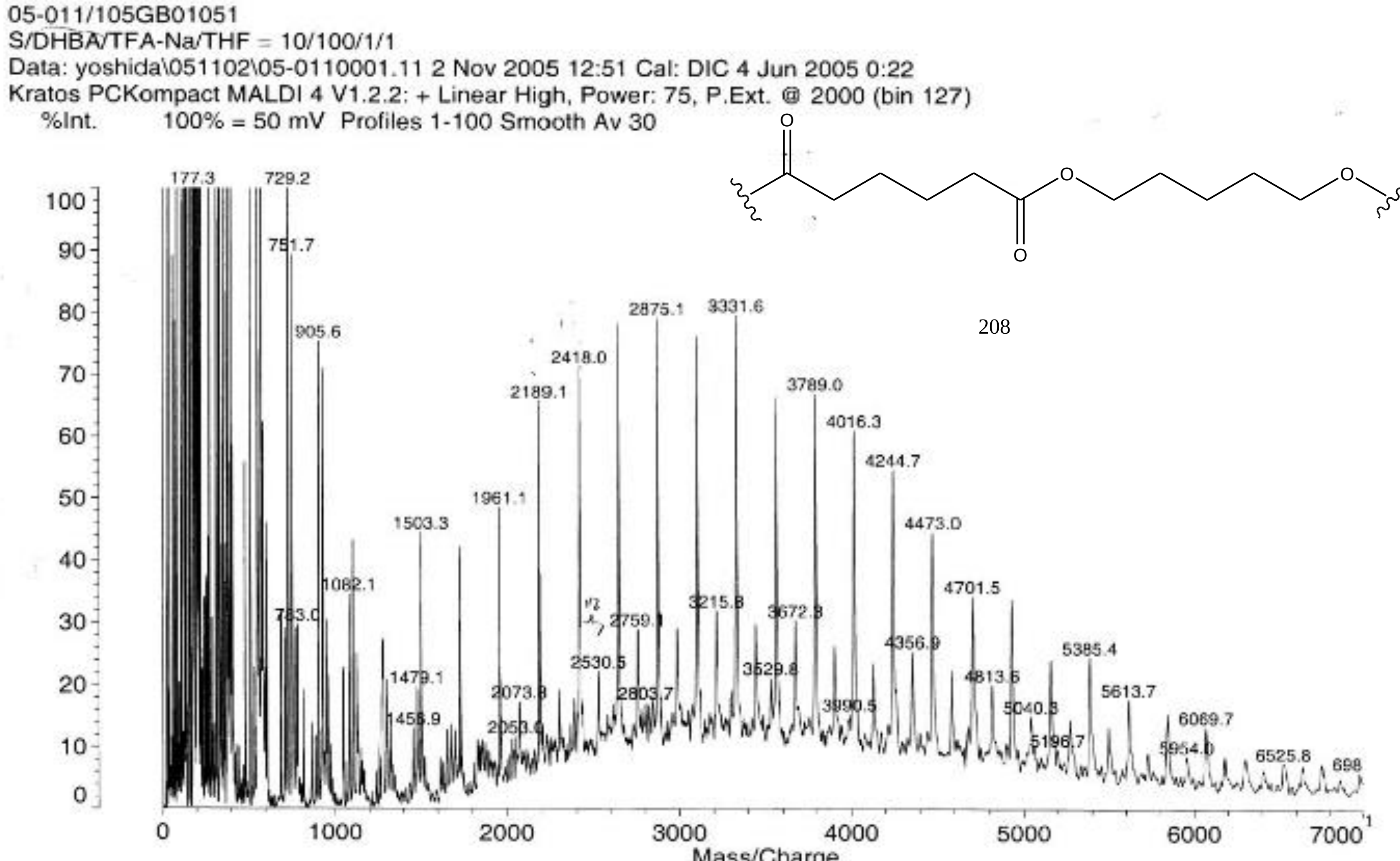


Sample 3 (exp. Mass = 448.1086 Da). Matrix: DHB



Matrix : DHBA
Cationizing agent : NaTFA
[M+Na]+
ΔM=120amu
★ : Matrix Peaks
OH
CH2
n
n=2
n=3
n=4
610.5
n=5
730.6
n=6
850.8
n=7
n=8
1091.3
n=9
1211.3
n=10
1331.6
n=11
1451.4
n=12
n=13
1692.4
n=14
1812.6
n=16
2053.4
n=17
n=18
n=19
n=20
n=21
n=22
22.7
361
567.0
771.1
Mass/Charge
0
500
1000
1500
2000
2500
3000
100
90
80
70
60
50
40
30
20
10

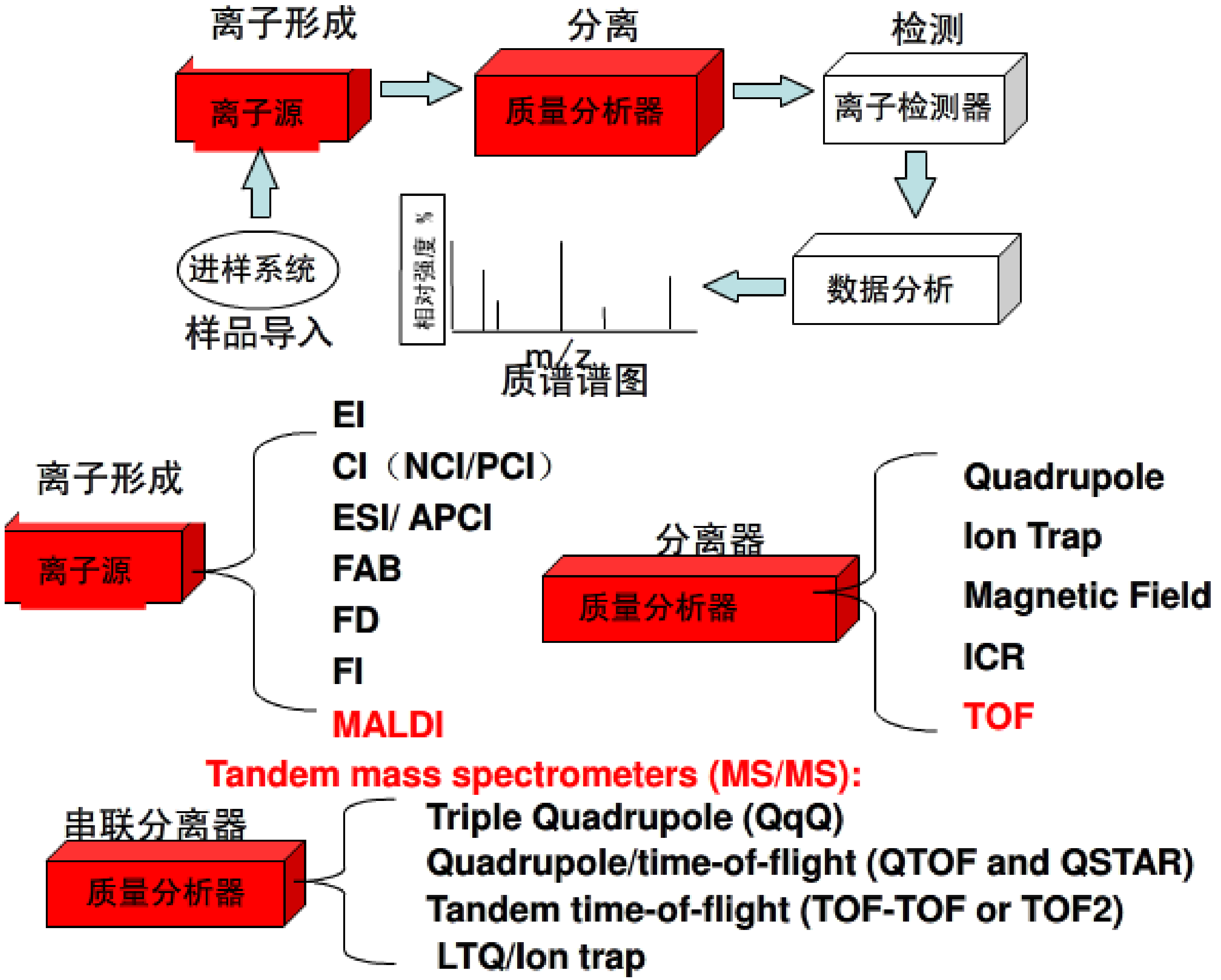
Polyester resin



208

质谱的基本结构

S


离子形成
分离
检测
离子源
质量分析器
离子检测器
进样系统
样品导入
相对强度 %
m/z
质谱谱图
数据分析
EI
CI（NCI/PCI）
ESI/ APCI
FAB
FD
FI
MALDI
离子形成
离子源
分离器
质量分析器
Quadrupole
Ion Trap
Magnetic Field
ICR
TOF
Tandem mass spectrometers (MS/MS):
串联分离器
质量分析器
Triple Quadrupole (QqQ)
Quadrupole/time-of-flight (QTOF and QSTAR)
Tandem time-of-flight (TOF-TOF or TOF2)
LTQ/Ion trap

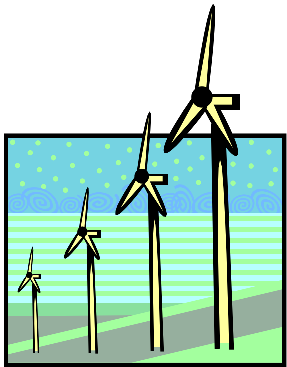
三、质量分析器

◆质量分析器是质谱计的核心。

◆不同类型的质量分析器构成不同类型的质谱计。

◆不同类型的质谱计其功能，应用范围，原理，

实验方法均有所不同。



Mass analyzers

Types of mass analyzers

- Magnetic
- Electrostatic

静态质谱仪

- Time of Flight（飞行时间）
- Quadrupole Mass Filter（四极滤质器）
- Quadrupole ion storage (ion trap)（离子阱）

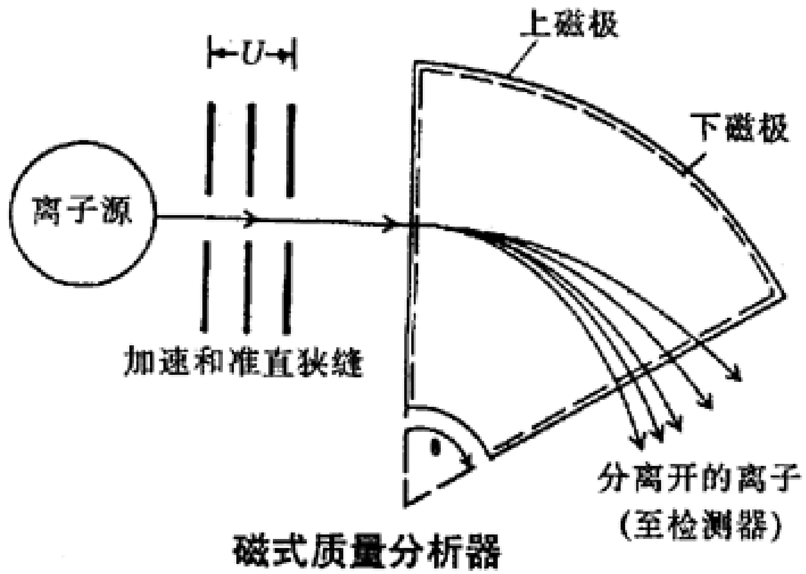
动态质谱仪

We'll look at the Quadrupole Mass Filter - it is commonly used in GC/MS systems.

磁分析器、飞行时间、四极杆、离子捕获、离子回旋

磁分析器

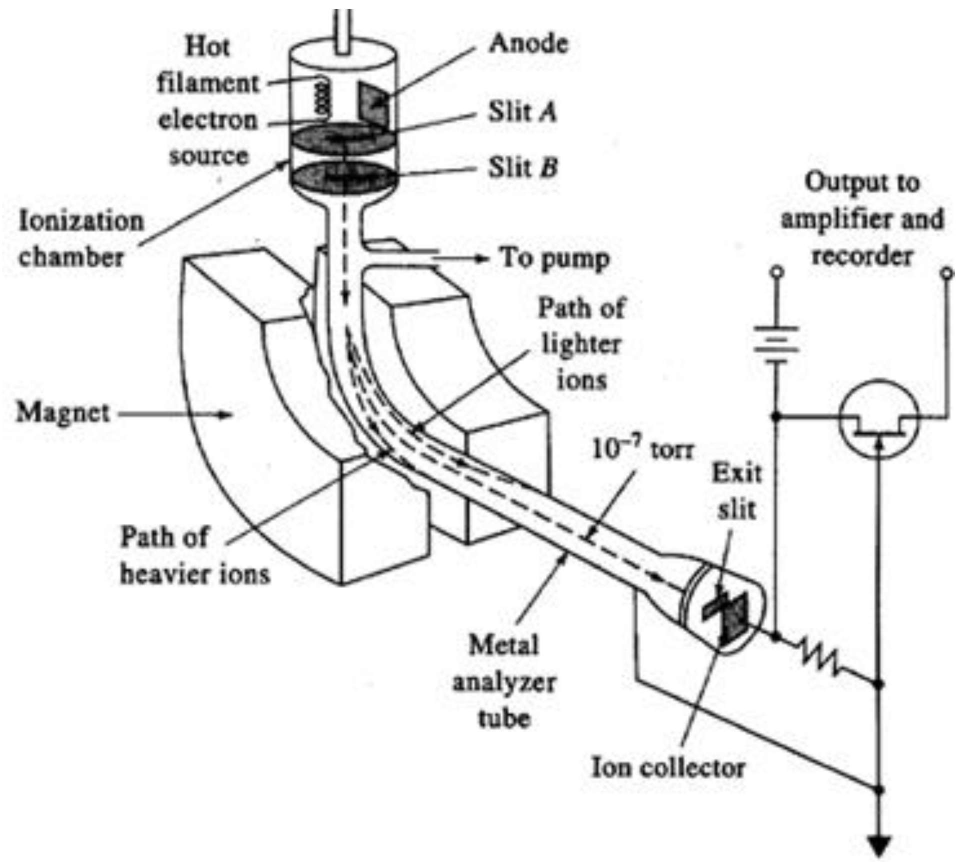
单聚焦型(**Magnetic sector spectrometer**)：仅用一个扇形磁场进行质量分析的质谱仪。



带电离子质量分析器，在磁场作用下，飞行轨道弯曲。

飞行时间分析器(Time of flight, TOF)

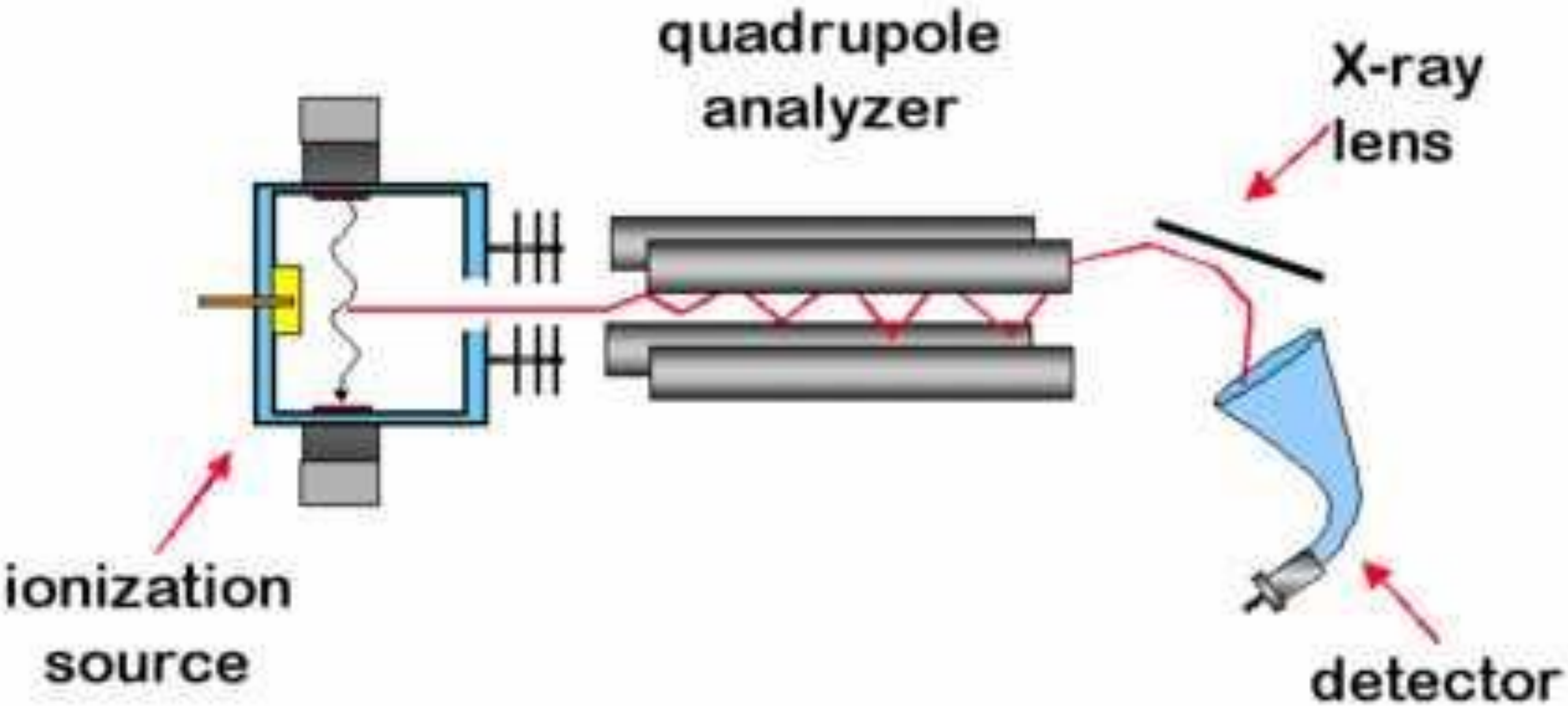
由于不同m/z的离子，其飞出漂移管的时间不同，因而实现了离子的分离。因为连续电离和加速使检测器产生连续输出而不能得到有效信息，因此实际工作中常采用相同频率的电子脉冲电离和脉冲电场加速，被加速的粒子经不同的时间到收集极上，并反馈到示波器上记录，从而得到质谱图。



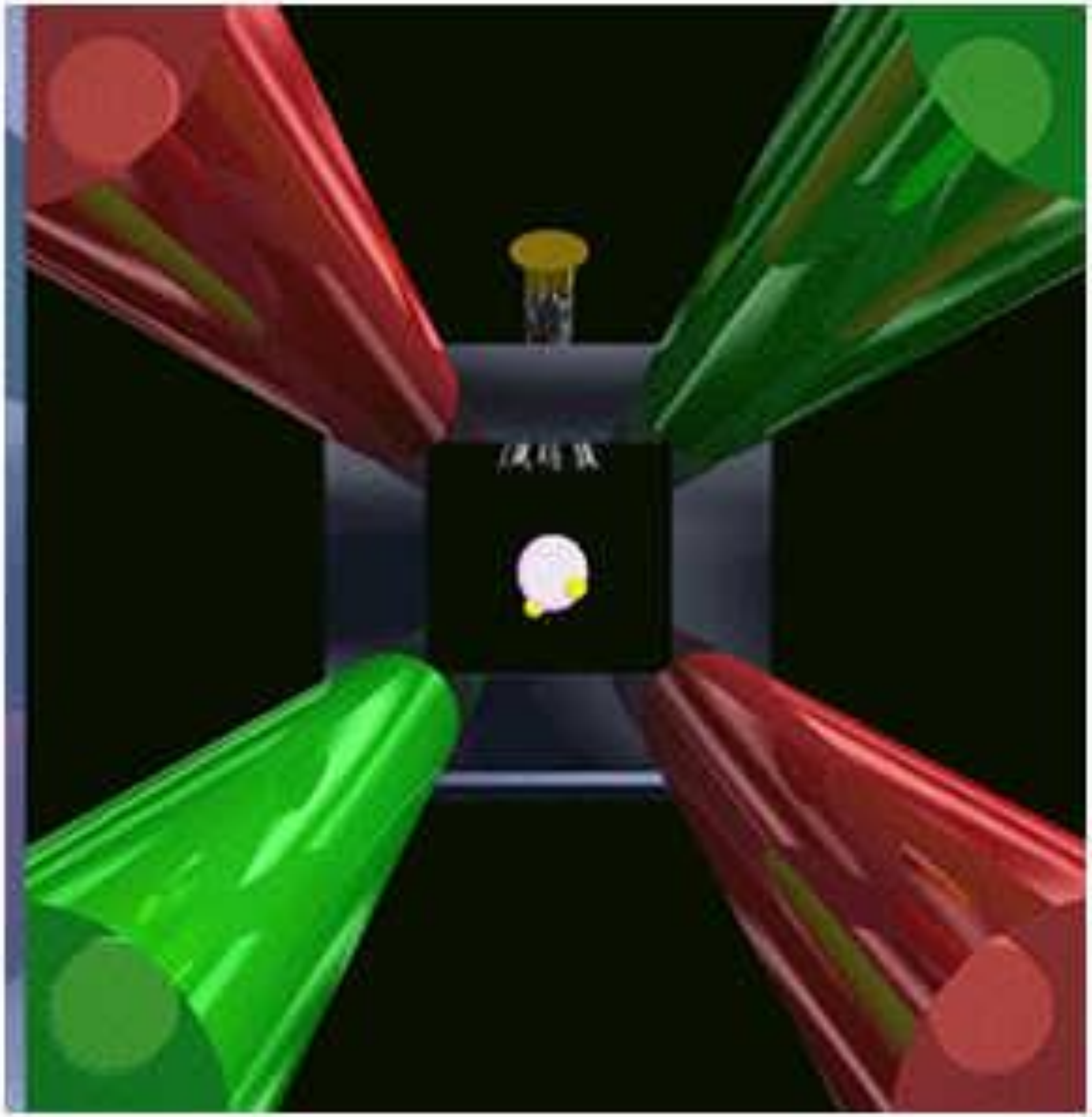
Quadrupole

关于四级杆质谱仪的最早文献时间为1950年中期。发明人沃尔夫冈 保罗 教授(Professor Wolfgang Paul)在1989年获得诺贝尔物理学奖

Schematic of a quadrupole MS system.



Quadrupole

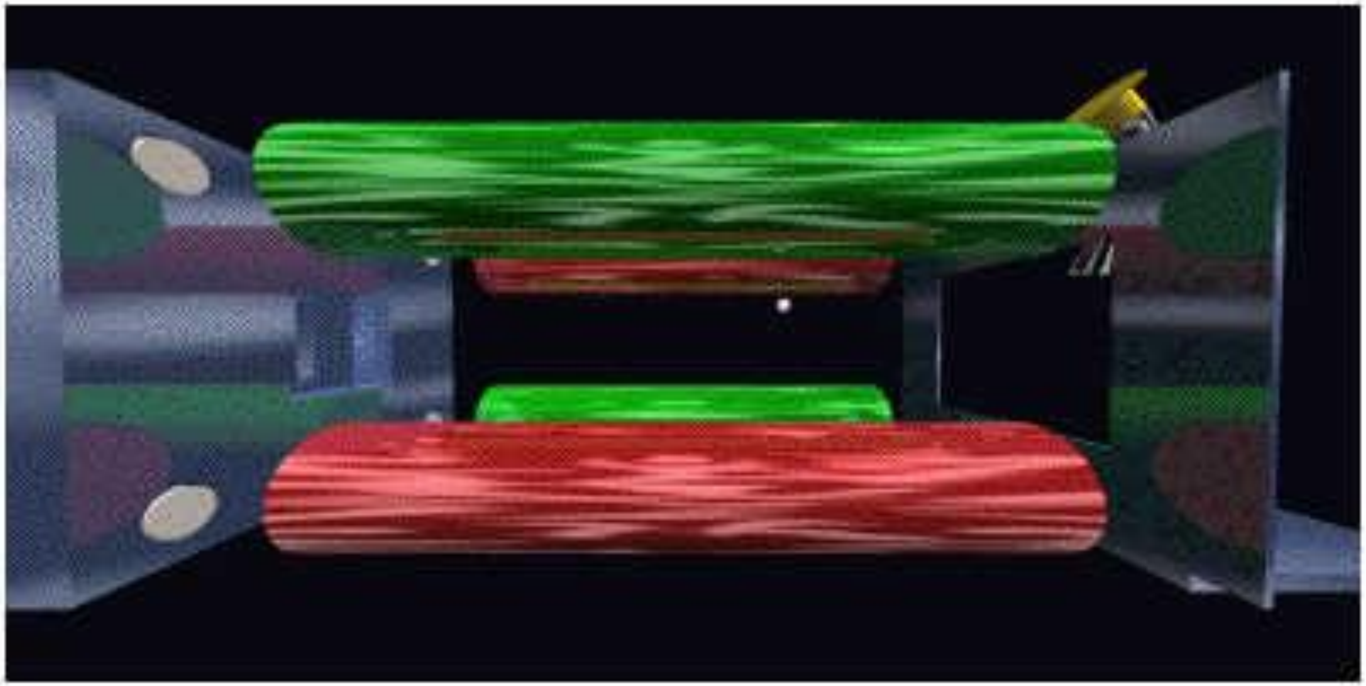


Ions are produced and accelerated.

They enter a region with four poles.

The potential of the poles varies resulting in ions of a specific mass reaching the detector.

Quadrupole



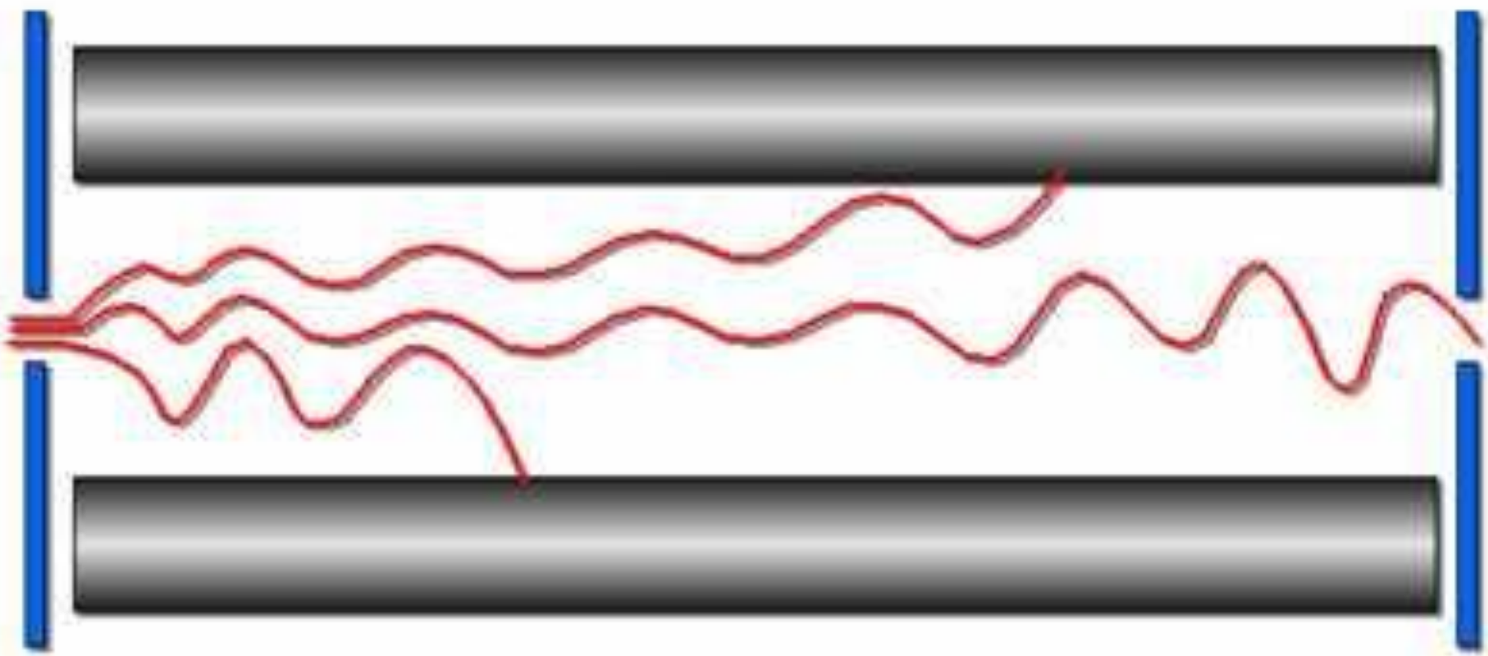
This type of analyzer is the most commonly used for GC/MS work. A large number of spectra are available for comparison to your unknowns.

在这样的电场环境下，离子会根据电场进行震荡。然而，只有特定荷质比的离子可以稳定的通过电场。当极杆上的电压被指定时，质量过小的离子会受到很大的电压影响，从而进行非常激烈的震荡，导致碰触极杆失去电荷而被真空系统抽走；质量过大的离子因为不能受到足够的电场牵引，最终导致碰触极杆或者飞出电场而无法通过质量选择器。

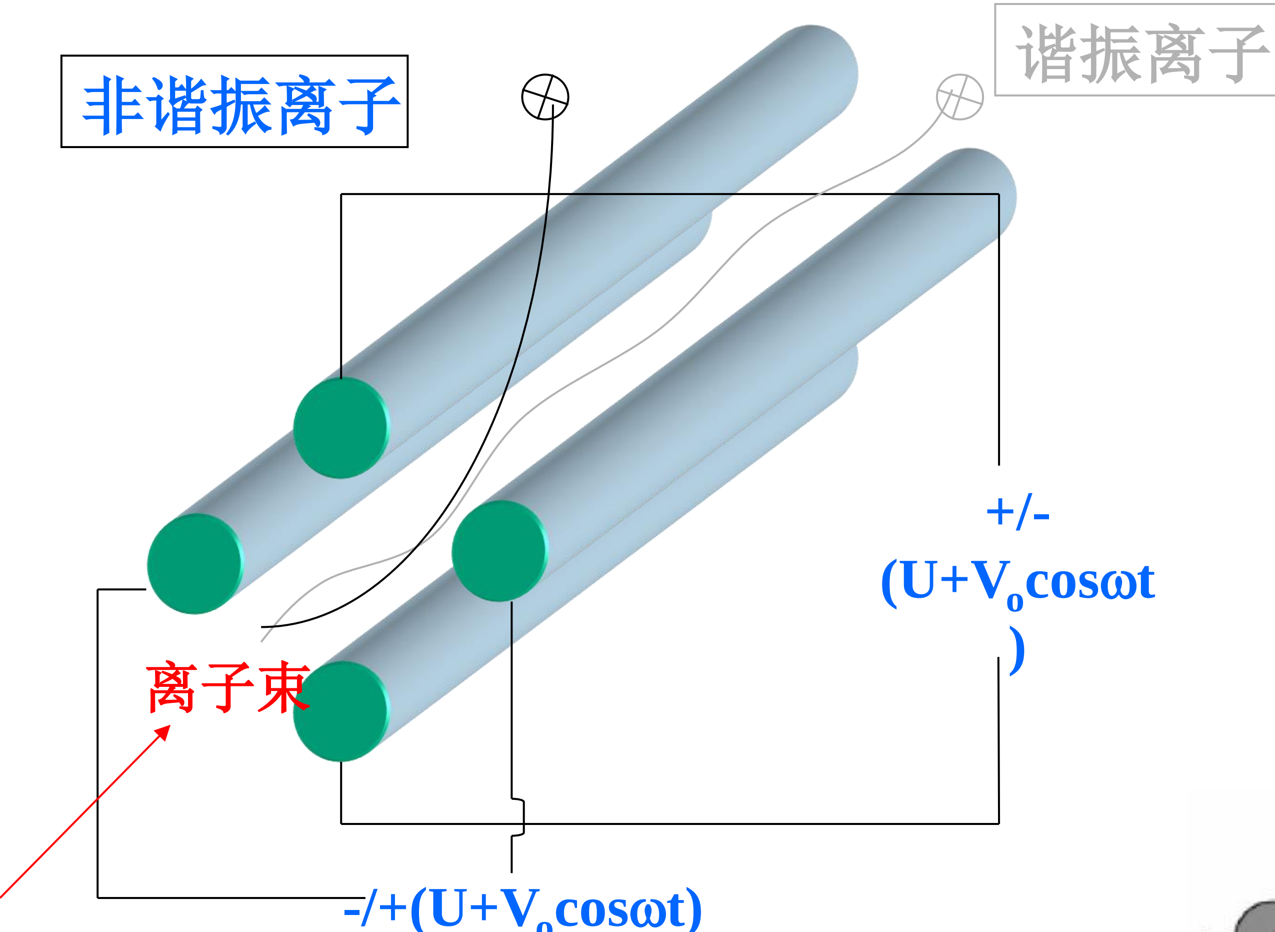
Quadrupole

At any set of conditions, only ions of a specific M/Z can successfully travel through the entire filter.

Others are drawn into the rods.

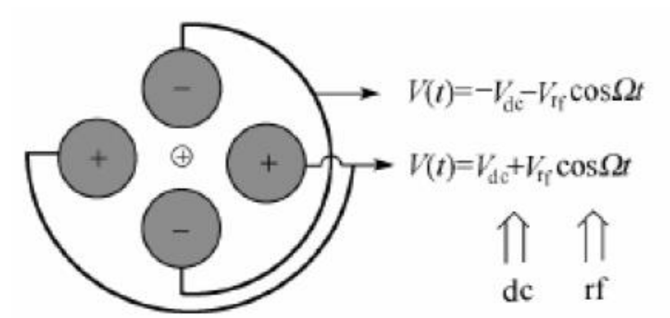


四极杆分析器



$$q_z = -q_z = \frac{2eV}{mr^2\omega^2}$$

$$a_z = -a_z = \frac{4eU}{mr^2\omega^2}$$

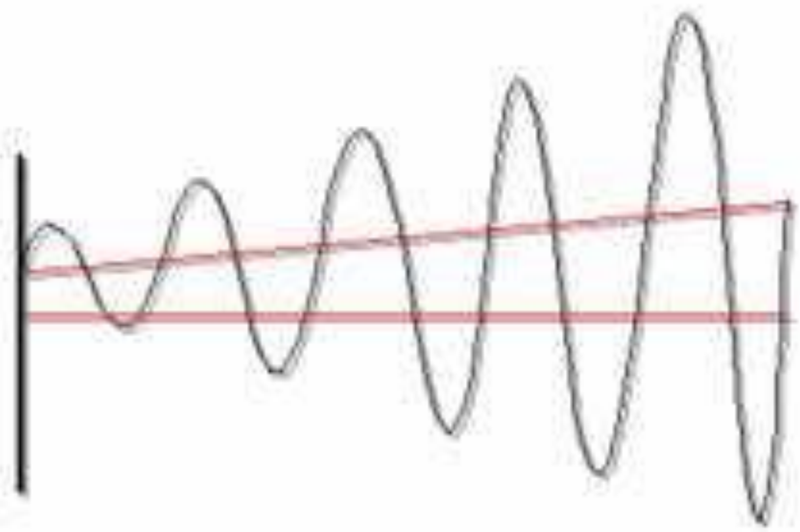


Quadrupole Scanning

When scanning over a mass range, the DC potential is varied linearly with the RF/DC ratio held constant.

反相交变电压/直流电

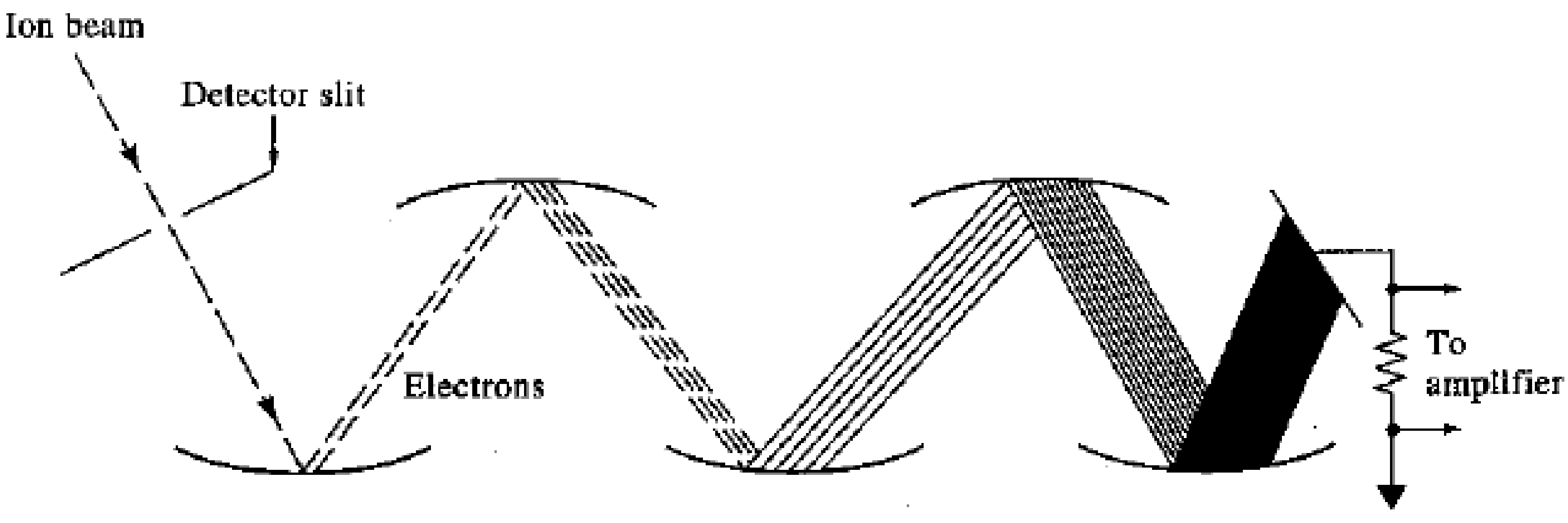
This is how entire mass spectra are obtained.



Mass range:
10 - 2000 M/Z
10 - 800 M/Z is typical

An entire mass spectra (10-800 M/Z) can be acquired in about one second.

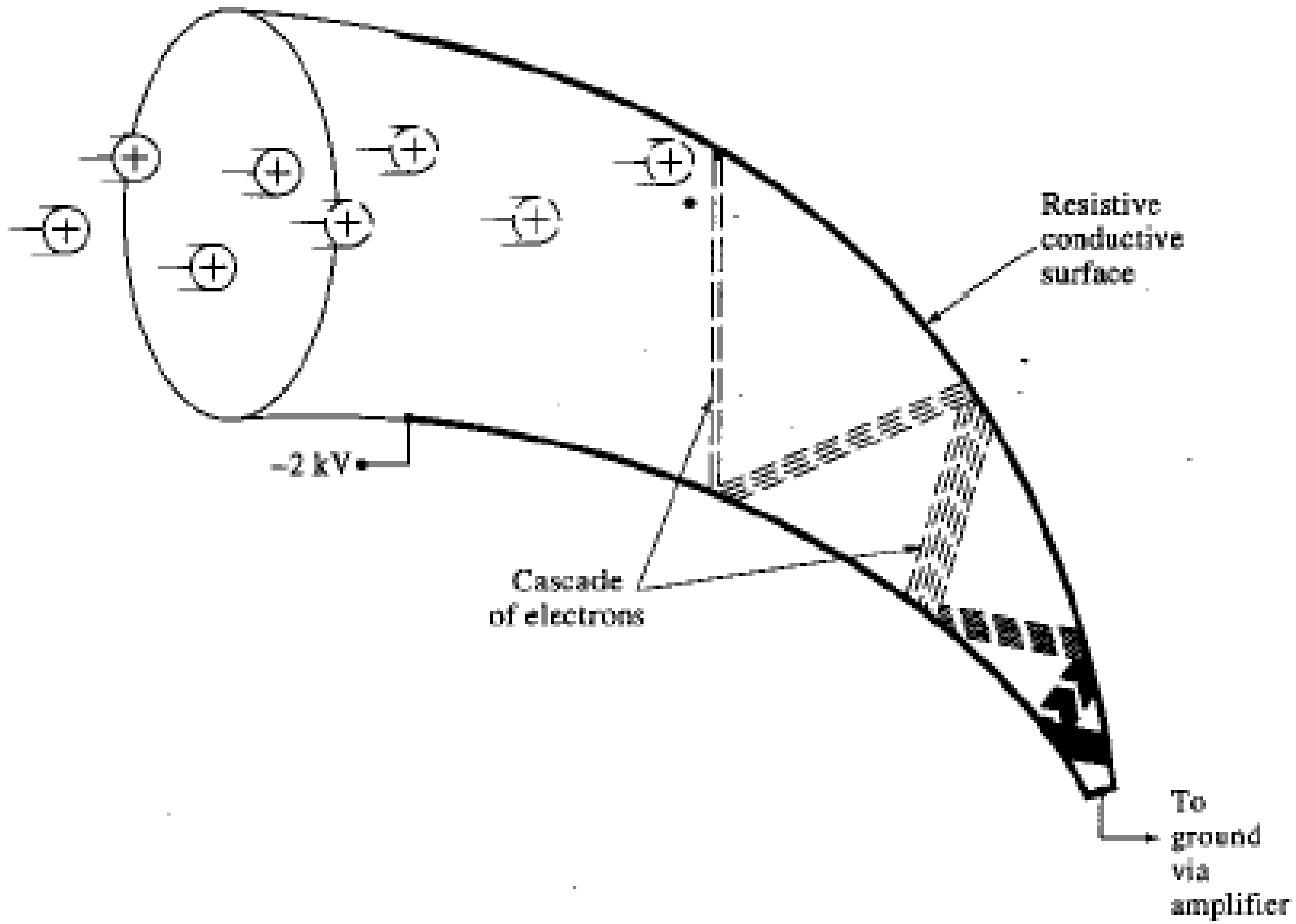
Ion detector



(a)

Cathode potential gradually changes positively

Discrete dynode(倍增器电极) electron multiplier

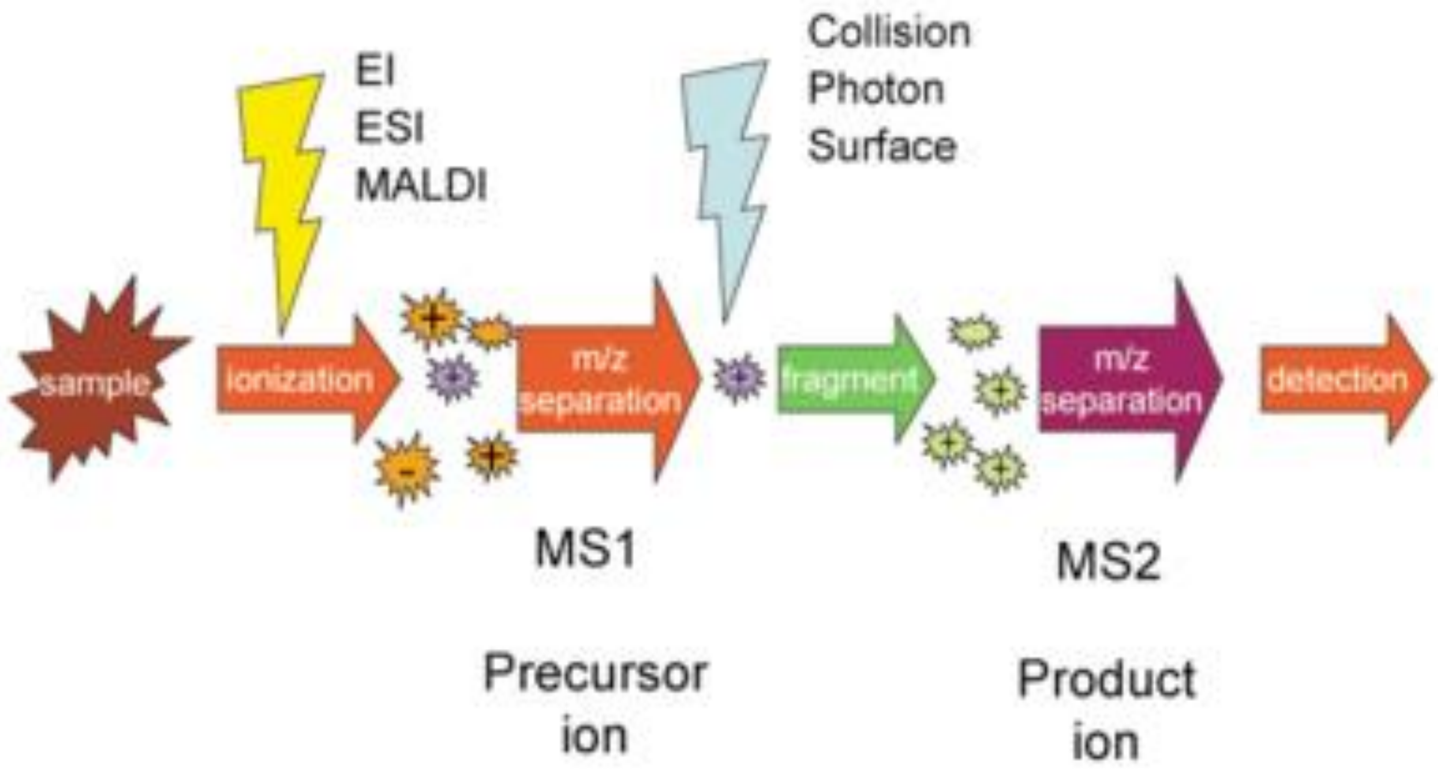


(b)

Figure 11-2 (a) Discrete dynode electron multiplier. Dynodes are kept at successively higher potentials via a multistage voltage divider. (b) Continuous dynode electron multiplier.
(Adapted from J. T. Watson, Introduction to Mass Spectrometry, *p. 247. New York: Raven Press, 1985. With permission.)*

串联质谱

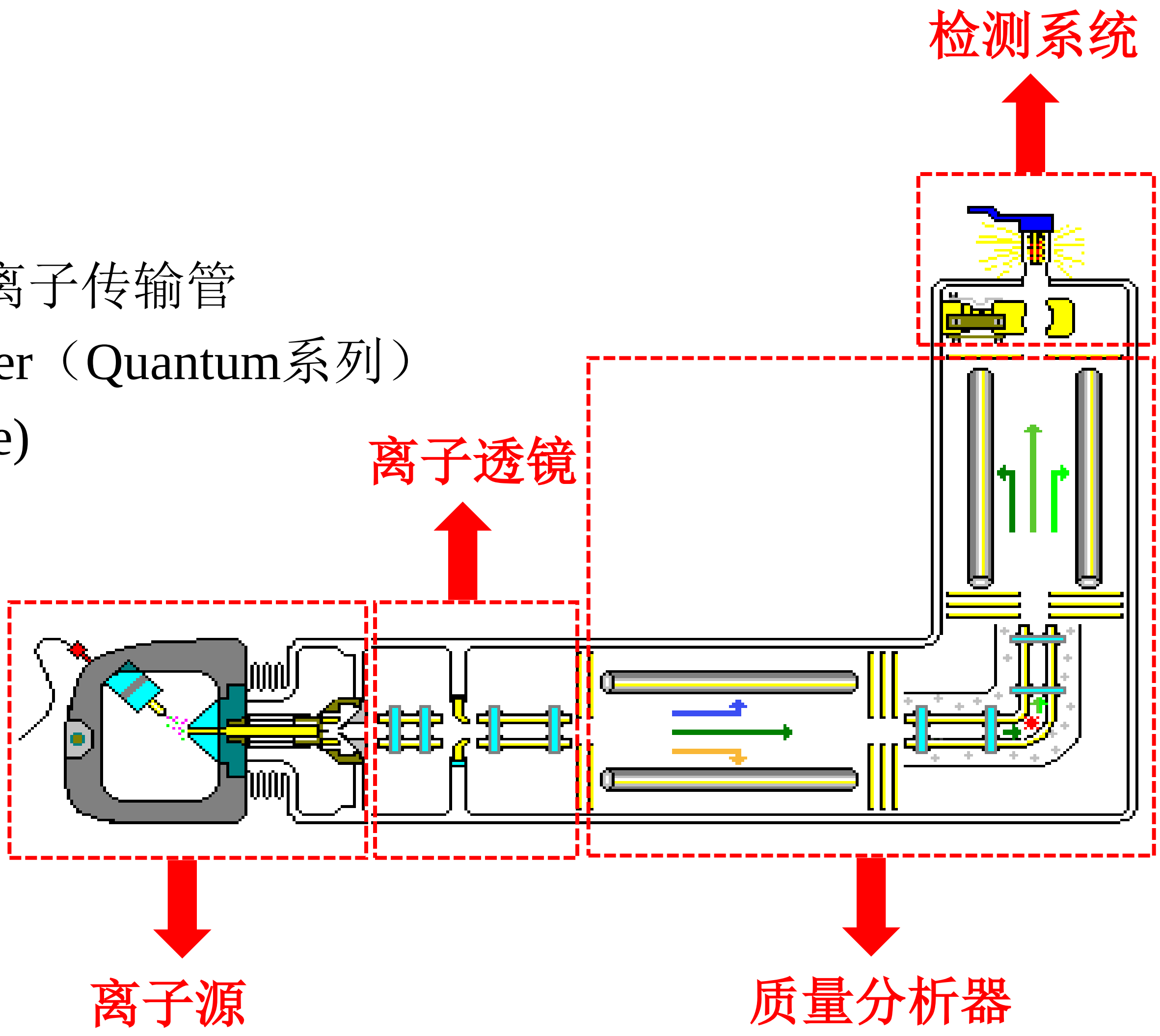
（QQQ）



EI
ESI
MALDI
Collision
Photon
Surface
sample
ionization
m/z separation
fragment
m/z separation
detection
MS1
MS2
Precursor ion
Product ion

TSQ三重四极杆的仪器结构

- Ion Max离子源
  - HESI-II/APCI
  - 吹扫挡锥
  - 可加热的金属离子传输管
  - T-lens和Skimmer（Quantum系列）
  - S-lens（Vantage)
- 离子透镜组件
  - Q0和Q00
- 质量分析器
  - Q1和Q3四极杆
  - Q2碰撞池
- 检测系统
  - 打拿极
  - 电子倍增器

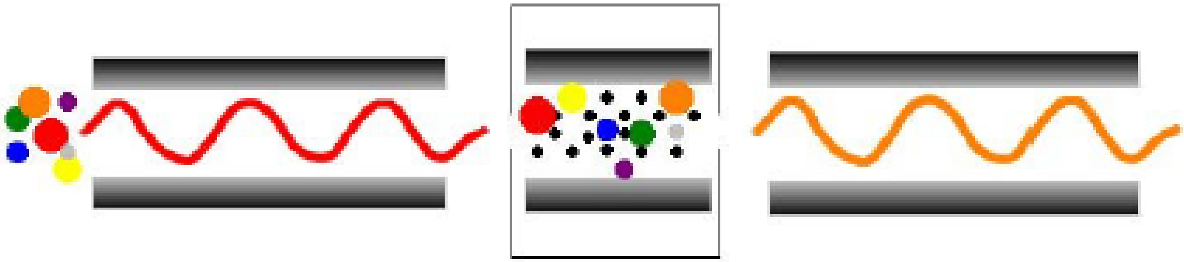


TSQ三重四极杆质谱的扫描模式

扫描模式	Q1	Q2	Q3	目的
全扫描	扫描	全部通过	全部通过	分子量信息
选择离子扫描	固定质荷比 m/z	全部通过	全部通过	定量
子离子扫描	固定质荷比 m/z	全部通过(+ CE)	扫描	结构信息
选择反应监测扫描	固定质荷比 m/z	全部通过(+ CE)	固定质荷比 m/z	目标定量
中性丢失	扫描	全部通过 (+ CE)	扫描	分析物筛选监测
母离子扫描	扫描	全部通过 (+ CE)	固定质荷比 m/z	分析物筛选监测

全扫描(Scan)

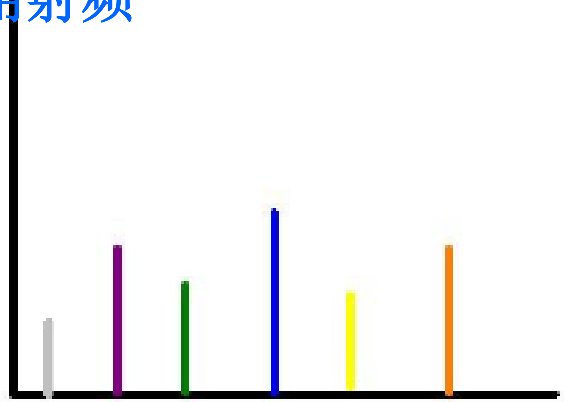
Q1全扫描模式——



Q1扫描

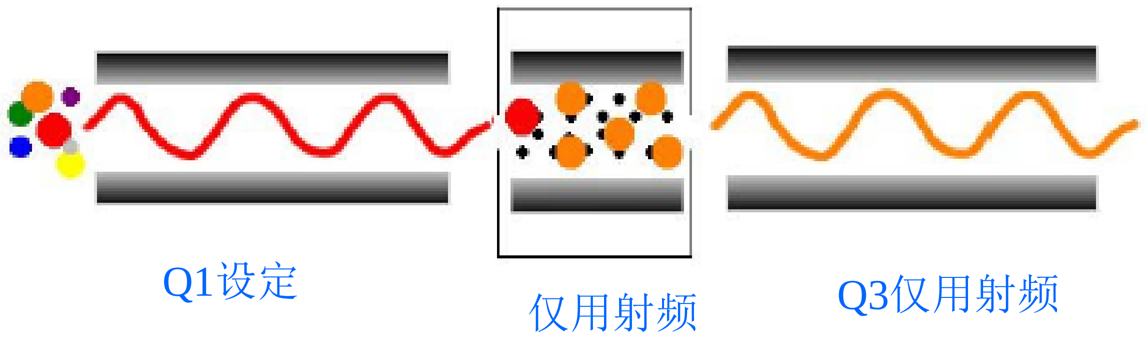
仅用射频

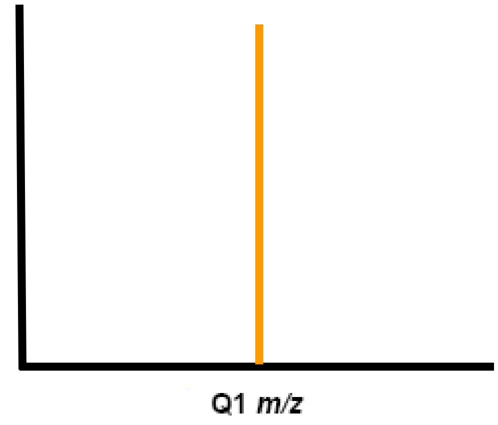
Q3仅用射频



选择离子扫描(SIM)

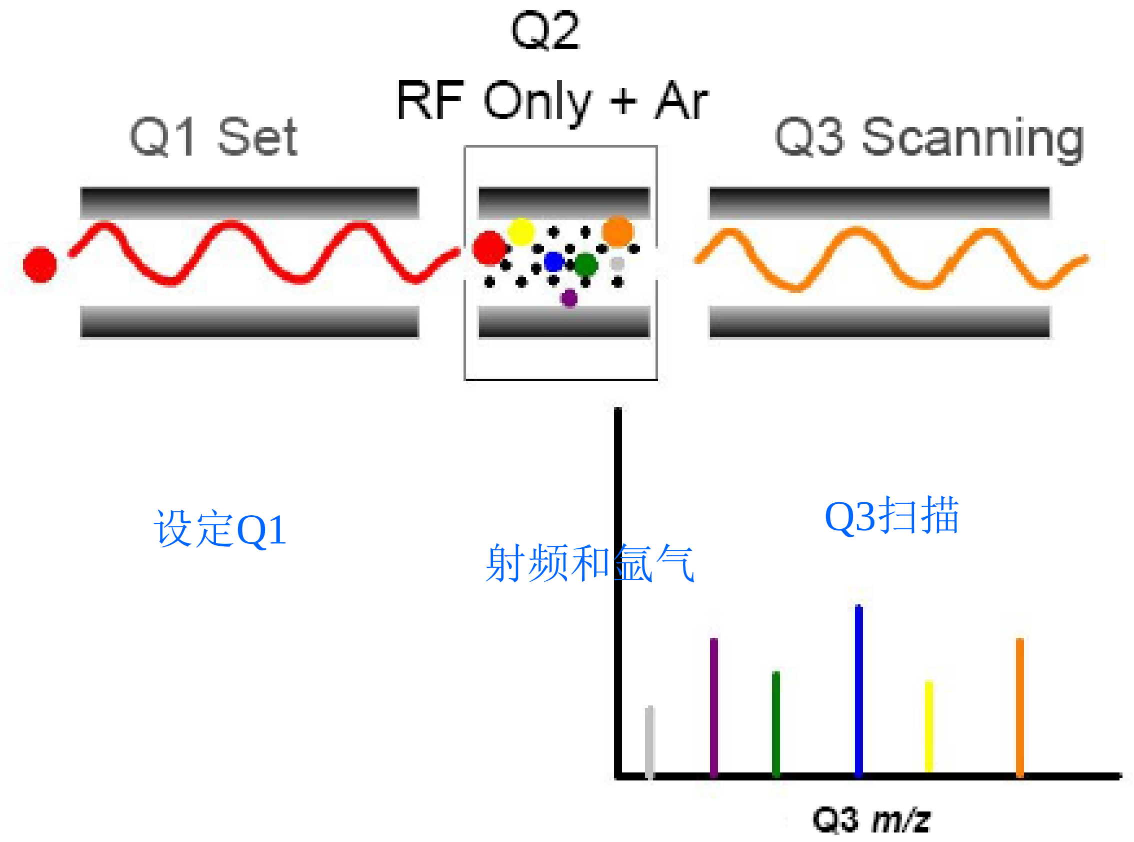
目的: 对特定离子进行扫描（定量）





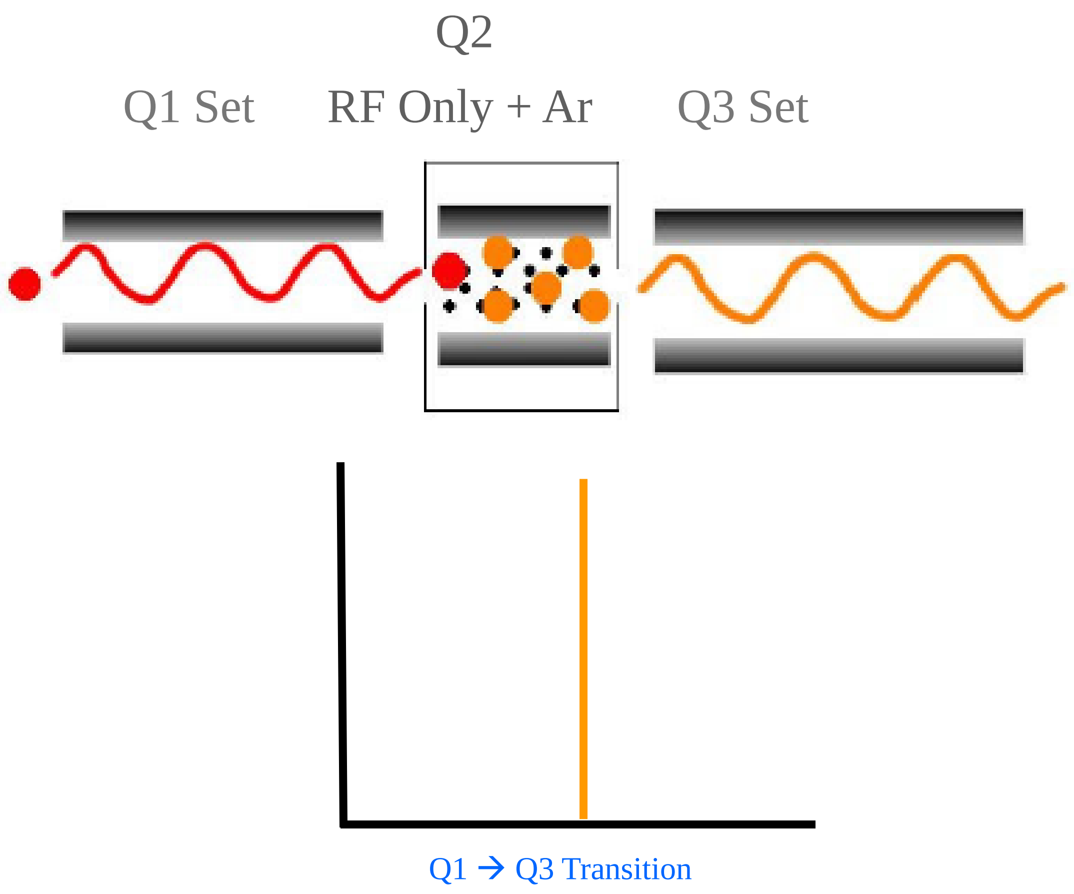
二级子离子全扫描

目的：对特定母离子的碎片离子进行全扫描



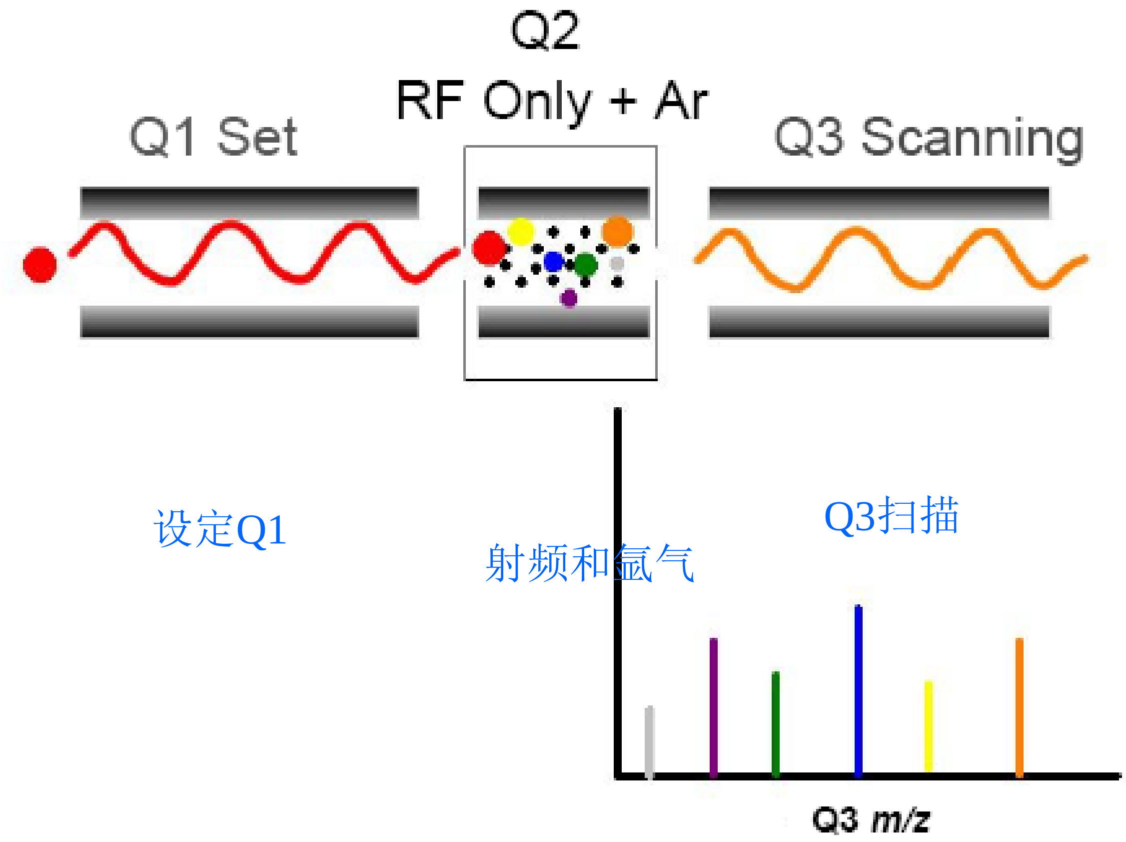
选择反应监测（SRM)

目的：对子离子进行定量扫描



二级子离子全扫描

目的：对特定母离子的碎片离子进行全扫描



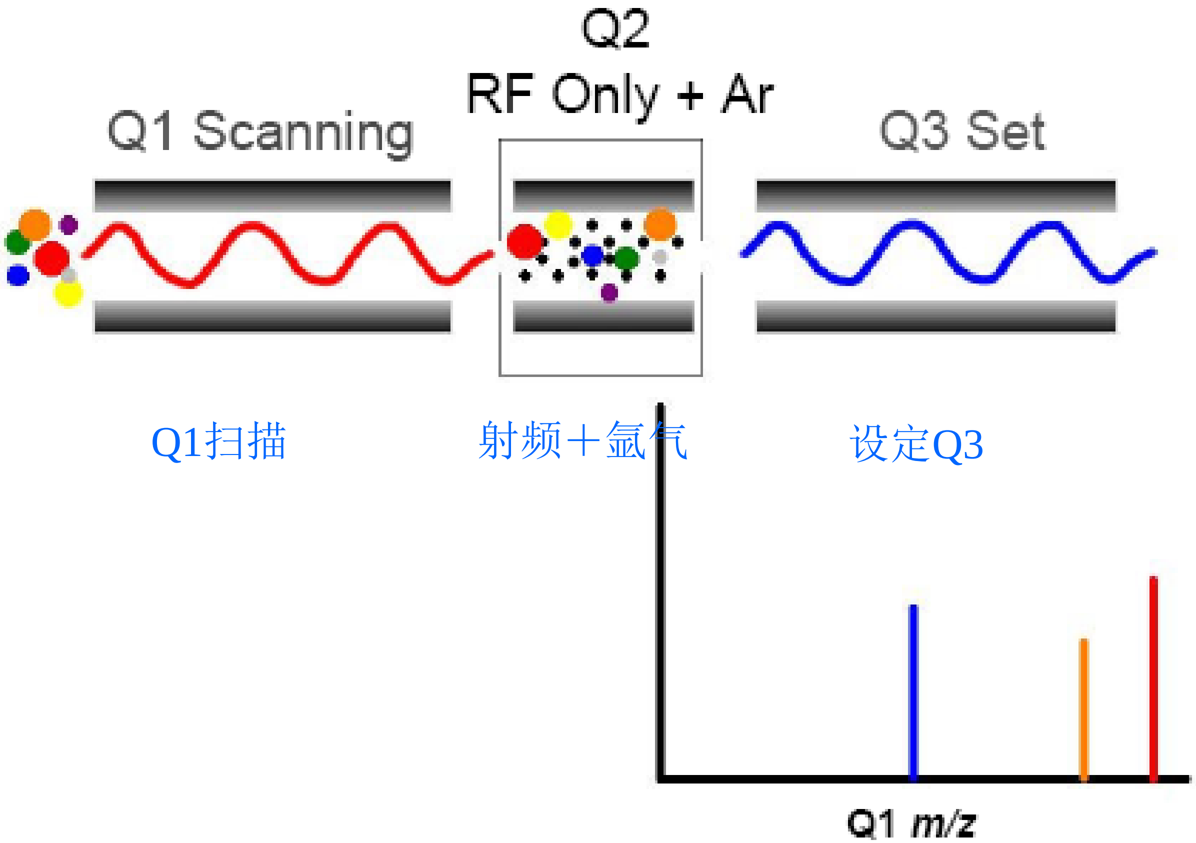
设定Q1

射频和氩气

Q3扫描

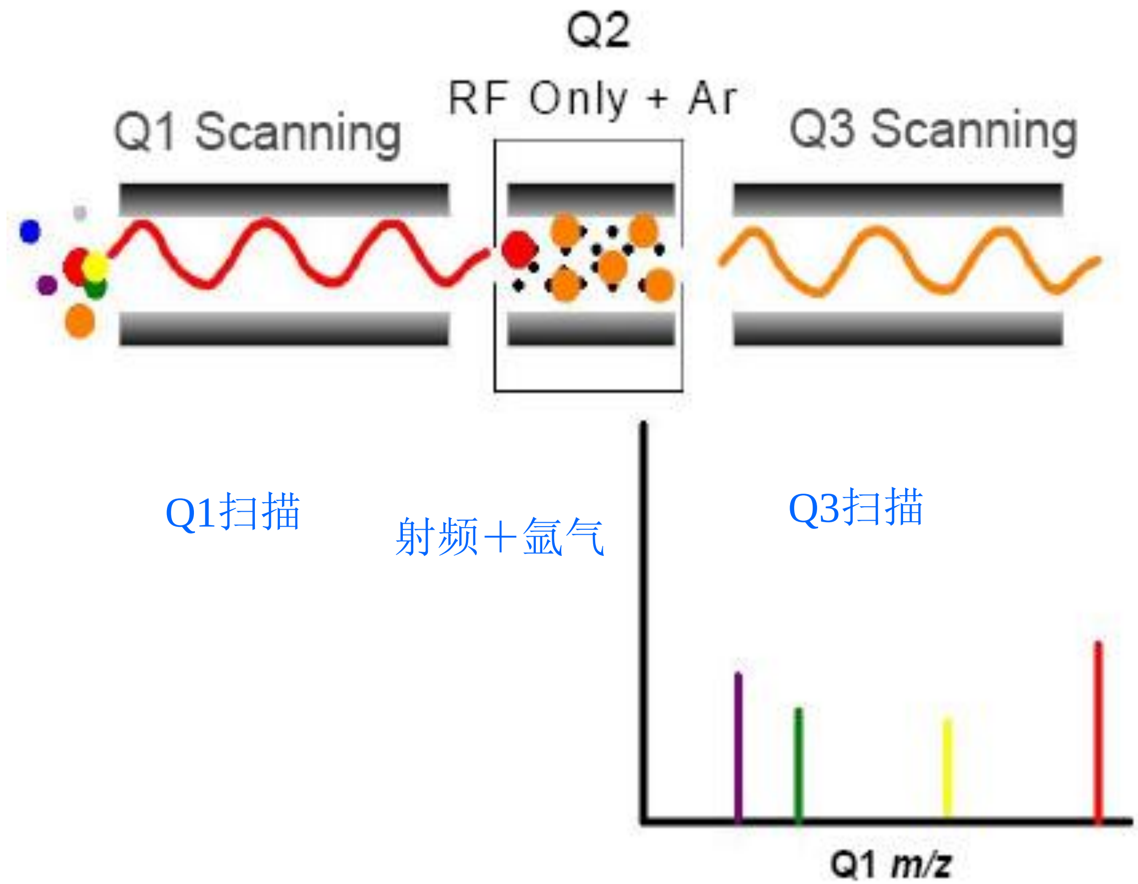
母离子扫描 (PS)

目的：检测产生特定碎片离子的母离子



中性丢失扫描（二级质谱扫描）

目的：化合物种类识别，例如磷酸化类化合物



TSQ三重四极杆质谱的特点及应用领域

特异性强

+

灵敏度高

适用范围

复杂且干扰严重的体系

多组分分析

痕量样品分析

以定量为主

食品安全

环境监测

临床法医毒物分析

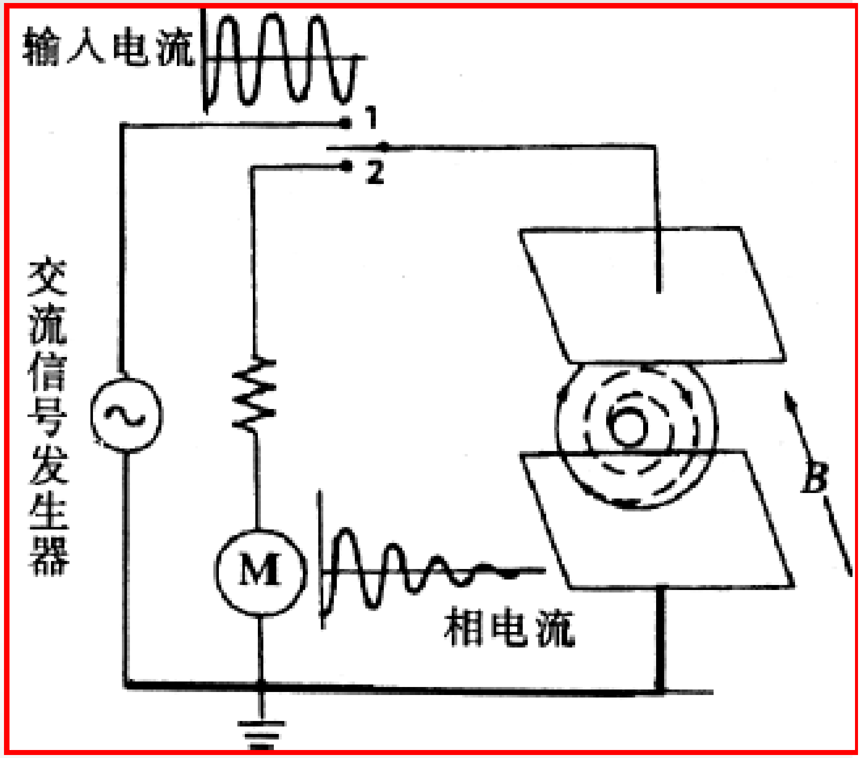
药物研发

Ion detection

5）离子回旋共振分析器(*Ion cyclotron resonance, ICR*)

过程:

处于磁场*B*中离子——回旋离子——吸收与*B*垂直的电场能量——当离子能量和吸收能量相等——共振——切断交变电场——回旋离子在电极上产生感应电流——感应电流衰减——记录该信号——通过*Fourier*变换将时域图转换为频域图(质谱图)。



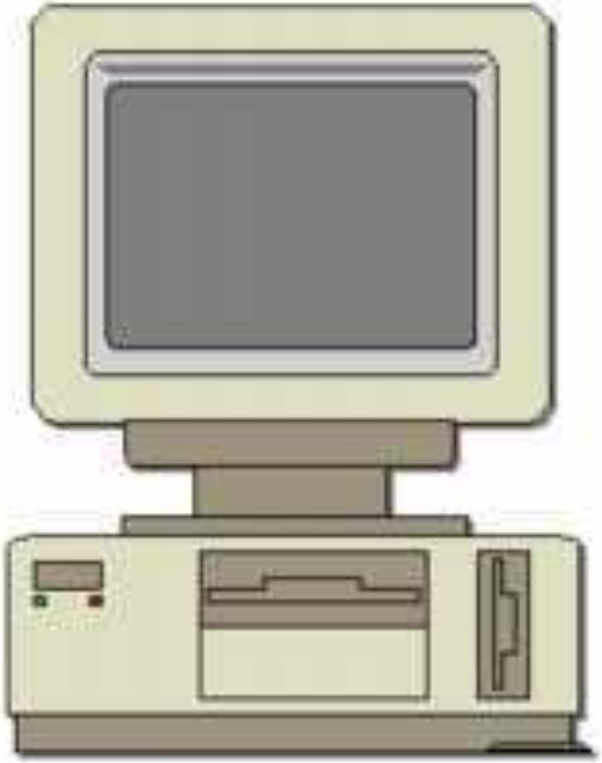
特点:

可用于分子反应动力学研究、扫描速度快，可与*GC* 联用、分辨率高、分析质量大、但仪器昂贵。

Data system

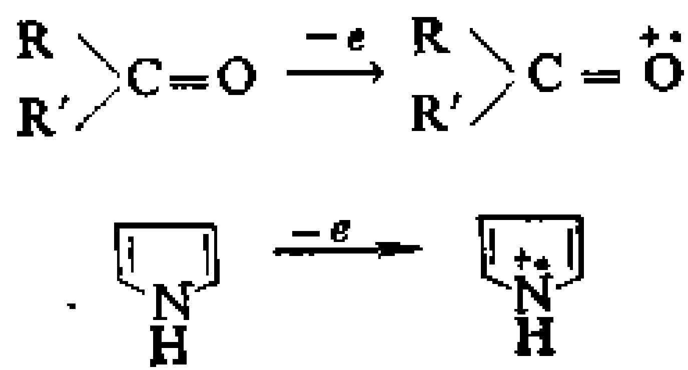
The data system is the heart of our MS system.

Modern MS systems rely on computers for rapid data collection, processing and identification of mass spectra.



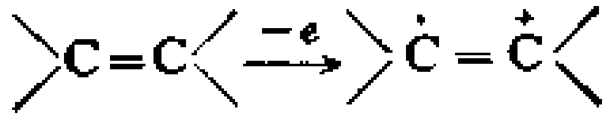
●分子离子峰(molecular ion peak)：分子失去一个电子形成分子离子，其质谱峰称为分子离子峰

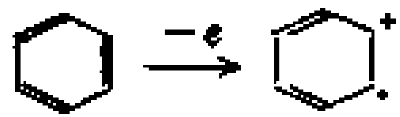
对于有机物，杂原子上未共用电子(n 电子)最易失去，其次是 π 电子，再其次是 σ 电子。所以对于含有氧、氮、硫等杂原子的分子，首先是杂原子失去一个电子而形成分子离子，此时正电荷的位置处在杂原子上，例如



上式中氧或氮原子上的"+·"表示一对未共用电子对失去一个电子而形成的分子离子。

含双键无杂原子的分子离子，正电荷位于双键的一个碳原子上：





当难以判断分子离子的电荷位置时可表示为"$]^{+\cdot}$"，例如

$$CH_3CH_2CH_3 \xrightarrow{-e} CH_3CH_2CH_3]^{+\cdot}$$

● **同位素峰(isotope ion peak)：**在质谱中由不同质量的同位素形成的峰，称为同位素离子峰。同位素峰的强度比与同位素的丰度比是相当的。

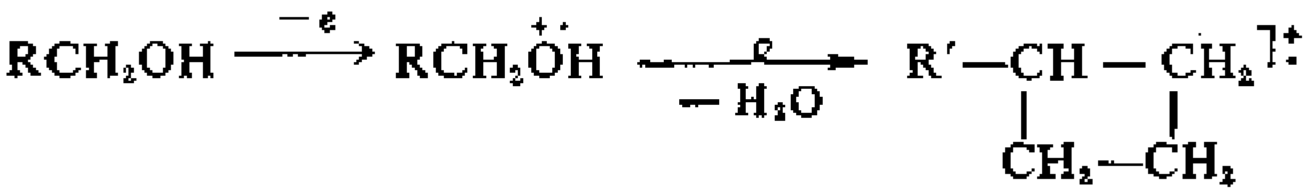
M+1, M+2峰等

●碎片离子峰（fragment ion peak）

●重排离子峰（rearrangement ion peak）

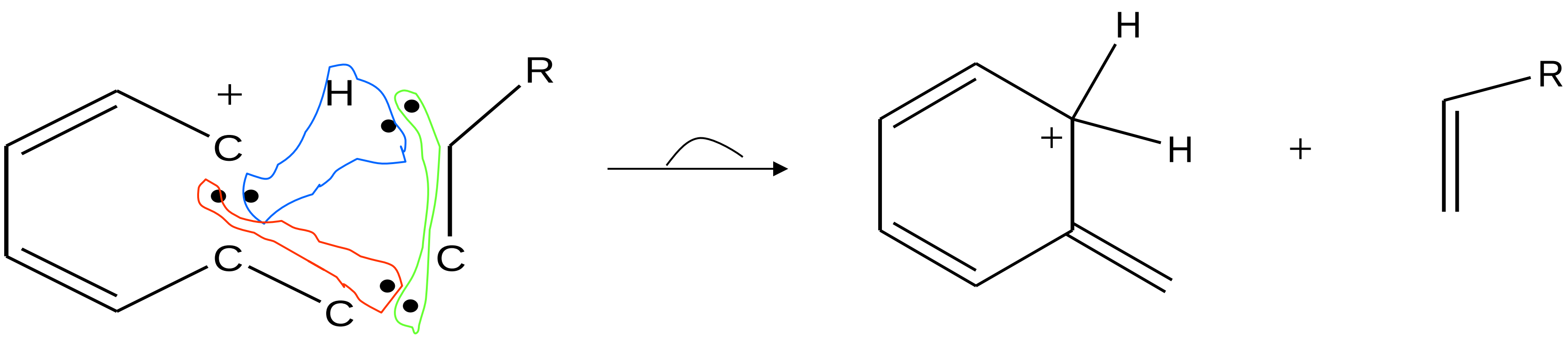
分子离子裂解为碎片离子时，有些碎片离子不是仅仅通过简单的键的断裂，而是还通过分子内原子或基团的重排后裂分而形成的，这种特殊的碎片离子称为重排离子

例如醇分子中引起脱水的重排：



上式中符号“$\xrightarrow{\ell}$”表示在裂解过程中发生了重排。

又如，当与化合物中**C=X(eg. C=O)**基团相连的键上有三个以上的碳原子，且γ 位上有**H**原子存在时，该**H**原子易于向缺电子的原子转移，从而脱去一个中性分子。例如：



含有双键的化合物如，醛，酮，链烯，酰胺，腈，酯，芳香化合物等均可出现此类重排(McLafferly rearrangement) 。

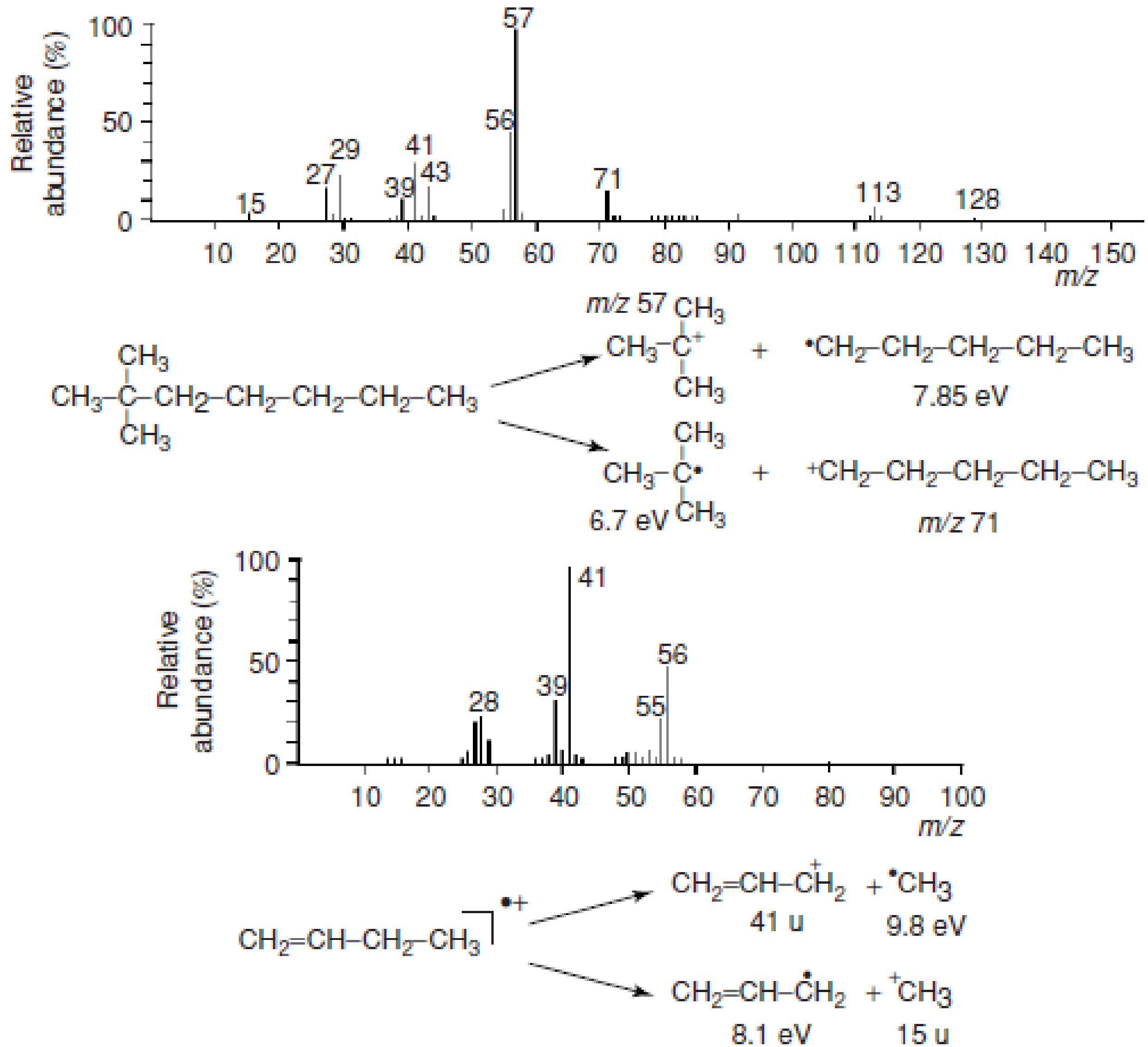
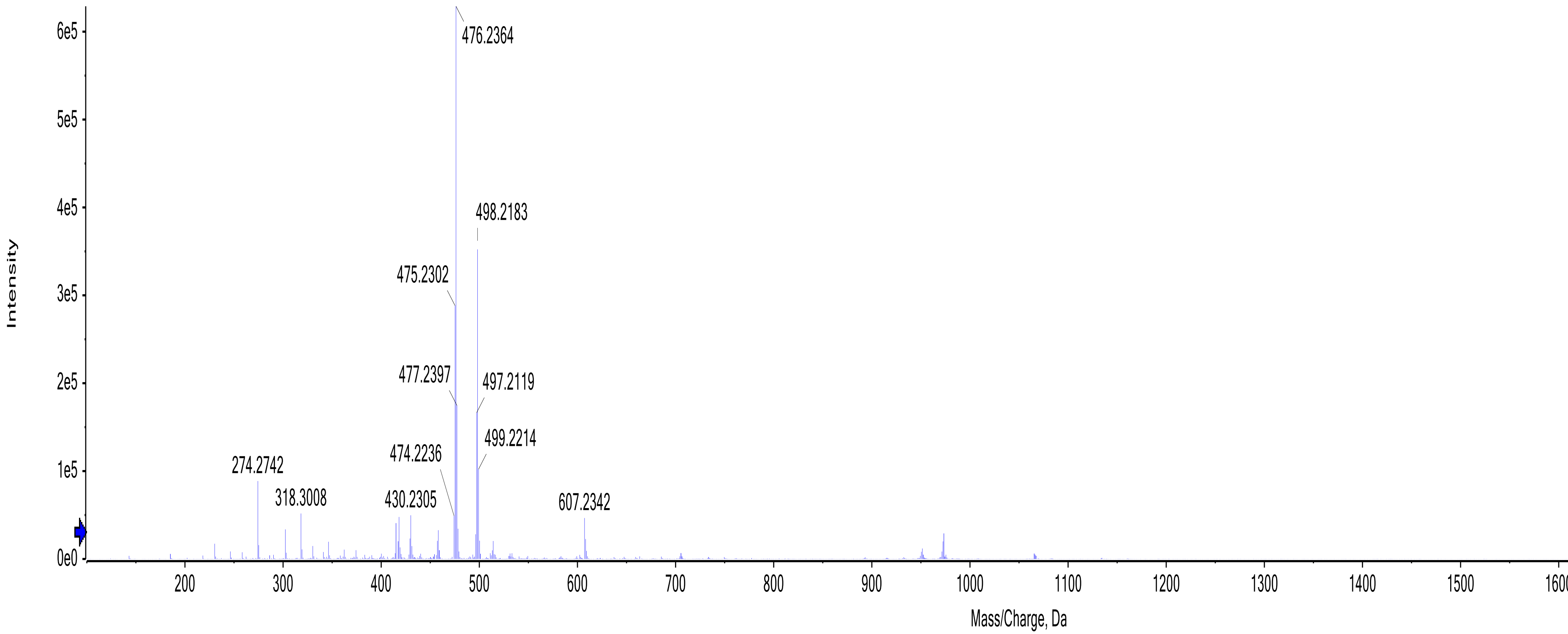
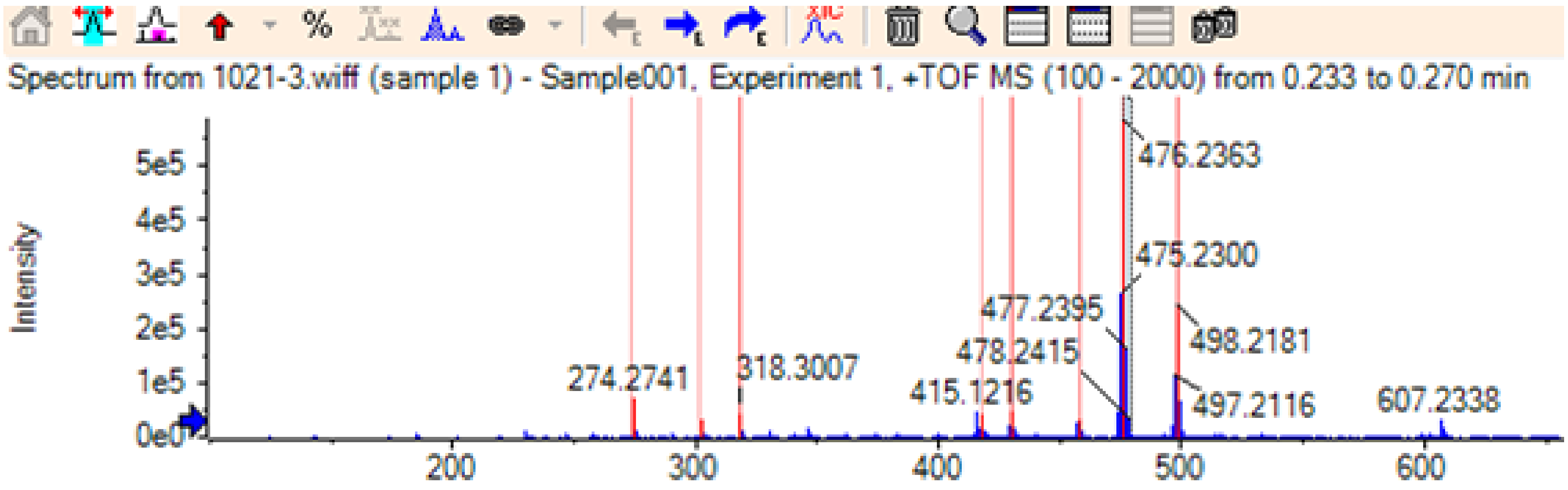


Figure 7.6
The ionization energy of the *t*-butyl radical is lower than that of the pentyl radical. Thus, the *t*-butyl ion is preferentially observed in the top spectrum. The same holds true for the comparison between the allyl and methyl radicals.

Spectrum from 1021-3.wiff (sample 1) - Sample001, Experiment 1, +TOF MS (100 - 2000) from 0.251 to 0.345 min





Found elemental compositions

Find Any

Find

Hit	Formula	m/z	RDB	ppm	MS Rank	MSMS ppm	MSMS Rank	Foun
1	*C19H24N16*	*476.2364*	*16.5*	*-0.3*	*1*			*NA/N*
2	C20H29N9O5	476.2364	11.0	-0.3	2			NA/N
3	*C21H36N2O10*	*476.2364*	*5.5*	*-0.3*	*3*			*NA/N*
4	*C6H32N14O11*	*476.2369*	*-1.5*	*-1.4*	*4*			*NA/N*
5	C4H29N17O10	476.2356	-1.0	1.5	5			NA/N
6	C36H29N	476.2373	23.0	-2.1	6			NA/N
7	C19H33N5O9	476.2351	6.0	2.5	7			NA/N
8	*C18H28N12O4*	*476.2351*	*11.5*	*2.5*	*8*			*NA/N*

MS Details | MSMS Details

Isotope cluster details

Peak	Use	m/z
0	☑	476.2363

Elements from

Elements to C50 H10

Mass tolerance (ppm)

Intensity tolerance (%)

#C/#heteroatoms greater

液相色谱质谱联用

离子化接口

- **液质联用的主要困难**

HPLC	MS
高压液相操作	要求高真空
液体进入离子源转变为大量气体	只允许有限的气体进入离子传输系统
常常使用无机盐缓冲盐	需采用挥发性缓冲盐
分子状态	离子状态

离子化方式

API（大气压离子化）